

A Tutorial of Resting-state fMRI/DTI-based Connectivity Analysis

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Please feel free to send feedbacks to my email: longspace82@gmail.com

My Github: <https://github.com/XiangyuLong/>

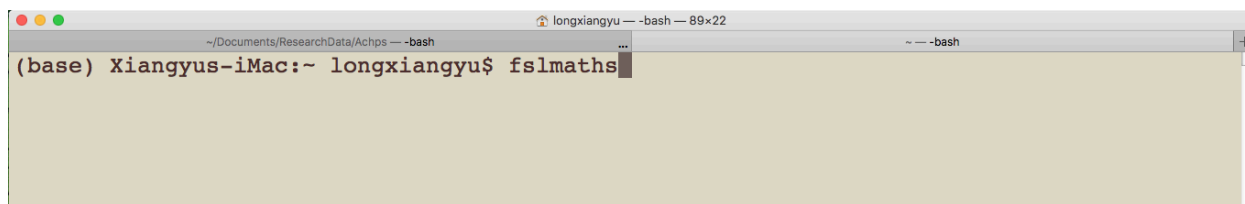
1. Softwares/Toolboxes requirement

Several software and MATLAB toolboxes are required for the data analysis and processing. Almost all lab computers pre-install the required software. But just in case, here are some ways to check if the specific software is installed or not on your lab computer:

For **MATLAB**, check if there was a MATLAB icon at the launchpad of Mac OS by pressing F4.

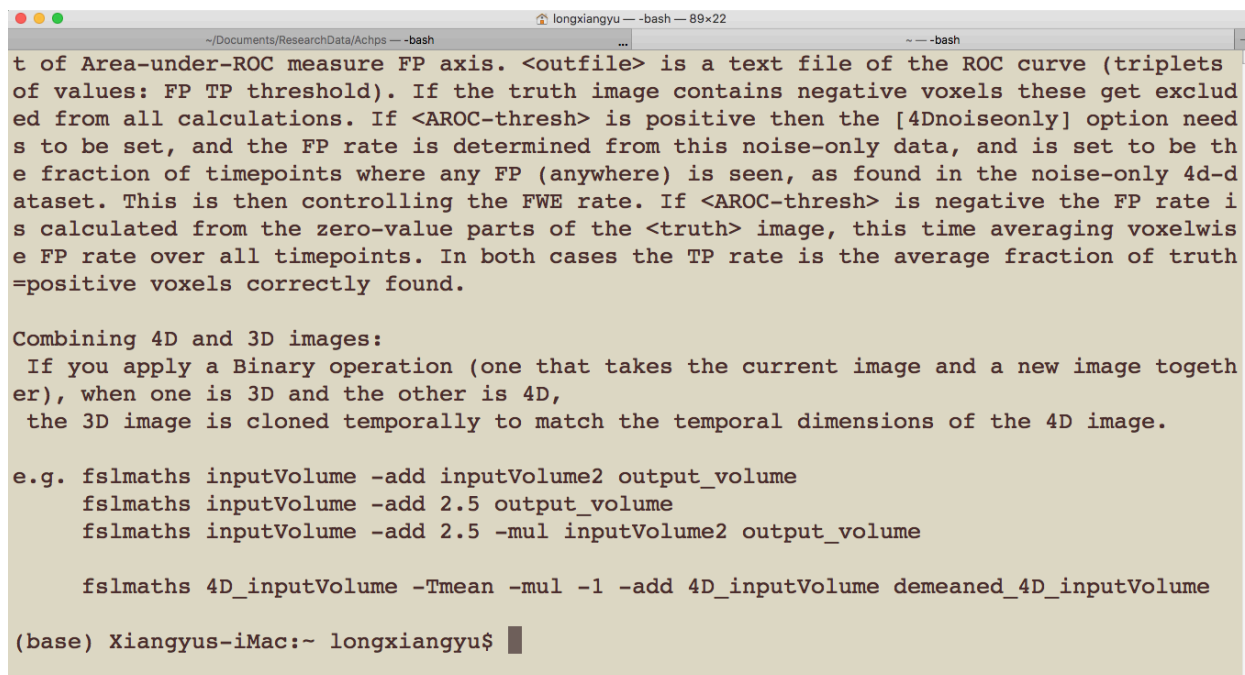
For **MRicron**, check if there was a Mricron icon at the launchpad of Mac OS by pressing F4.

For **FSL**, open a Terminal (In Mac OS, click Applications in Finder--->Utilities--->double click Terminal), then type “fslmaths” and press return in the terminal as follows:



```
longxiangyu -- bash -- 89x22
~/Documents/ResearchData/Achps -- bash
(base) Xiangyu-iMac:~ longxiangyu$ fslmaths
```

If a lot of messages were shown after press return as follows:



```
longxiangyu -- bash -- 89x22
~/Documents/ResearchData/Achps -- bash
t of Area-under-ROC measure FP axis. <outfile> is a text file of the ROC curve (triplets
of values: FP TP threshold). If the truth image contains negative voxels these get exclud
ed from all calculations. If <AROC-thresh> is positive then the [4Dnoiseonly] option need
s to be set, and the FP rate is determined from this noise-only data, and is set to be th
e fraction of timepoints where any FP (anywhere) is seen, as found in the noise-only 4d-d
ataset. This is then controlling the FWE rate. If <AROC-thresh> is negative the FP rate i
s calculated from the zero-value parts of the <truth> image, this time averaging voxelwis
e FP rate over all timepoints. In both cases the TP rate is the average fraction of truth
=positive voxels correctly found.

Combining 4D and 3D images:
If you apply a Binary operation (one that takes the current image and a new image togeth
er), when one is 3D and the other is 4D,
the 3D image is cloned temporally to match the temporal dimensions of the 4D image.

e.g. fslmaths inputVolume -add inputVolume2 output_volume
fslmaths inputVolume -add 2.5 output_volume
fslmaths inputVolume -add 2.5 -mul inputVolume2 output_volume

fslmaths 4D_inputVolume -Tmean -mul -1 -add 4D_inputVolume demeaned_4D_inputVolume
(base) Xiangyu-iMac:~ longxiangyu$
```

It means FSL is installed on your computer.

For **AFNI**, it is a similar procedure as for FSL, but type “3dcalc” instead of “fslmaths”.

One tip here, generally every time when a Terminal is opened, it will display the software installed in the current computer, something like this:

```
## cd /Users/longxiangyu/Documents/ResearchData/Misc/temp/rest ##
Last login: Fri Sep 13 14:29:58 on ttys004
----- freesurfer-Darwin-lion-stable-pub-v5.3.0 -----
Setting up environment for FreeSurfer/FS-FAST (and FSL)
FREESURFER_HOME    /Applications/freesurfer
FSFAST_HOME        /Applications/freesurfer/fsfast
FSF_OUTPUT_FORMAT  nii.gz
SUBJECTS_DIR       /Applications/freesurfer/subjects
MNI_DIR            /Applications/freesurfer/mni
FSL_DIR            /Users/longxiangyu/Documents/Softwares/fsl
(base) Xiangyus-iMac:~ longxiangyu$
```

If AFNI or FSL was installed in your computer before, but they are no longer be “configured” due to system re-install or users changing, you can simply add the directory of the software package back to the computer by the following steps:

- 1) Type “cd” in a Terminal;
- 2) Type “open .bash_profile” or “open .bashrc” or “open .zshrc” if any of these files exist in your computer;
- 3) If any of those “.bash_profile”, “.bashrc”, or “.zshrc” are not in your computer, simply create a text file and rename it to “.bash_profile”, “.bashrc” or “.zshrc” ;
- 4) Add “export PATH=\$PATH:/your_AFNI_or_FSL_directory/” to the “.bash_profile” and save;
- 5) Close the current Terminal and open a new one, then the new Terminal will load the “.bash_profile” and install AFNI or FSL in your computer;

You can find “/your_AFNI_or_FSL_directory/” by searching for “3dcalc” or “fslmaths” in Finder.

Tips: If the output from AFNI was +tlrc not +orig with the MNI NIFTI dataset, try the following command in the terminal:

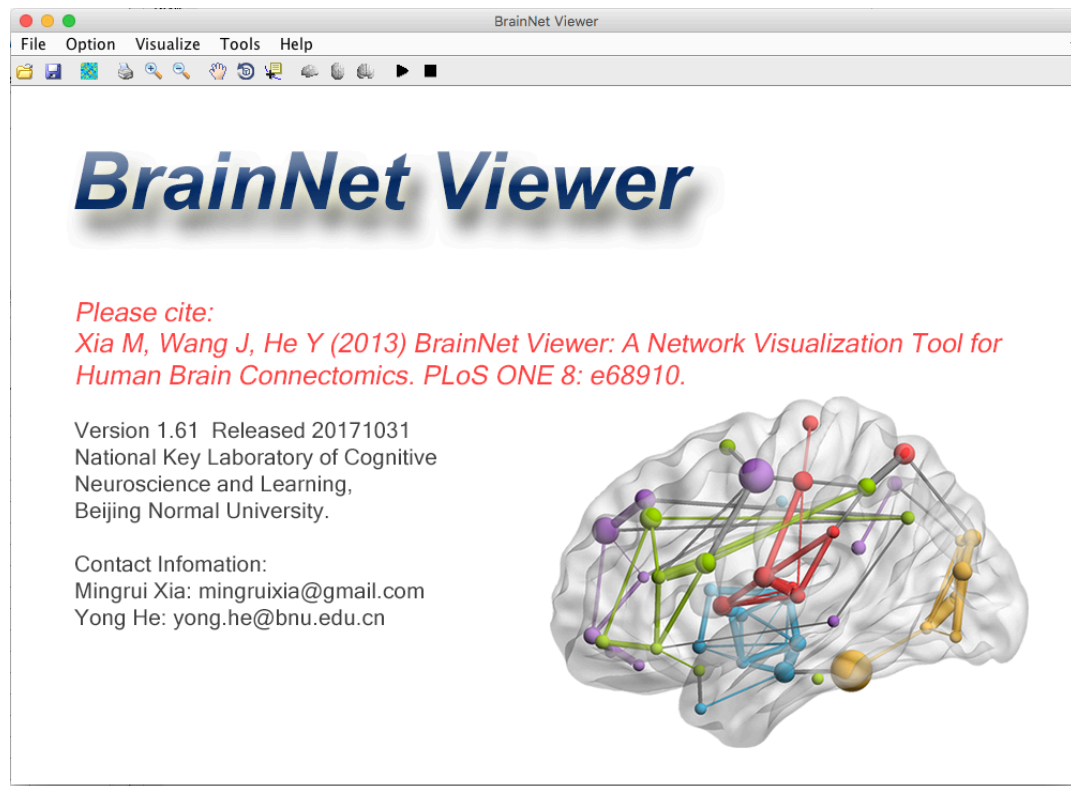
```
3drefit -view orig -space ORIG yourdata.nii
```

For **BrainNet Viewer** or **GRETNA toolbox**, open MATLAB, then type “BrainNet” or “gretna” in the command line in MATLAB:

Command Window

```
fx >> BrainNet|
```

if a GUI was popped up:



then the toolbox was installed.

For **ExploreDTI**, type "MainExploreDTI" and see if a GUI will pop up.

For the **NIFTI toolbox**, type "help load_nii" in the MATLAB command window.

If any of the software or toolbox were not installed, here are links for all software/toolboxes:

MATLAB: UofC software distribution

MRICron: <https://www.nitrc.org/projects/mricron>

AFNI: <https://afni.nimh.nih.gov/>

FSL: <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki>

BrainNet Viewer toolbox: <https://www.nitrc.org/projects/bnv/>

GRETNA toolbox in MATLAB (might not work well and haven't been updated since 2017):
<https://www.nitrc.org/projects/gretna>

Brain Connectivity Toolbox in MATLAB (basically a command line version of GRETNA, recommended if you are familiar with coding in MATLAB): <https://sites.google.com/site/bctnet/>

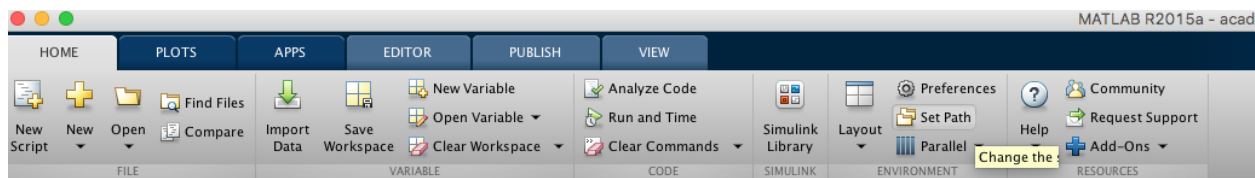
ExploreDTI toolbox: <http://www.exploredti.com/>

ANTs toolbox: <http://stnava.github.io/ANTs/>

NIFTI tools:

<https://www.mathworks.com/matlabcentral/fileexchange/8797-tools-for-nifti-and-analyze-image>

Please download them on your computer, then use “Set Path” in MATLAB to add them in your MATLAB environment (generally with subdirectory option):



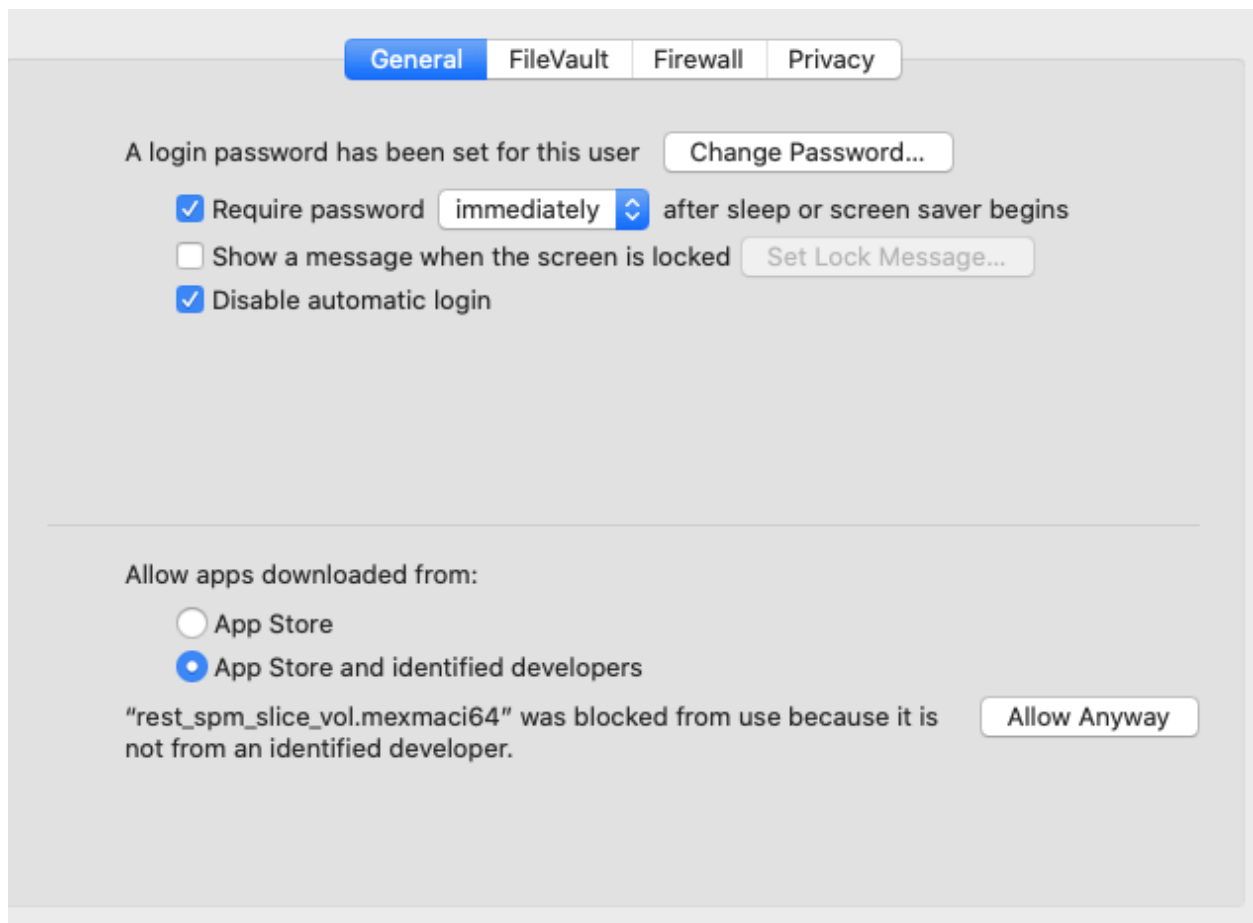
Generally all software have their own manuals/tutorials, please check them as well when going through this tutorial.

Also some MATLAB official toolboxes might be required, such as Image Process toolbox or Statistical toolbox, Curve Fitting, and Optimization toolboxes, please download them from the official MATLAB website if there's an error message shown during the usage of other toolboxes such as ExploreDTI.

One more tip for MAC users: If your MAC OS is very new like MAC OS 12+, sometimes you might see some pop ups saying the toolboxes/files were not allowed to run due it was not from the official Apple store, such as in following, a mexmaci64 file was not allowed to be used:



You can fix it by entering System Preferences in the MAC OS, then go to "Security & Privacy", and at the lower part of the interface of the General tab, you will see the related programs were blocked right after you click the "cancel" button from the pop up window, then click "Allow anyway". This will allow the MAC OS to run certain programs in future.



2. Pre-process resting-state fMRI data

1) The first step is to convert Dicom (*.dcm) format data to NIFTI format (*.nii or *.nii.gz). You can use “dcm2nii” from MRICron in Terminal, type the following command in Terminal:

```
cd /your_local_directory_of_dicom/  
/Applications/mricron/dcm2nii *.dcm
```

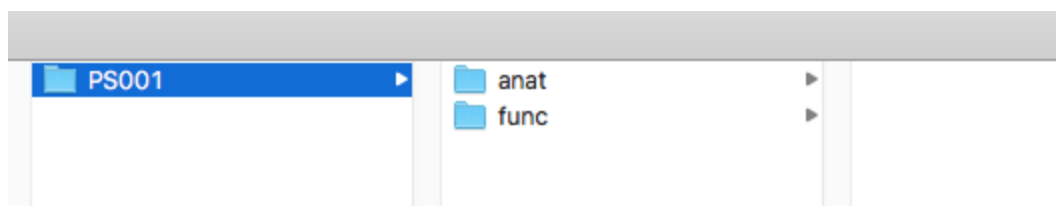
Please change the “/Applications/mricron/dcm2nii” accordingly with your computer settings. If there was an error message “Argument list too long”, then could try to run dcm2nii for the folder:

```
/Applications/MRICron.app/Contents/Resources/dcm2niix -z y -o your_output_directory -f rest  
./your_local_directory_of_dicom/
```

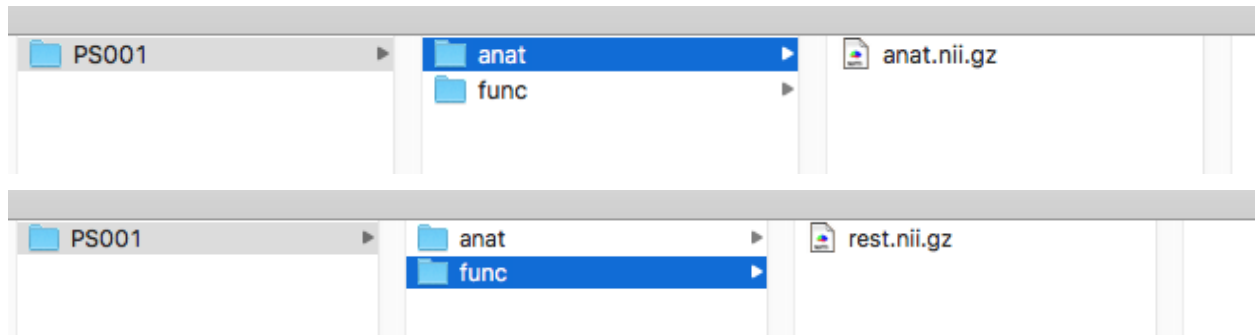
Just to make sure there are only Dicom files in the folder, and no other types of files in the folder. “-f rest” here means the name of the output files will start with “rest”, you could change it accordingly.

This procedure should be run for both T1 and fMRI Dicom images. It will create something like “2xxxx.nii.gz” for fMRI and “2xxxx.nii.gz”, “o2xxxx.nii.gz” and “co2xxxx.nii.gz” for T1 (or the name after the option “-f”). The file “co2xxxx.nii.gz” is a neck removal image, and could be used in the following analysis.

2) The next step is to construct a file structure for the pre-processing. Create a folder with an identifiable name, such as “PS001”, then create a subfolder named “rest” and another subfolder named “anat” as follows:



Then copy “co2xxxx.nii.gz” from T1 dicom images folder to the “anat” folder and renamed it as “anat.nii.gz”; copy “2xxxx.nii.gz” from fMRI dicom images folder to the “func” folder and renamed it as “rest.nii.gz” as follows:



3) There are several bash scripts and MATLAB codes to pre-process resting-state fMRI (rs-fMRI) data as constructed above. The bash scripts should be run in Terminal, and the MATLAB codes should be run in MATLAB. You can copy those files from the lab server and paste them to your local hard drive. The description of the pre-processing could be found in this paper <https://www.sciencedirect.com/science/article/pii/S2213158219304292?via%3Dihub>:

Here are the list of bash scripts and MATLAB codes for pre-processing:

`gbip_restfMRI_36regs.sh` : T1 image segmentation, slice timing and head motion correction for fMRI images, image coregistration between T1, standard MNI template and fMRI images, generating nuisance signals;

`gbip_restfMRI_create36regs.m` : create nuisance signals regressors

`gbip_restfMRI_36regs_filter_smooth.sh` : nuisance signals removal, band-pass filtering, spatial smoothing, transform to standard MNI space;

`gbip_restfMRI_run36regs.m` : how to create nuisance signals for multiple dataset

`gbip_restfMRI_runscripts.sh` : how to run the scripts for multiple dataset

If the “sh” or shell file can not be executed in your own computer, e.g., some error message of “Permission denied”, run the following command in the Terminal at the directory of the shell file:

```
chmod +x gbip_restfMRI_36regs.sh
```

This command will let the shell file be executable, so run this for all shell files which gives the “Permission denied” error.

Generally a loop command is used to run the analysis for multiple datasets or participants. To do so, a text file including a list of folders’ names is required. Assume there were two folders/participants, a text file named “datalist.txt” was created:



Then use the following codes in Terminal to run the scripts step by step for each participant:

4) The procedure of pre-process fMRI data:

First step, run `gbip_restfMRI_36regs.sh`. This script is a combination of the old scripts and needs the input of TR (unit: seconds), the directory of all folders/participants (“/your_data_directory/” in the example below), the standard brain template (skull-stripped) including whiter matter and cerebrospinal fluid (CSF) mask, the volumes range (the first time point starts with 0) and the threshold of head motion framewise displacement (FD). Run the following codes by copying and pasting all of them in a Terminal:

```
subject_list=/your_local_directory/data_list.txt
subjects=$( cat ${subject_list} )
for subject in $subjects
do
/your_local_directory/gbip_restfMRI_36regs.sh ${subject} 2 /your_data_directory/
/your_local_directory/nihpd_sym_04.5-08.5_nifti/nihpd_sym_04.5-08.5_brain.nii
/your_local_directory/nihpd_sym_04.5-08.5_nifti/nihpd_sym_04.5-08.5_csf.nii
/your_local_directory/nihpd_sym_04.5-08.5_nifti/nihpd_sym_04.5-08.5_wm.nii 0 239 0.3
done
```

Tips: If you are using `zsh` shell in the latest MacOS, you need to change the first three lines of this batch code to the following:

```
subject_list=/your_local_directory/data_list.txt
for subject in $( cat ${subject_list} )
```

“`${subject}`” is the name of the individual folder, which has the “anat” and “func” folders

Parameter “2” is the TR, the unit is seconds. So if TR of your data is 1010ms, the input should be 1.01.

“/your_data_directory/” is the directory which includes all individual folders.

In this example, a 4.5-8.5 years old brain template is applied. You can find them at the lab server (/Volumes/catherine_team/Long/BrainTemplates/). You can also use other brain templates. Here are some pediatric templates at different age stages:

<http://www.bic.mni.mcgill.ca/ServicesAtlases/NIHPD-obj1>. The anatomical brain has to be skull-stripped T1 image. The CSF and white matter mask can be binary masks.

Parameters "0 239" means volumes from 0 to 239 are included, i.e., 240 volumes. You can change the volume number according to your own data. The volume number of the image can be checked by opening with MRICron.

The threshold of FD is 0.3mm, which is a common head motion threshold for resting-state fMRI.

Tips: The value stored in "rest_mcf_abs_mean.rms" is a half of the frame-wise displacement (FD), so it times two could be used to report the FD.

Tips: For slice timing, the default setting is "interleaved", if you want to change it, find the command "slicetimer" in the gbip_restfMRI_36regs.sh file, then change the option "-odd" to others. If you want to check the slice timing order of the images, either ask technicians or use "dicominfo" in MATLAB, use MRICrogl from MRICron to get the json file. Then use "gbip_restfMRI_36regs_custom_st.sh" to run the pre-processing with a customized slice order file.

More tips on slice timing: The json file should be in the same folder of the DICOM/NIFTI files after running "dcm2niix".

This file can be opened in the text editor, and there is a row named "Slice timing" with a series of numbers. Those numbers are the "time stamps" when scanning each slice. So the rows of these numbers are the same as the number of slices of fMRI data.

Then you can create an empty array in MATLAB, such as:

```
A = [];
```

Then open "A" in the workspace in MATLAB, copy those numbers in the workspace, then use command "sort":

```
[ia ib] = sort(A);
```

The output "ib" is the order of the numbers, then you can see if "ib" is something like "1 3 5 7 92 4 6 8", this order is "interleaved" which is the default setting of my preprocessing script. If "ib" looks like something else, then you need to use the "gbip_restfMRI_36regs_custom_st.sh" to do the pre-processing.

Second, check the `gbip_restfMRI_create36regs.m` in MATLAB. You can also modify this script on multiple dataset/participants. More details can be found in `gbip_restfMRI_run36regs.m` as below:

```
% 36 regressors
clear;
allid = dir('/your_data_directory/*');
allhm = zeros(length(allid),1);
tr=2;
for i = 4 : length(allid)
    datadirectory = [datadir, '/',allid(i).name, '/func/regressors/'];
    allhm(i) = gbip_restfMRI_create36regs(datadirectory,tr);
end
size(find(allhm==1))
% remove high head motion data, <4min
for i = 1 : length(allhm)
    if allhm(i) == 1
        unix(['mv ',datadir,'motiondata/',allid(i).name,' ',datadir,'lessthan4/']);
    end
end
```

This is just an example of processing multiple data folders, so you can modify it accordingly. For instance, in this m-file, you can see a loop with “for i = 4 : length(allid)” to loop all participants/folders inside “/your_data_directory/”. The “i” starts at 4 because sometimes the 4th filename is the first folder in the OS system, so it can be changed accordingly. “datadir” can be defined as the directory of your data. “for i = 4 : length(allid)” means the code will access the from the 4th sub-directory to the last sub-directory in the mother folder.

The data directory should look like this, only contains the folders with data, nothing else:



Then this m-file will launch another m-file called “gbip_restfMRI_create36regs” to create a 36 regressors file (please check the methods part in) named “36regsandspikes.1D” in the ‘/func/regressors/’ directory. The input of “gbip_restfMRI_create36regs” are the data directory, in this case, ‘/PS001/func/regressors/’, and TR, which unit is seconds.

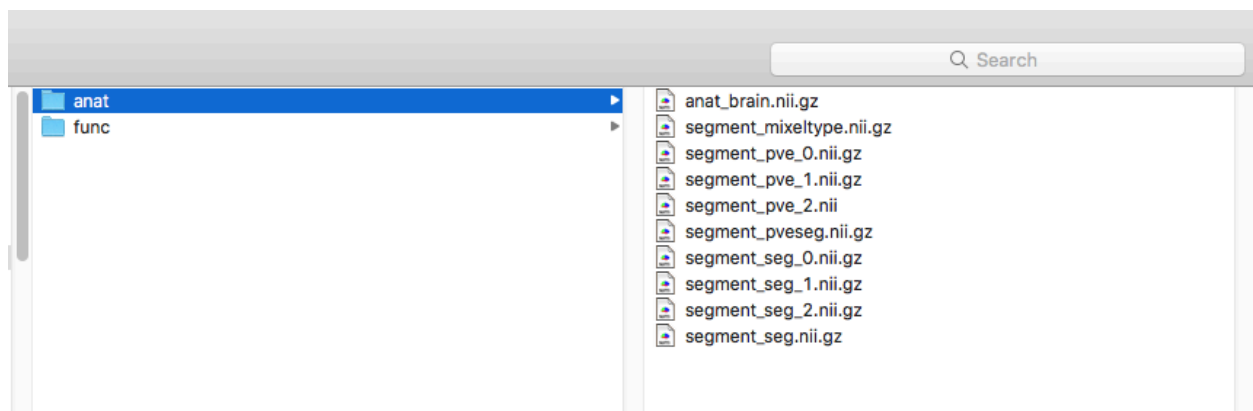
The output of “gbip_restfMRI_create36regs” (e.g., allhm in the previous m-file) is a binary value, 0 or 1, 1 means the time length is less than 4 minutes after removing high motion time points, then this dataset will be too noisy to be available. The second loop of “gbip_restfMRI_run36regs.m” is just to remove the noisy dataset/folder out of the original directory. In a word, this is just an example, please modify and run the codes above accordingly, such as creating a folder named “lessthan4” :)

Third step, after finishing running “gbip_restfMRI_create36regs.m”, here is the final step of pre-processing, including band-pass filtering, spatial smoothing and nuisance signals removal. Run gbip_restfMRI_36regs_filter_smooth.sh as follows in a Terminal:

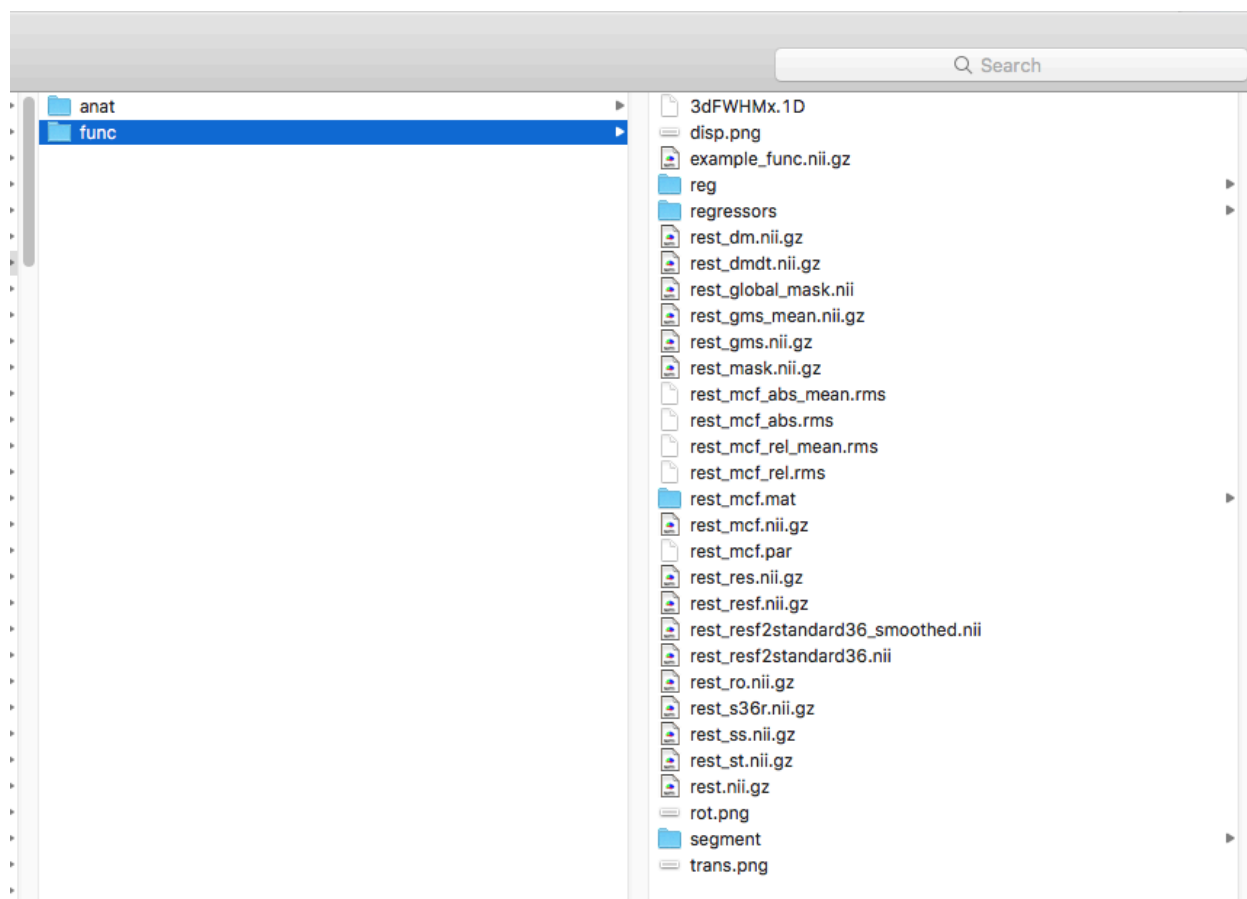
```
subject_list=/your_local_directory/data_list.txt
subjects=$( cat ${subject_list} )
for subject in $subjects
do
/your_local_directory/gbip_restfMRI_36regs_filter_smooth.sh ${subject} 4 /your_data_directory/
done
```

The input of gbip_restfMRI_36regs_filter_smooth.sh are the smoothness parameter and your data directory. The smoothness parameter unit is in millimeters. So in this example, 4 mm FWHM was applied on the fMRI data. This parameter could be changed, generally around 2 times of the voxel size, but not too high, since higher smoothing will introduce pseudo-positive correlations (i.e., functional connectivity) between fMRI signals.

After all the steps, the folder “anat” for each participant/dataset should look like this:



And the “func” folder should be like this:



Here, the file “rest_resf2standard36_smoothed.nii” is the final data that can be used in the second level analysis, such as functional connectivity or connectome analysis.

Also, if any error message appeared, then better run the process again, just delete everything except rest.nii.gz and anat.nii.gz.

Tips: There might be errors when regressing out the 36 regressors due to their auto-correlations. Some participants' noise signal regressors (36regsandspikes.1D) might include data with linear relationships, so they can not be regressors due to autocorrelations. So the participants with this error don't have the "rest_s36r.nii.gz" file in their "func" folders.

One solution is to remove the signals which show highly correlated with other signals. To do so, I generally run a self-correlation of this file in MATLAB, such as:

```
cd /your_data_folder/subject01/func/regressors/;
load 36regsandspikes.1D;
a = corr(X36regsandspikes);
a(a==1) = 0; % remove self-correlation elements
a(a<0.95) = 0; % remove lower correlation coefficients
```

Then I open "a" in workspace, check which row/column has high correlation coefficient such as "0.96" or "0.99", then I remove that column for "a":

```
allcovs = X36regsandspikes;  
allcovs(:,10) = []; % Assume column 10 shows high correlation with other columns
```

Then backup the old 1D file (change the raw 36regsandspikes.1D to something like "36regsandspikes_backup.1D"), save the new "allcovs":

```
save(['36regsandspikes.1D'],'allcovs','-ASCII','-DOUBLE','-TABS');
```

Then re-run the "gbip_restfMRI_36regs_filter_smooth.sh" script to see if this issue was solved.

3. Create first-level maps from the pre-processed rs-fMRI data

3.1 Here I will introduce how to create regional homogeneity (ReHo), amplitude of low frequency fluctuations (ALFF), eigenvector centrality (EC) maps from the pre-processed rs-fMRI data. You can check this paper for more details: <https://www.ncbi.nlm.nih.gov/pubmed/28461057>.

For EC maps, a MATLAB toolbox named "fastECM" should be installed: <https://github.com/amwink/bias/tree/master/matlab/fastECM>. You can also copy the folder "fastECM" from the lab server:

For ReHo and ALFF analysis, two MATLAB codes are needed: gbip_reho.m and gbip_alff.m.

The commands in MATLAB to use those codes look like this:

```
gbip_reho(ADataDir, voxelsize, AMaskFilename, AResultFilename);
```

```
gbip_alff(ADataDir,voxelsize, ASamplePeriod, HighCutoff, LowCutoff, AMaskFilename,  
AResultFilename);
```

```
fastECM(ADataDir,0,0,1,16,AMaskFilename);
```

Here, "ADataDir" is the name of fMRI data, such as
"/your_data_directory/rest_resf2standard36_smoothed.nii";

“voxelsize” is the voxel size in millimeters, such as “3”;

“AMaskFilename” is the name of a binary brain mask, means it will only calculate the non-zero voxels in the mask; This mask could be created based on the data or a template. The resolution and dimension of this mask must be the same as the fMRI data;

“AResultFilename” is the name of output, such as “PS001_reho.nii”;

“ASamplePeriod” is TR in seconds, such as “2”;

“HighCutoff” and “LowCutoff” are sort of fixed values. They should be “0.08” and “0.009”. This is the band of the low frequency fluctuations;

After generated the spatial maps, such as “PS001_reho.nii”, a further standardization step is general needed. This step will standardize the individual ReHo (or ALFF and ECM) map to z-score so the maps could be compared across populations. This step could be done in by AFNI in the Terminal, the codes look like this:

```
3dcalc -a PS001_reho.nii -expr '(a-mean(a))/stdev(a)' -prefix PS001_reho_z.nii
```

The PS001_reho.nii image must be masked out the voxels outside the brain. Then the final image “PS001_reho_z.nii” can be used for further statistical analysis.

3.2 Create seed-based functional connectivity maps

This section will introduce how to create seed-based functional connectivity (FC) maps. Only two elements are required for such analysis: a region-of-interest (ROI, also known as “seed” region) and a 4-D fMRI data.

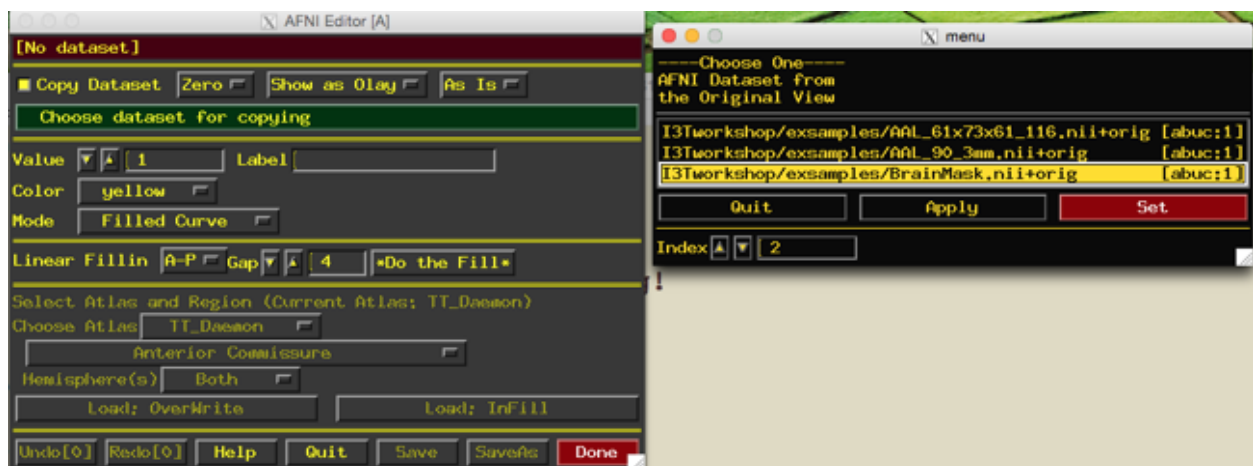
3.2.1 Create a ROI mask

There are many ways to create an ROI mask. One way is to simply draw a region on a volumetric map. I’m using AFNI to do such thing:

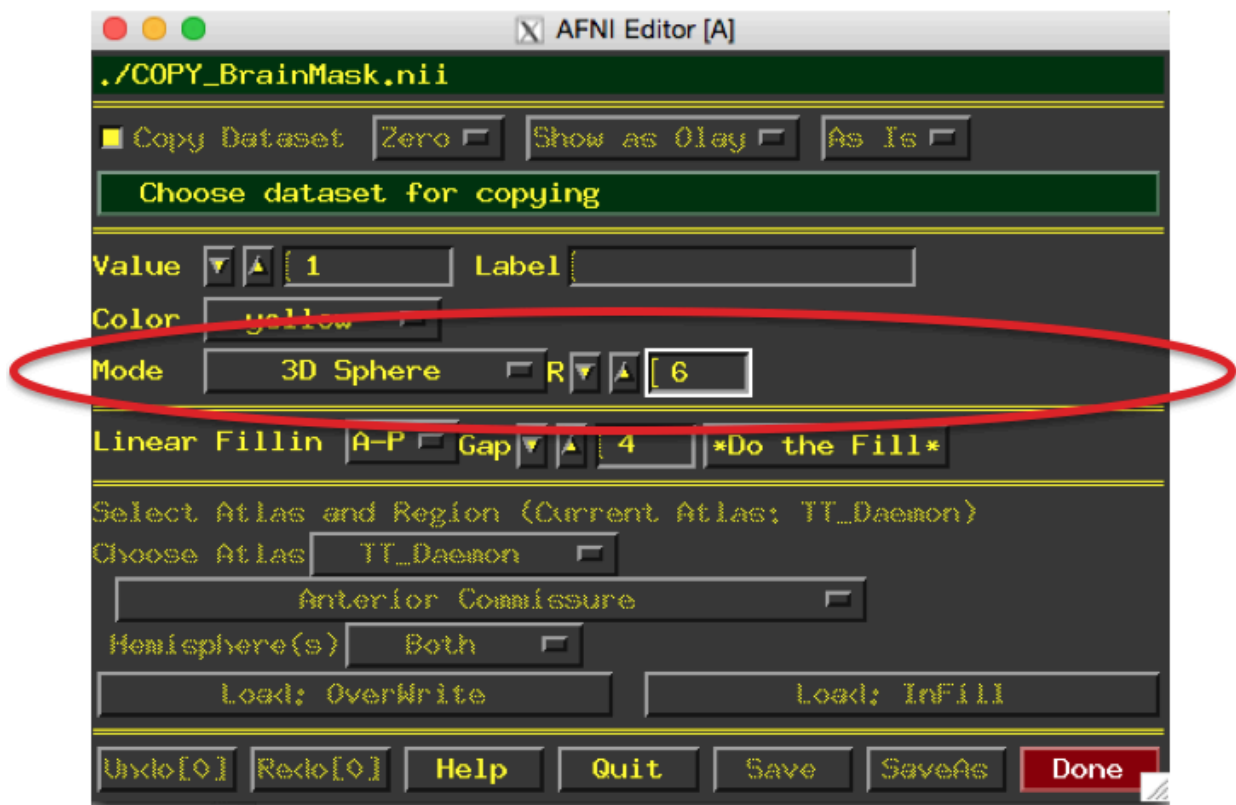
Select “Define Datamode ->” then “Plugins” then “Draw Dataset” in AFNI:



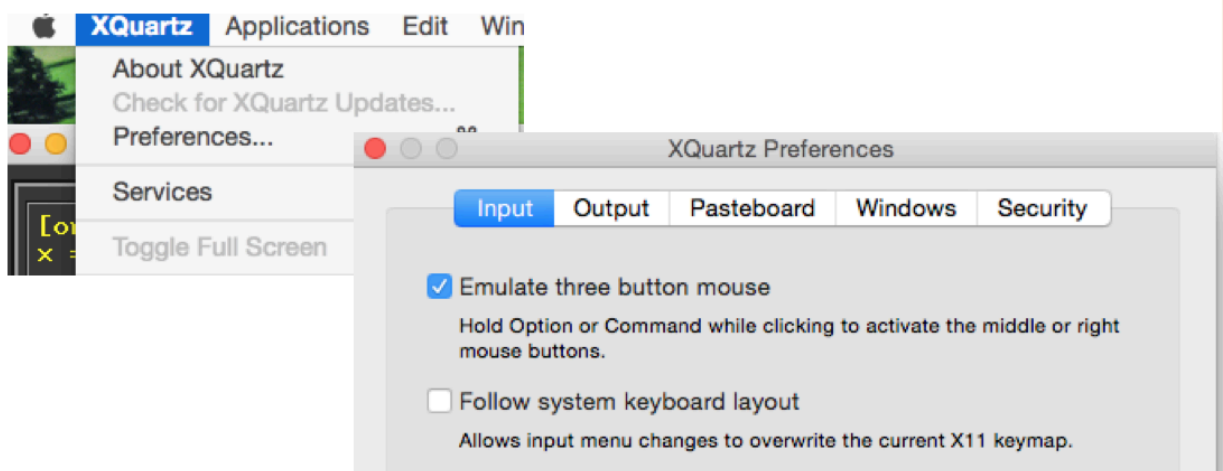
Then click “Choose dataset for copying” and select a NIFTI image which is in the same space with the 4-D fMRI data in a pop-up window, here a “BrainMask.nii” was selected:



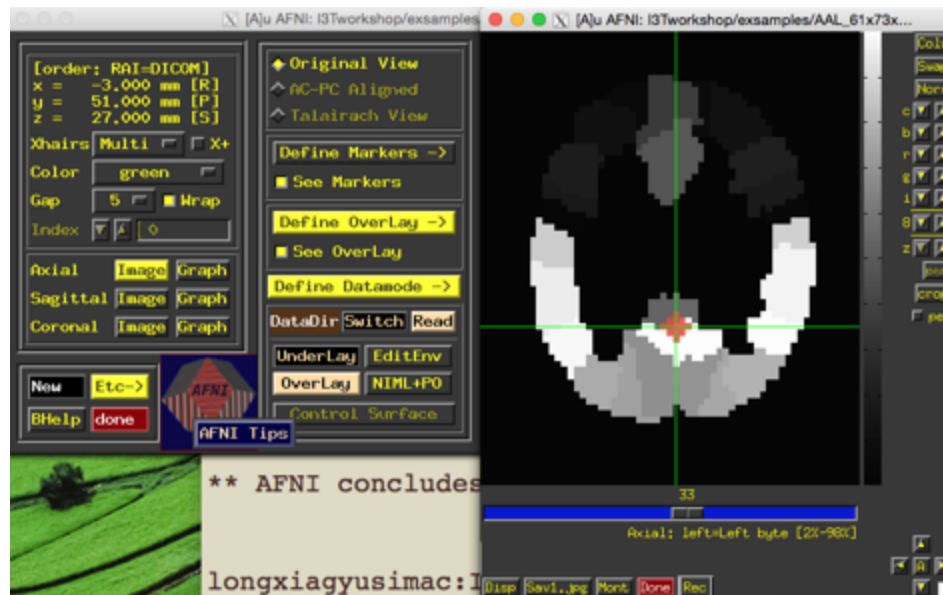
Then change the “Mode” to “3D Sphere”. This option will create a sphere as an ROI mask. “R” is the radius of the ROI in millimeters:



Then the middle-button of the mouse (the computer one, not the animal) needs to be enabled. To do so, click “XQuartz” at the AFNI menu then “Preferences...”, then select “Emulate three button mouse”:



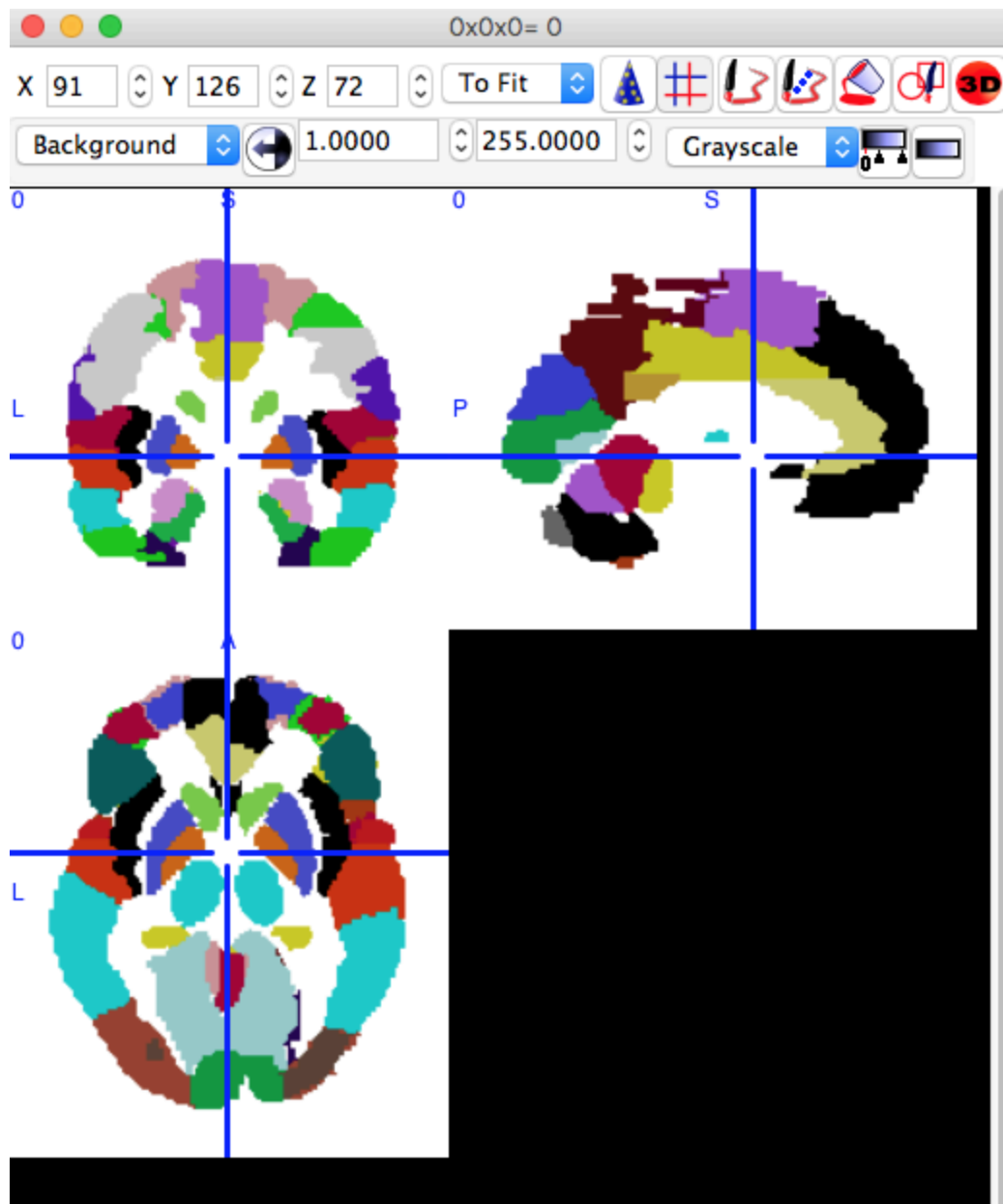
Now you can use the combination of “Option” button in MAC OS and right click as the middle mouse buttons. The center of the middle mouse button clicking will be the center of ROI. You can click “A” at the right corner of the viewer GUI to make the green cross closed:



Then click “SaveAs” in the Draw Dataset window and input a name for the ROI mask, here I named it “PCC_ROI.nii”. You can find this file in the directory when opened AFNI:



Another way to create an ROI mask is selected from an atlas, such as Automated Anatomical Labeling (AAL) atlas. There is one in MRICron if selects “File” then “Open templates” then “aa.nii.gz”:



This atlas includes 116 brain regions. The list of the regions can be found at `'/your_MRIcron_directory/mricron/mricron.app/Contents/MacOS/templates/aal.nii.txt'` (also `aal.nii.gz` or at lab server). We can select any region as an ROI for the analysis. Here is the AFNI command to select an AAL region, in this example, the 11th region from the atlas:

```
3dcalc -a aal.nii.gz -expr 'within(a,10.5,11.5)' -prefix AAL_11.nii
```

The option `'within(a,10.5,11.5)'` means a value fall into the range between 10.5 and 11.5, which is exactly "11" in this case, so you can change this range to other values for different regions,

such as if you want to extract a region with the value of 20, then use “within(a,20.1,21.7)”, any range includes value 20 is okay in this option.

Then this AAL_11.nii needs to be in the same space with the 4-D fMRI data. One way to transform this AAL_11.nii to the space of the 4-D fMRI data is to use FLIRT in FSL:

```
flirt -ref template_of_fMRI_data.nii -in MNI_T1.nii -out AAL2template -omat AAL2template.mat  
-cost corratio -dof 12 -interp trilinear
```

This step needs the T1 template in the MNI space that uses in spatial normalization of 4-D fMRI data and a T1 image in MNI space, because the original AAL atlas aal.nii.gz is in MNI space. This step will output a transformation matrix “AAL2template.mat” that can be used to transform any MNI space images to the spatial normalized 4-D fMRI space. The command above can do such transformation:

```
flirt -ref template_of_fMRI_data.nii -in AAL_11.nii  
-out AAL_11_fMRI -applyisoxfm 3 -init AAL2template.mat -interp trilinear
```

Here “3” is the voxel size of 4-D fMRI data, the unit is millimeter. The output AAL_11_fMRI.nii.gz will be the ROI mask for the functional connectivity analysis.

Another way is to use 3dresample if the functional fMRI data was already in the standard MNI space:

```
3dresample -input AAL_11.nii -master template_of_fMRI_data.nii -prefix AAL_11_fMRI.nii
```

Please always double-check if the ROI file is a binary image (0 value outside the region, 1 value within the region). You can run additional step to make sure of it:

```
3dcalc -prefix AAL_11_fMRI_bin.nii -a AAL_11_fMRI.nii -expr 'step(a)'
```

3.2.2 Perform seed-based functional connectivity analysis

After the ROI mask is created, first is to extract the averaged time series within the ROI from 4-D fMRI data using a AFNI command “3dmaskave”:

```
3dmaskave -quiet -mask PCC_ROI.nii 4D_fMRI.nii.gz >>seed.1D
```

The averaged time series within the ROI is stored in “seed.1D”. It is a AFNI format text file, so you can load it in MATLAB or just open it with a text file editor.

Then the FC map can be generated by a AFNI command “3dTcorr1D”:


```
3dTcorr1D -pearson -mask BrainMask.nii -Fisher -prefix Corr_example_z.nii 4D_fMRI.nii.gz seed.1D
```

BrainMask.nii should be a gray matter mask. The output is Corr_example_z.nii contains the correlation coefficients between the time signal of each voxel and the seed time signal. The command 3dTcorr1D can also be used to run the voxel-wise correlation between a 4-D dataset and any variables, such as age or cognitive scores, just replace seed.1D with a file containing the variables you want to correlate with. “-Fisher” means Fisher’s z transformation, because the correlation coefficient is not normally distributed, so it should be transformed to the normal distribution for further statistical analysis.

The final file Corr_example_z.nii can be used for the second-level analysis.

4. Run statistical analysis on the first-level maps

After creating the first-level maps such as ReHo or ALFF for all participants/dataset, some group-level analysis can be run on those maps, such as t-tests or correlation analysis across population. For AFNI commands, type the name of the command and press return in a Terminal to get the help information.

4.1 One-sample t-test

One-sample t-test can show the within-group effect of the measurements, such as which areas had higher ReHo values than other areas. The AFNI command in a Terminal looks like this:

```
3dttest -prefix one_sample_t.nii -base1 0 -set2 PS001_FC.nii PS002_FC.nii
```

The output one_sample_t.nii contains two volumes, the second volume is the t-value map.

4.2 Two-sample t-test

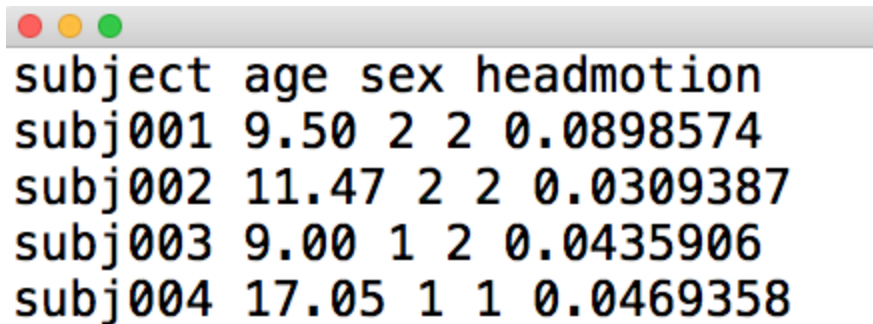
Two-sample t-test is used for the group comparison. The AFNI command in a Terminal looks like this:

```
3dttest++ -prefix patient-controls \  
-mask brain_mask.nii \  
-Clustsim \  
-setA patient \  
subj001 P001.nii \  
subj002 P002.nii \  

```

```
-setB control \
subj003 C001.nii \
subj004 C002.nii \
-covariates covariates.txt
```

The “covariates.txt” is a text file contains covariates that want to be removed. The order of the participants in “covariates.txt” must be exactly the same with the command. It looks like this:



subject	age	sex	headmotion	
subj001	9.50	2	2	0.0898574
subj002	11.47	2	2	0.0309387
subj003	9.00	1	2	0.0435906
subj004	17.05	1	1	0.0469358

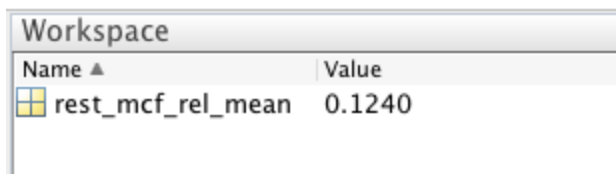
The “subj001” stuff should be the same with the 3dttest++ command.

The output “patient-controls” is a AFNI format image containing all statistical maps.

Here is a tip to get the averaged head motion parameters. Framewise displacement (FD) can be used as the representation of the averaged head motion. To get this, simply find the file named “rest_mcf_rel_mean.rms” in the ‘./func/’ folder, then load this file in MATLAB:

```
load rest_mcf_rel_mean.rms;
```

Then you can find a variate named “rest_mcf_rel_mean” in MATLAB Workspace:



Workspace	
Name ▲	Value
rest_mcf_rel_mean	0.1240

The value of “rest_mcf_rel_mean” is nearly half of FD. So this value can be used as a representation of head motion status for this participant/dataset.

4.3 Perform multiple comparison correction analysis

Tips: You could skip this step since the latest 3dttest++ gives the outputs of the NN3 1D file as well.

Multiple comparison correction is used to correct false-positive errors induced by the vast number of comparisons among voxels in images.

1) To estimate smoothness parameters on brain images, The AFNI command 3dFWHMx is used as follows:

```
3dFWHMx -acf -input rest_resf2standard36_smoothed.nii -detrend -mask brain_mask.nii >>
fhwmx_acf.txt
```

This step should run on all 4D fMRI dataset used in the analysis. The output “fhwmx_acf.txt” contains a set of parameters look like this:

0	0	0	0
0.550139	10.2576	21.8584	25.7435
0	0	0	0
0.593653	10.9226	18.8554	25.8287
0	0	0	0
0.585681	10.4109	18.152	24.6838
0	0	0	0
0.659679	11.0735	19.1242	26.1646
0	0	0	0
0.54679	11.606	19.1874	27.1151
0	0	0	0

The non-zero rows include the estimated smoothness parameters for x, y and z dimensions (more details can be found in the help document of 3dFWHMx by typing 3dFWHMx in a Terminal).

2) The smoothness parameters across all datasets will be averaged and input to the following command 3dClustSim:

```
3dClustSim -mask brain_mask.nii -acf 0.806823 3.12514 7.61553 -pthr 0.01 -nodec -iter 2000
-prefix brain_mask_acf_0p01 -quiet
```

The parameters “0.806823 3.12514 7.61553” are the averaged smoothness parameters from the first three values in the output of 3dFWHMx. “0.01” is the voxel-wise p-value. This command will output the certain cluster size at voxel-wise $p < 0.01$. The output are set of text files (*.1D, but can be opened by text editor) named with “brain_mask_acf_0p01”:

```

brain_mask_acf_0p01.NN1_1sided.1D
brain_mask_acf_0p01.NN1_2sided.1D
brain_mask_acf_0p01.NN1_bisided.1D
brain_mask_acf_0p01.NN2_1sided.1D
brain_mask_acf_0p01.NN2_2sided.1D
brain_mask_acf_0p01.NN2_bisided.1D
brain_mask_acf_0p01.NN3_1sided.1D
brain_mask_acf_0p01.NN3_2sided.1D
brain_mask_acf_0p01.NN3_bisided.1D

```

Generally we used the file that is named with “NN3_bisided.1D”. It means the cluster will be voxels connected corner by corner and it’s two-sided (higher and lower) statistics. The content of brain_mask_acf_0p01.NN3_bisided.1D looks like this:

```

# 3dClustSim -mask brain_mask.nii -acf 0.806823 3.12514 7.61553 -pthr 0.01 -nodedc -iter
2000 -prefix brain_mask_acf_0p01 -quiet ./brain_mask_acf_0p01.0000.sdat ./
brain_mask_acf_0p01.0001.sdat ./brain_mask_acf_0p01.0002.sdat ./
brain_mask_acf_0p01.0003.sdat
# bi-sided thresholding
# Grid: 60x72x60 3.00x3.00x3.00 mm^3 (56528 voxels in mask)
#
# CLUSTER SIZE THRESHOLD(pthr,alpha) in Voxels
# -NN 3 | alpha = Prob(Cluster >= given size)
# pthr | .10000 .09000 .08000 .07000 .06000 .05000 .04000 .03000 .02000 .01000
# -----
0.100000 1113 1155 1205 1259 1316 1396 1493 1622 1805 2068
0.090000 924 968 1010 1057 1119 1187 1262 1361 1519 1817
0.080000 774 798 838 883 928 987 1053 1146 1276 1532
0.070000 634 663 696 723 763 805 867 946 1060 1257
0.060000 511 530 550 581 619 657 699 760 869 1031
0.050000 405 420 436 456 479 513 548 603 667 823
0.040000 312 324 337 351 369 391 421 457 520 615
0.030000 223 234 244 258 271 288 307 334 378 454
0.020000 148 154 162 172 181 191 208 228 256 315
0.015000 112 117 122 129 137 147 159 177 199 243
0.010000 79 82 86 90 96 103 113 123 143 172
0.007000 59 61 64 68 72 77 84 94 107 131
0.005000 46 47 49 52 55 59 64 73 82 104

```

3) pick up the value under the “alpha = 0.05” column and the row with “pthr = 0.01”, in the above case is “103”. This number is the cluster size that under the corrected threshold at 0.05. It’s the number of voxels, so the volume of this cluster-wise threshold is 103 times the cubic of voxels ($3 \times 3 \times 3 = 27$), equals to $103 \times 27 = 2781 \text{ mm}^3$.

Then calculate the t-value at voxel-wise $p < 0.01$ with command cdf in the Terminal:

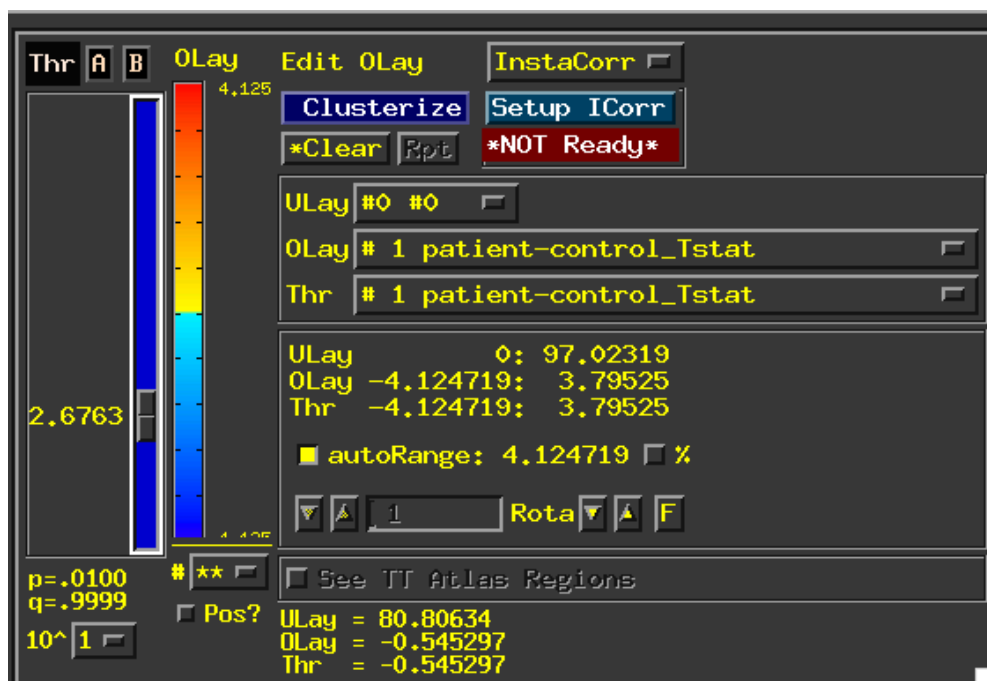
```
cdf -p2t fitt 0.01 10
```

Then an output “t = 3.16927” will be printed in the Terminal. “10” is the degree of freedom, which is the summary of two sample size minus 2;

Then, Input the related parameters in the command 3dmerge to get the thresholded statistical map:

```
3dmerge -prefix patient-controls_mc.nii -1clust 4.2 2781 -1thresh 3.16927  
patient-controls+orig.[1]
```

The “[1]” means it is the t-map (in the latest version of AFNI, it’s called Z_scr or something like that instead of Tstat) from the statistical package patient-controls+orig files, you could check about it in AFNI as follows:



The output patient-controls_mc.nii is the statistical map after multiple comparisons correction. It is the final map of the group comparison.

4.4 Report the results

Generally, there are several things that should be reported in the paper. The first one is the details of the regions/clusters detected in the patient-controls_mc.nii image.

You can use the following AFNI command to get the details of the regions/clusters:

```
3dclust 4 100 patient-controls_mc.nii
```

“4” and “100” above means radius = 4 mm and any clusters/regions with more than 100 mm cubic volumes. Since the results were already corrected based on the cluster size and it’s generally more than 100, any clusters/regions in the patient-controls_mc.nii will be detected. The output of this command looks like this:

```
++ 3dclust: AFNI version=AFNI_24.1.00 (Apr  2 2024) [64-bit]
++ Authored by: RW Cox et alii
++ *** Consider using program 3dClusterize instead of 3dclust ***
#
#Cluster report for file cbf_bmi_glm_t_0p05.nii
#[Connectivity radius = 4.00 mm  Volume threshold = 104.00 ]
#[Single voxel volume = 8.0 (microliters) ]
#[Voxel datum type      = float ]
#[Voxel dimensions      = 2.000 mm X 2.000 mm X 2.000 mm ]
#[Coordinates Order     = RAI ]
#Mean and SEM based on Absolute Value of voxel intensities:
#
#Volume  CM RL  CM AP  CM IS  minRL  maxRL  minAP  maxAP  minIS  maxIS  Mean  SEM  Max Int  MI RL  MI AP  MI IS
#-----
904  -27.0  61.6  -12.9  -36.0  -16.0  50.0  74.0  -22.0  -6.0  2.2336  0.0177  -2.9138  -34.0  56.0  -12.0
856  25.3   4.2  -25.4  16.0  36.0  -6.0  18.0  -34.0  -12.0  2.2582  0.0193  -3.0231  26.0   4.0  -30.0
832  -42.9  -9.0   5.5  -48.0  -34.0  -18.0  -2.0  -6.0  16.0  2.5007  0.0391  -3.9509  -44.0  -10.0  4.0
768  58.2  41.1   5.7  60.0  54.0  28.0  52.0  4.0  16.0  2.3418  0.0222  2.8027  56.0  48.0  0.0
```

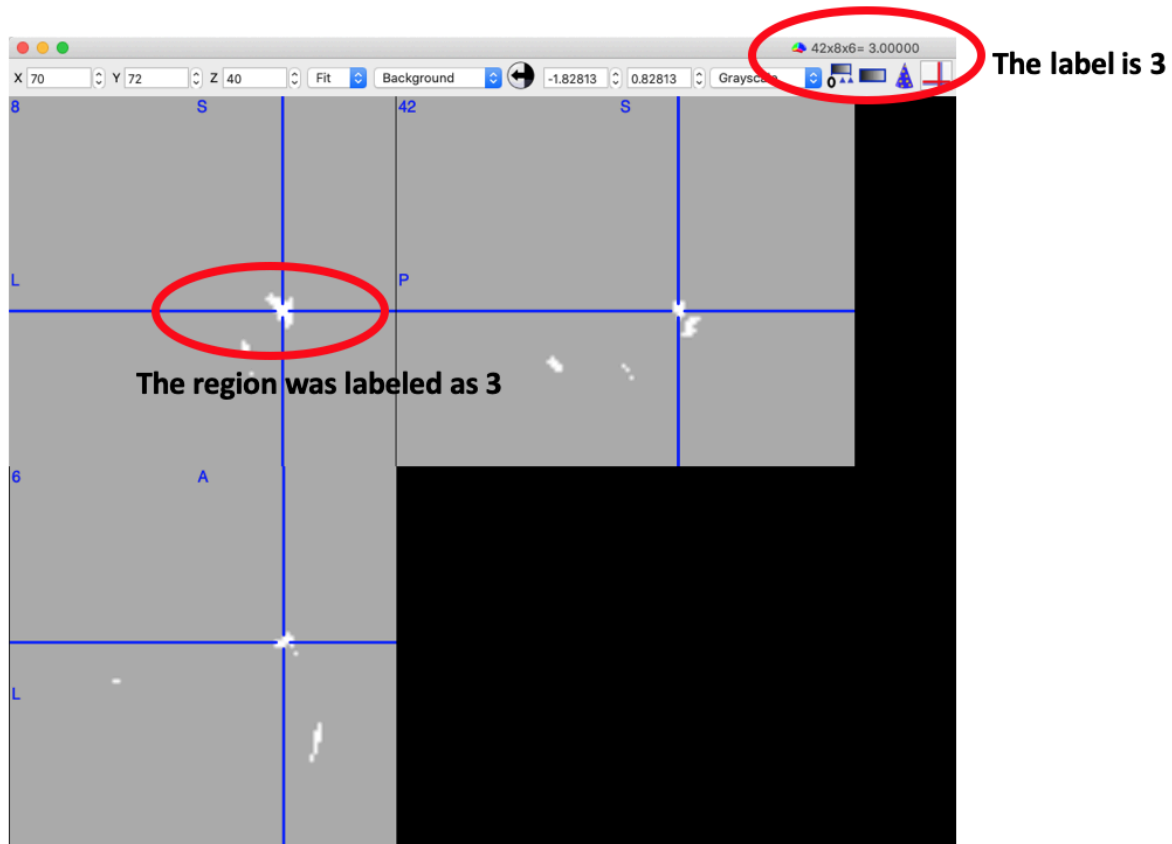
You can type “3dclust” in the terminal and get the information of what those columns mean. Generally, we report “Volume”, “Max Int”, and “MI RL MI AP MI IS” columns. Those are the size of the detected cluster/region for each row, the maximum t-scores/z-scores within this region and the coordinates of the voxel with the maximum t-scores/z-scores. Just be careful that the “MI AP” column might be opposite to the real MNI coordinates system, so the coordinates of the first region above should be “-34 -56 -12” instead of “-34 56 -12”. Just double-check if this region was located in the posterior or anterior of the brain, since the “y-axis” in the MNI coordinates system, negative y value means posterior part of the brain. In the x-axis, the negative x values mean the left brain, and the negative z values in the z-axis mean the inferior of the brain.

The next thing is to report or draw a bar figure that shows group-level mean and standard deviations of the correlation coefficients within the detected regions/clusters. The reason for doing this is to show the change visually and find out the real difference between groups.

First, “labeled” the clusters/regions detected in the patient-controls_mc.nii Image.

```
3dclust -savemask patient-controls_mc_labeled.nii 4 100 patient-controls_mc.nii
```

The output image `patient-controls_mc_labeled.nii` includes the regions which were labeled from 1 to the number of regions in the `patient-controls_mc.nii` image. You can use MRICron to open it and check the regions you would like to extract:



Next, extracting the region (of interested) from `patient-controls_mc_labeled.nii`, for instance the 3rd region:

```
3dcalc -prefix region_3.nii -a patient-controls_mc_labeled.nii -expr 'within(a,2.5,3.5)'
```

This procedure will output an image "region_3.nii" which only includes the third labeled region from `patient-controls_mc_labeled.nii`. "2.5" and "3.5" here is a range that only includes "2". That's the meaning of the "within" option of 3dcalc, and of course this range could be "2.3" and "3.1" or "2.9" and "3.8", any range that only covers the value 3 should be fine. So if you want to extract region 1 from the image, use "within(a,0.5,1.5)".

Now we need to get the correlation coefficients from the individual FC maps. The following command will first "concatenate" the individual's FC maps within each group.

```
3dTcat -prefix control_group_FC.nii control_001_FC.nii control_002_FC.nii control_003_FC.nii
```

Finally, we can calculate the averaged FC values within region 3 from previous steps.

```
3dmaskave -mask region_3.nii control_group_FC.nii >> averaged_FC_region_3_control.1D
```

The output averaged_FC_region_3_control.1D contains the averaged FC values across participants in the control group. After doing the same procedure for the patients group, you can draw the bar figure of mean and standard deviations of two groups with the software you are familiar with.

Also, you might notice that the procedure above can also allow you to run other analyses, such as the correlation between FC values and age/behavioral/cognitive scores across participants.

Besides, you can use 3dROIstats to get the average values within all ROIs in an atlas or an output which the clusters were labeled:

```
3dROIstats -mask atlas.nii FC.nii >> FC_atlas_mean.txt
```

The output FC_atlas_mean.txt indicates the average values within certain labeled ROIs. You can add “-quiet” here to remove the names of the labels in the output text file as well:

```
3dROIstats -quiet -mask atlas.nii FC.nii >> FC_atlas_mean.txt
```

If the number of participants were huge like over 100, you might notice that the 3dTcat command will be very long, for example:

```
3dTcat -prefix control_group_FC.nii control_001_FC.nii control_002_FC.nii  
control_003_FC.nii control_004_FC.nii ..... control_100_FC.nii
```

In this case, you can use a batch “for” loop to get the values from the images:

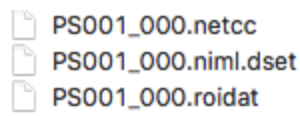
```
for id in 001 002 003 .....100  
do  
3dROIstats -mask atlas.nii $(id)_FC.nii >> FC_atlas_mean.txt  
done
```


5. Create connectivity matrix with pre-processed rs-fMRI data

An AFNI command 3dNetCorr is used to create a connectivity matrix for the graph-theory based analysis.

```
3dNetCorr -inset rest_resf2standard36_smoothed.nii -in_rois AAL90.nii -fish_z -mask  
brain_mask.nii -prefix PS001
```

The output will be three files:



- PS001_000.netcc
- PS001_000.niml.dset
- PS001_000.roidat

The file “PS001_000.netcc” contains the connectivity matrix. A MATLAB code gbip_readnetcc will be used to extract the connectivity matrix from the netcc file. The MATLAB codes look like this:

```
CM = gbip_readnetcc('PS001_000.netcc');  
CM(CM==4)=0; % This step will remove the diagonal elements  
save('PS001','CM');
```

Those codes will extract the connectivity matrix from PS001_000.netcc and save it to a new mat-file named PS001.mat. The “gbip_readnetcc.m” is a function from my Github website.

For the functional connectivity matrix, generally one or a set of thresholds should be applied to remove the low connectivity edges, this procedure could be done like this in MATLAB:

```
r_thr = gbip_rcutoff(0.05,dsize);  
z_thr = atanh(r_thr);  
load('PS001.mat');  
CM(CM<z_thr) = 0;  
CM(CM>0) = 1; % binarize the matrix, could be undone to get the weighted matrix  
save('PS001_thr','CM');
```

The “dsize” is the number of time points of fMRI data, such as 240. “0.05” means the p-value of the threshold was set at 0.05 level. The thresholded mat-file (i.e., ‘PS001_thr.mat’ in the

example) can be used to calculate graph-theory based metrics later in the GRETNA toolbox. The “gbip_rcutoff.m” is a function from my Github website.

6. Pre-process diffusion tensor imaging (DTI) data

The pre-processing of DTI data needs a MATLAB toolbox named ExploreDTI. Some MATLAB official toolboxes might be required, such as Image Process toolbox, Statistical toolbox, Curve Fitting, and Optimization toolboxes, please download them from the official MATLAB website if there's an error message shown during the usage of ExploreDTI.

The following procedures can be done with multiple datasets as well.

1) Convert Dicom to NIFTI

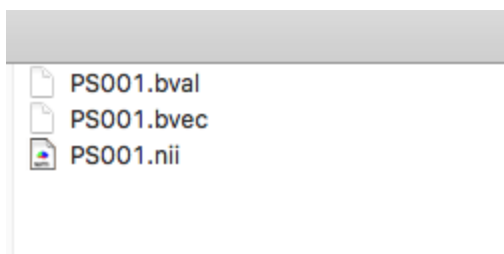
Similar to fMRI data analysis, dcm2nii is used to convert Dicom files to NIFTI format in the Terminal:

```
/Applications/mricron/dcm2nii *.dcm
```

Then, copy the file named something like “2xxxxxx.gz”, “2xxxxx.bval” and “2xxxxxx.bvec” to a new folder and rename them with something simple, such as “PS001.nii.gz”, “PS001.bval”, “PS001.bvec” if the data id is PS001. Then unzip the nii.gz file by using “gunzip” in the Terminal:

```
gunzip PS001.nii.gz
```

The output will be PS001.nii:



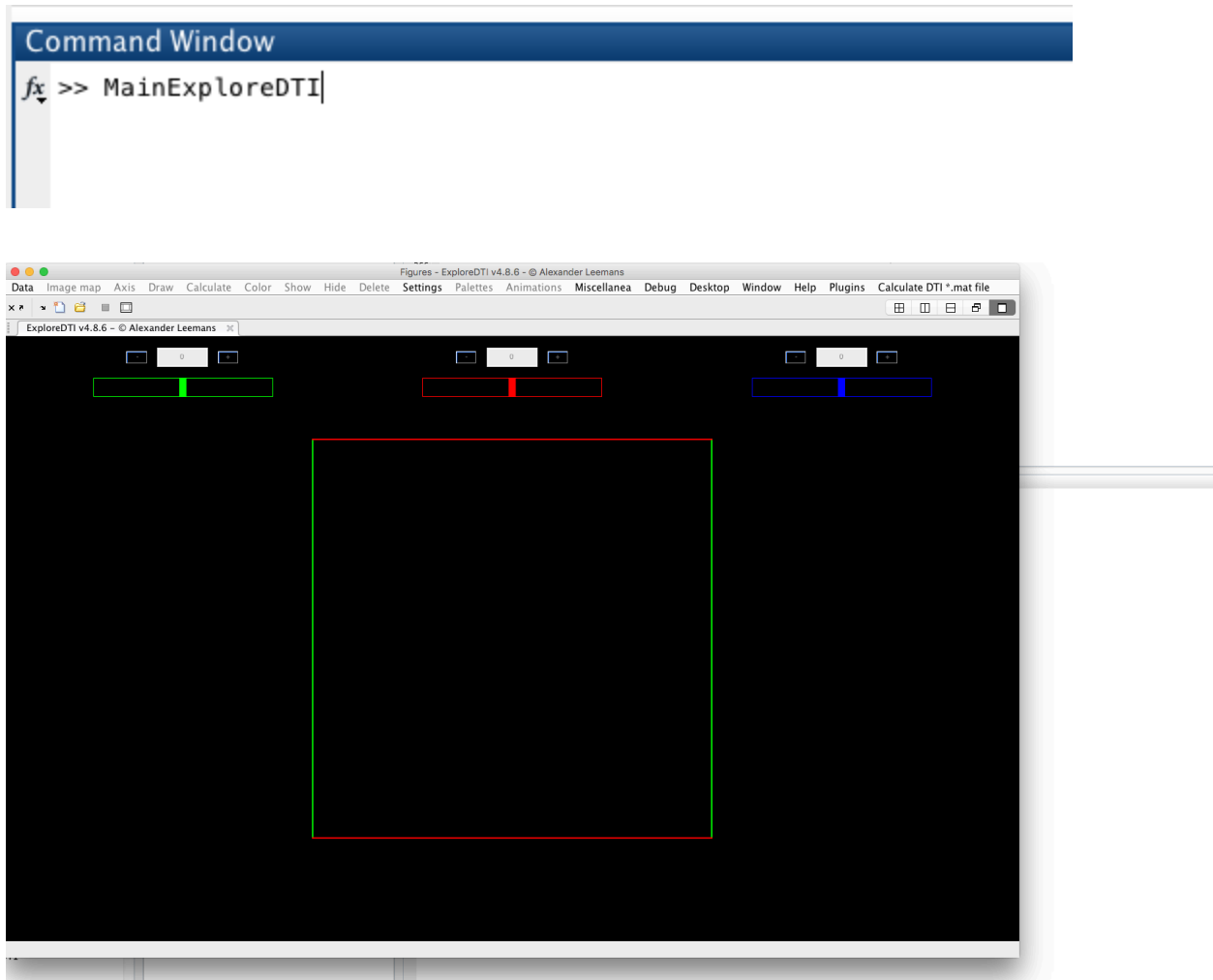
Tips: You could find the b-value in the .bval file, simply by opening it with a text editor, it generally looks like this:

“0 0 0 0 0 750 750 750 750”

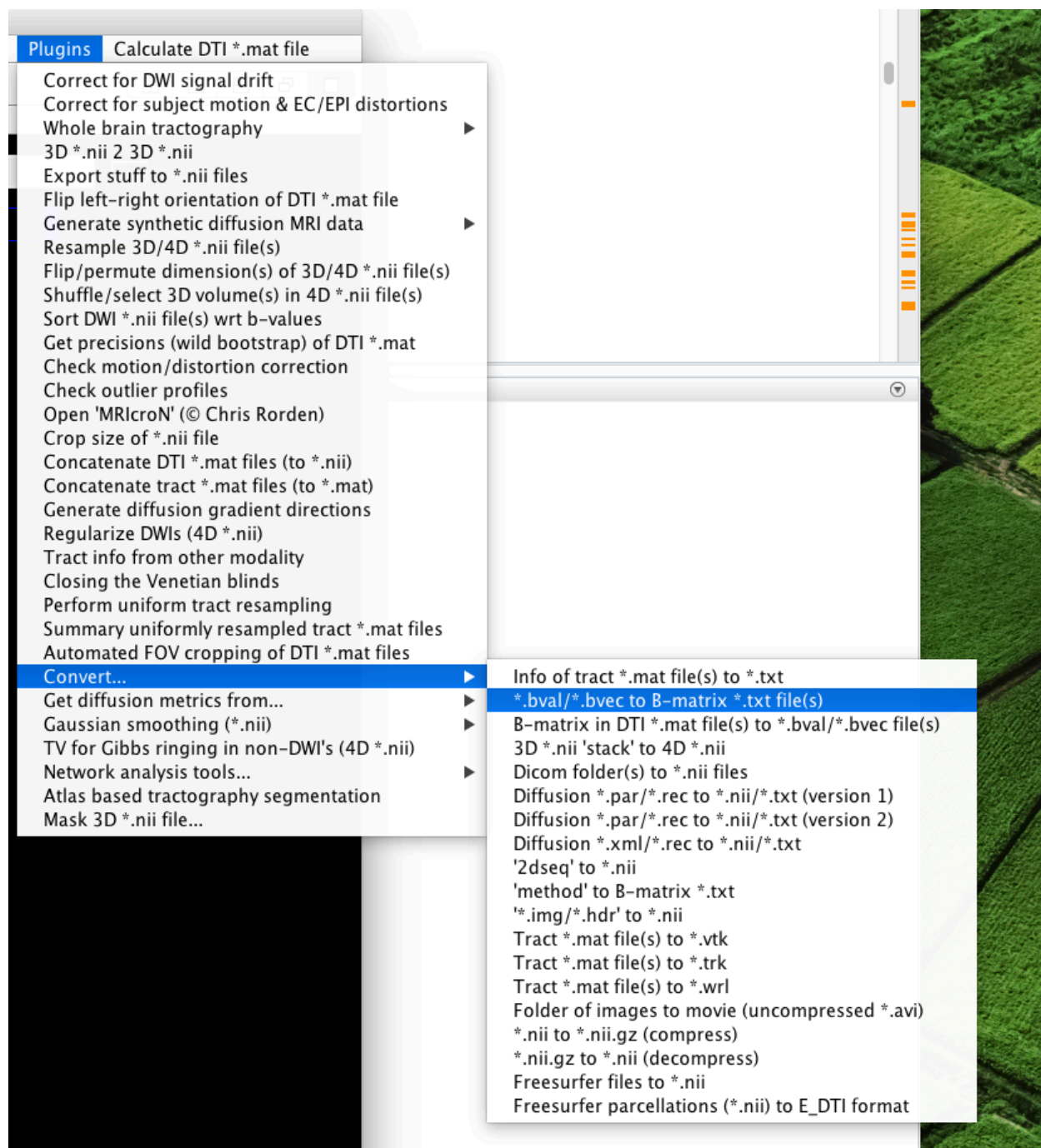
It means this DTI data has 5 non-DWI images and 4 DWI images, and the number of directions is 4. Consulting with the MRI technicians who scan the data is still suggested.

2) Convert bval/bvec file to a single text file

Open an ExploreDTI GUI in MATLAB by typing “Main” in the MATLAB command window:



Click “Plugins” then “Convert” then select “ *.bval/*.bvec to B-matrix *.txt file(s)” in ExploreDTI:

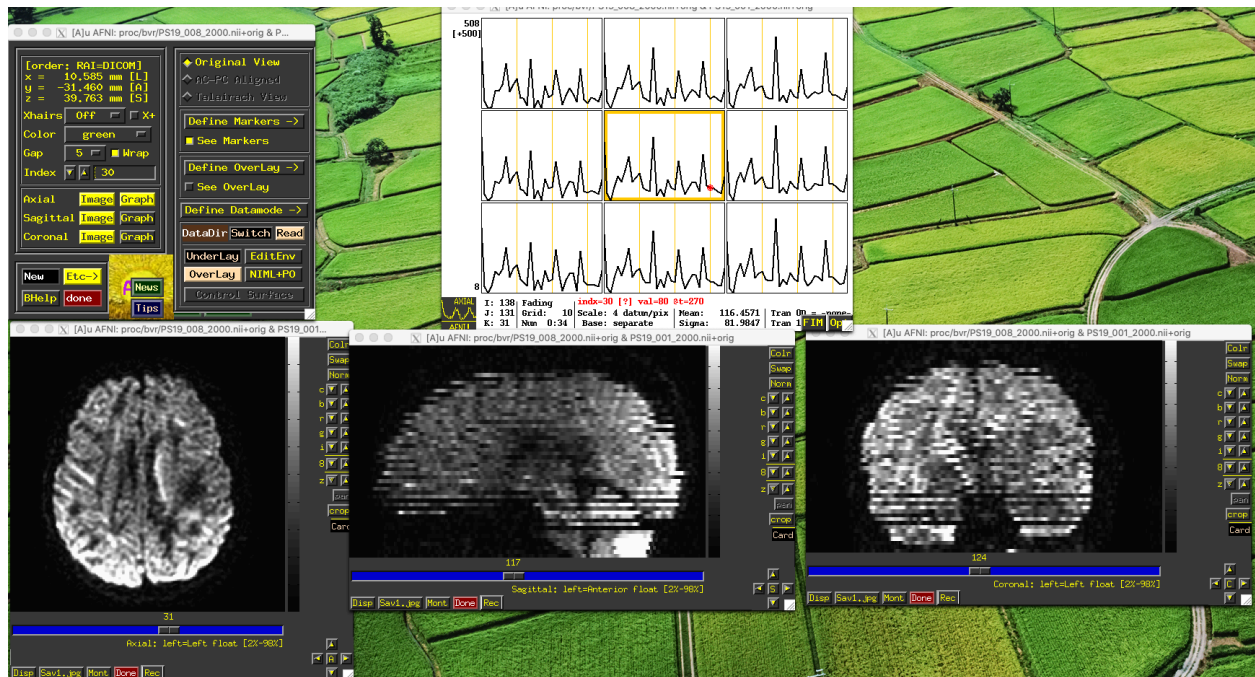


This step will merge bval and bvec file into a single text file, for example, PS001.txt

2.5) Optional: Remove the volumes with bad quality

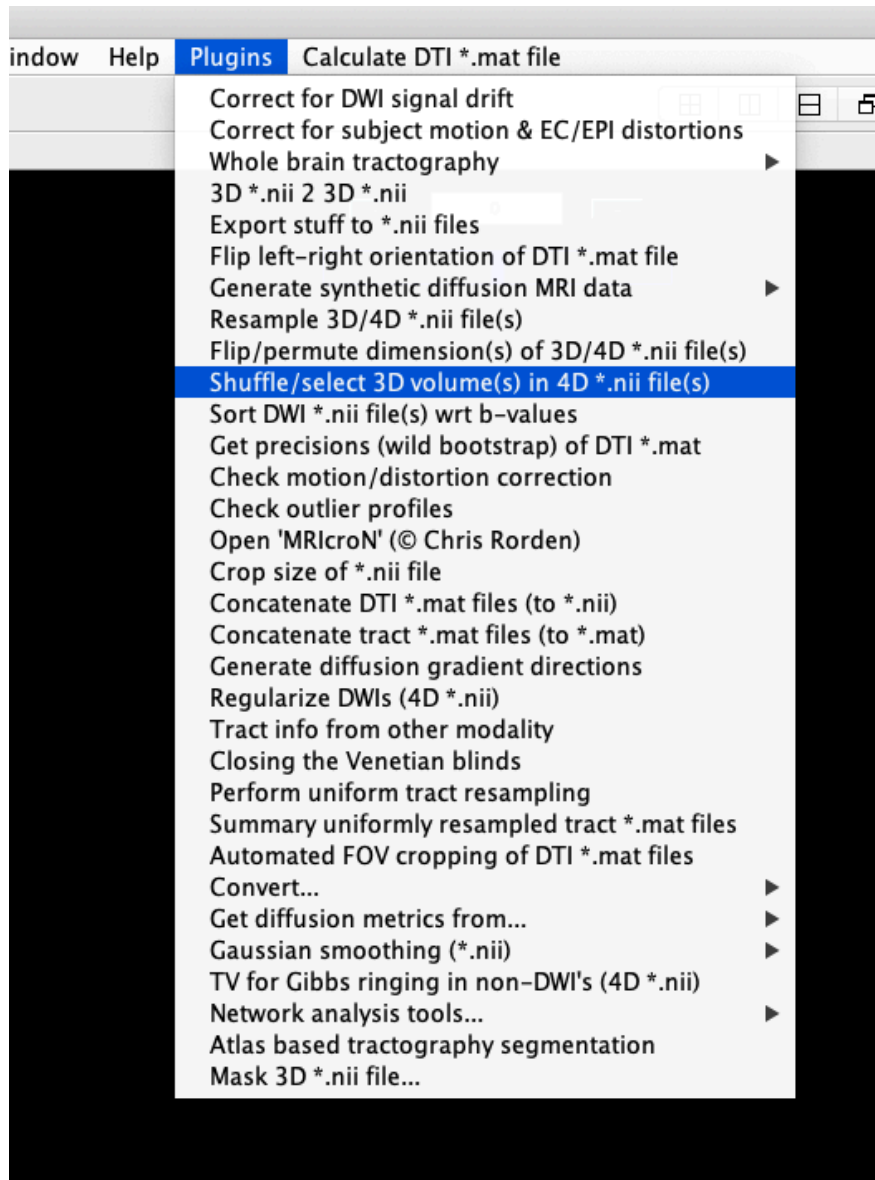
This step would remove the volumes with bad quality in the DTI data (the text file will be revised as well). The “bad quality” depends on the professional experience with the motion artifacts, so it could be varied across researchers who are processing the DTI data.

For example, by using AFNI and Graph” interface (ie., the 3X3 grid UI, pressing left or right button in keyboard in this UI will switch the volumes across the whole session), the following DTI volume was considered as a bad volume due its sheering/distortional looking:



In the Graph UI, you could see “indx=30”. Since AFNI counts the volume from “0”, but Explore DTI counts the volume from “1”, so the input index for this volume is “31”.

Then back to Explore DTI, select “Shuffle/select 3D volume(s) in 4D *.nii file(s)” from the Plugins



Then a new window will pop up as below. Then input the index of the **good** volumes except the 30th volume (could be multiple, in this example the total number of volumes is 35) and the suffix for the new file (generally we use BVR which is short for “brain volume removal”), and press the OK button, the toolbox will ask you to select the DTI file and the output directory of the new file. After saving the new file, you can see a new file was created in the directory with the suffix “_BVR”.

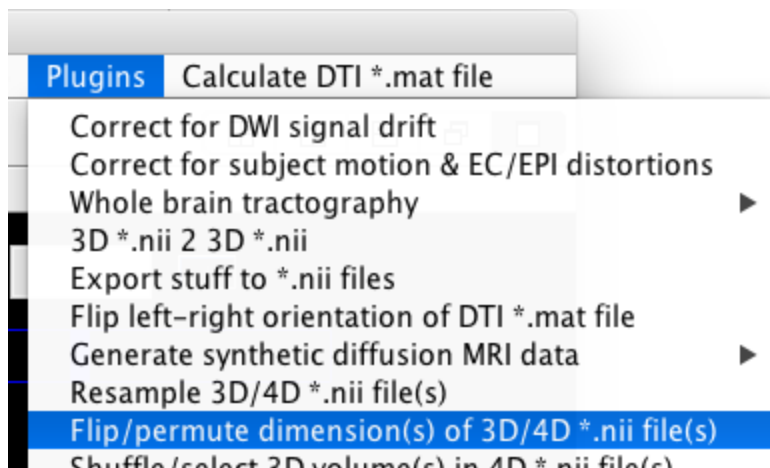


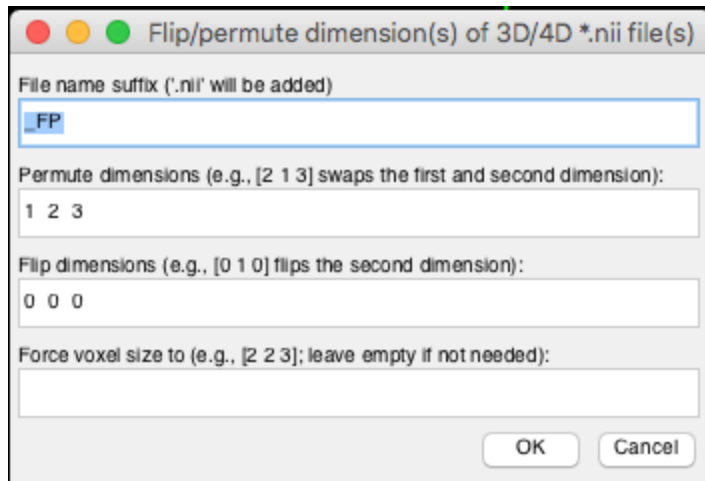
Again, this step depends on the visual inspections of the bad volumes, and so far it's hard to select all the “correct” volumes perfectly. It could be helpful with other methods to identify those volumes automatically.

3) Flip/Permute image dimensions

This step will put the NIFTI images into ExploreDTI and make sure the images would not left/right flipped in later steps.

To do so, select “Plugins” then “Flip/permute dimension(s) of 3D/4D *.nii file(s)”



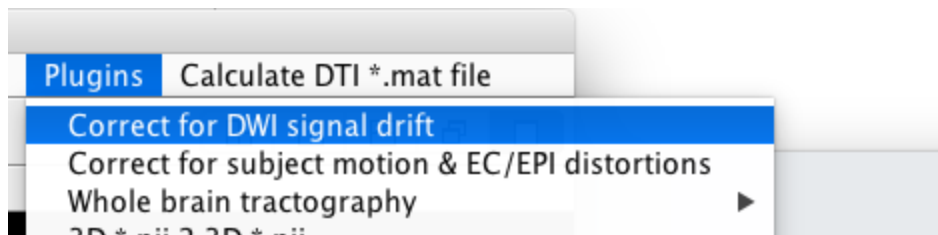


Then select the NIFT file, for example, PS001.nii, then the output will be PS001_FP.nii. **Open PS001_FP.nii with MRICron to make sure PS001_FP.nii has the same left-right orientation with the original image PS001.nii.** If not, change the Flip dimensions to “0 1 0” to make sure the PS001_FP.nii image has the same orientation with the raw data. **Then also change the previously created text file PS001.txt to the same prefix name, i.e., PS001_FP.txt.**

Tips: From this point, please delete the previous files after each step was done, in this case, delete PS001.nii.

4) Correct signal drift (only for multiple b0 images distributed among the whole volumes)

Select “Plugins” then “Correct for DWI signal drift”, the input is from the previous step, PS001_FP.nii and PS001_FP.txt:



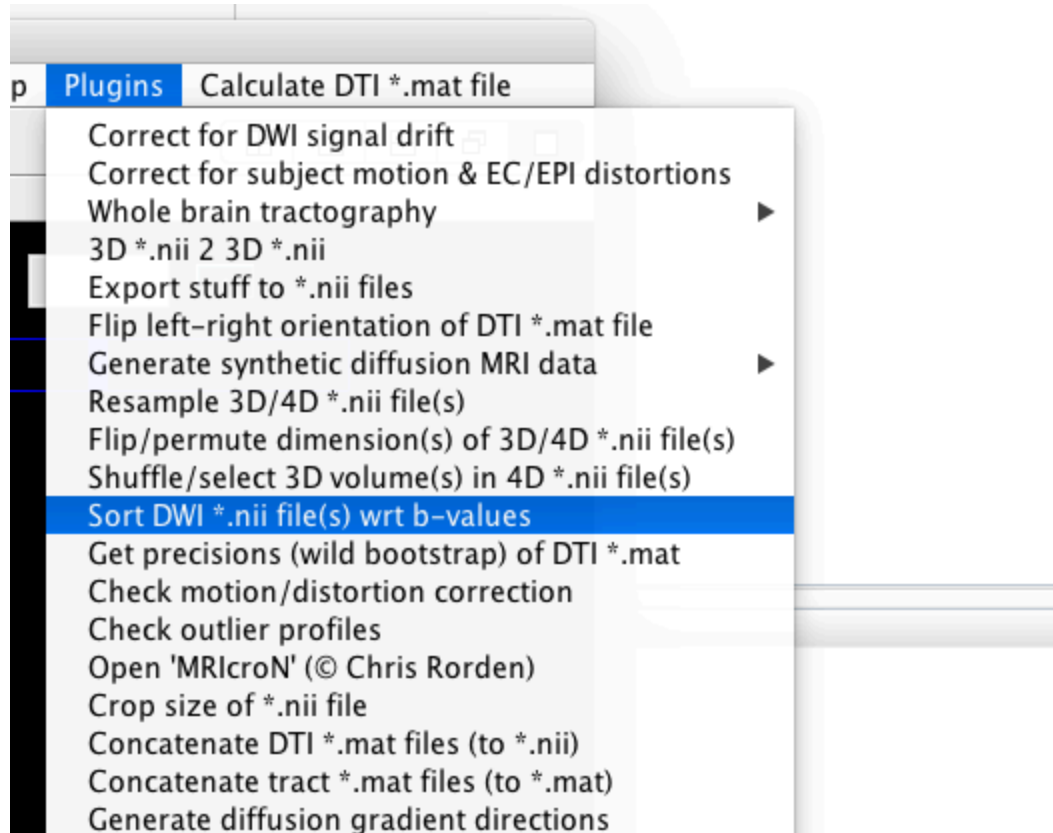
The output will be “PS001_FP_sdc.nii”, “PS001_FP_sdc.txt” and “PS001_FP_sdc.png”.

Tips: If b0 images were at the beginning of the DTI volume, this step might affect quality of the data and the final output (i.e., FA maps) might look bad. If so, this step could be skipped. Signal drift correction works better if b0 images were distributed across the whole volume.

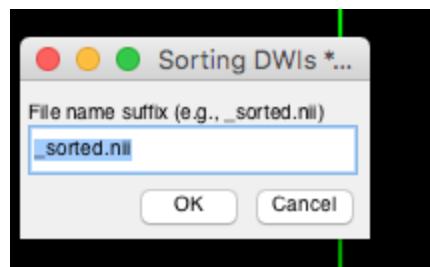
Tips: Please delete PS001_FP.nii.

5) Sort b0 volumes in the DTI volume

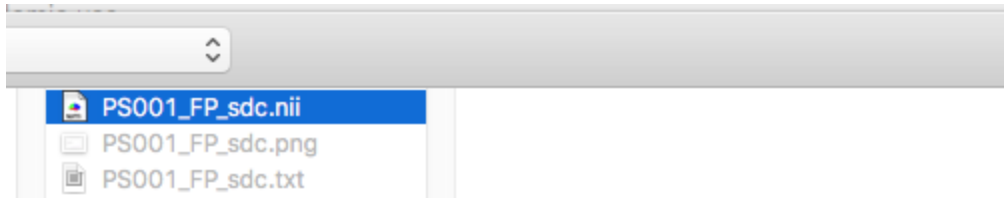
This step will put all b0 volumes at the beginning of the 4-D DTI volume. Select “Plugin” then “Sort DWI *.nii file(s) wrt b-values”. The input is from the previous step, PS001_FP_sdc.nii and PS001_FP_sdc.txt:

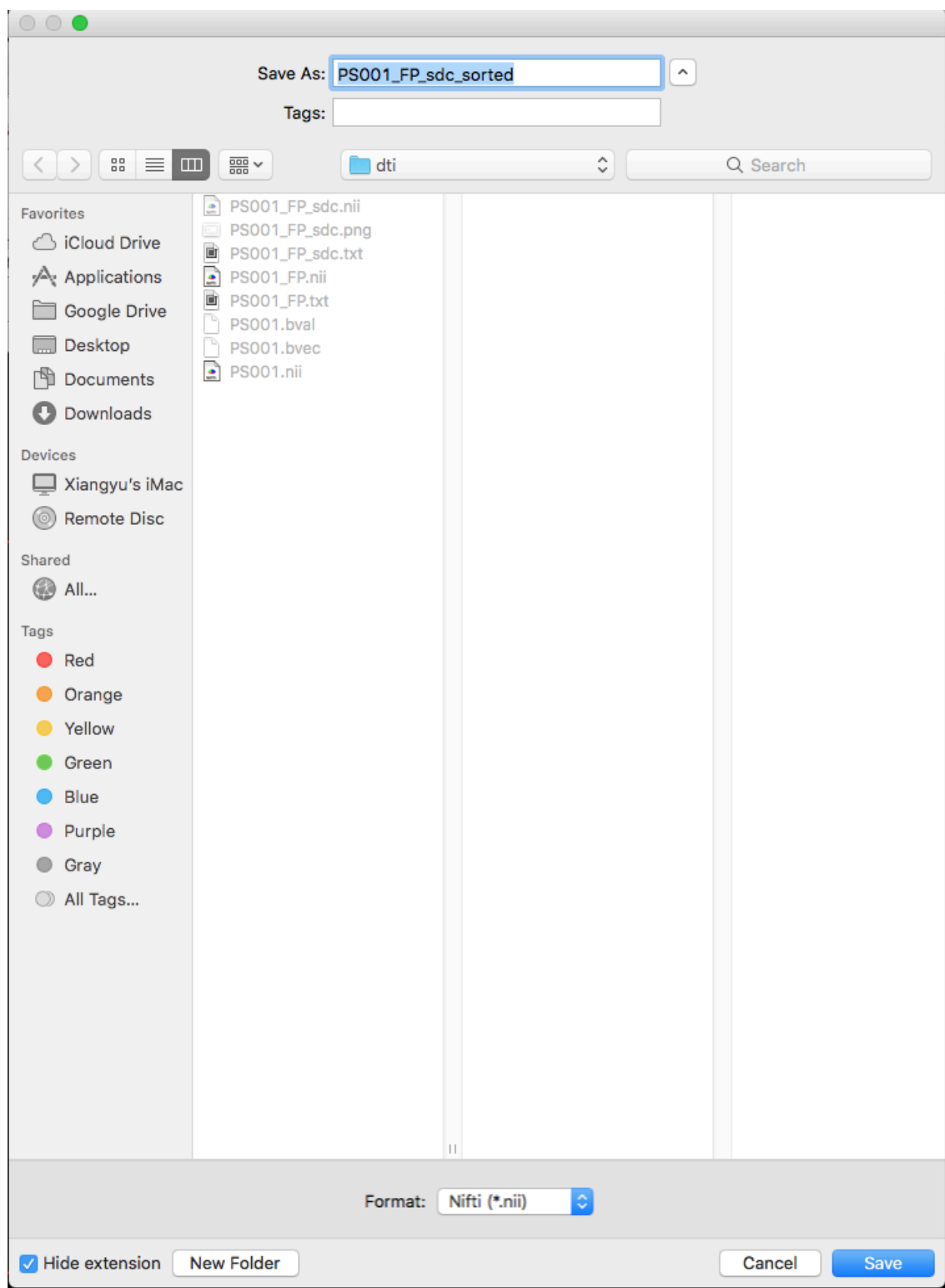


A small window will pop up, click “OK”:



Then select “PS001_FP_sdc.nii” file and click “Open” then click “Save” with the default naming:



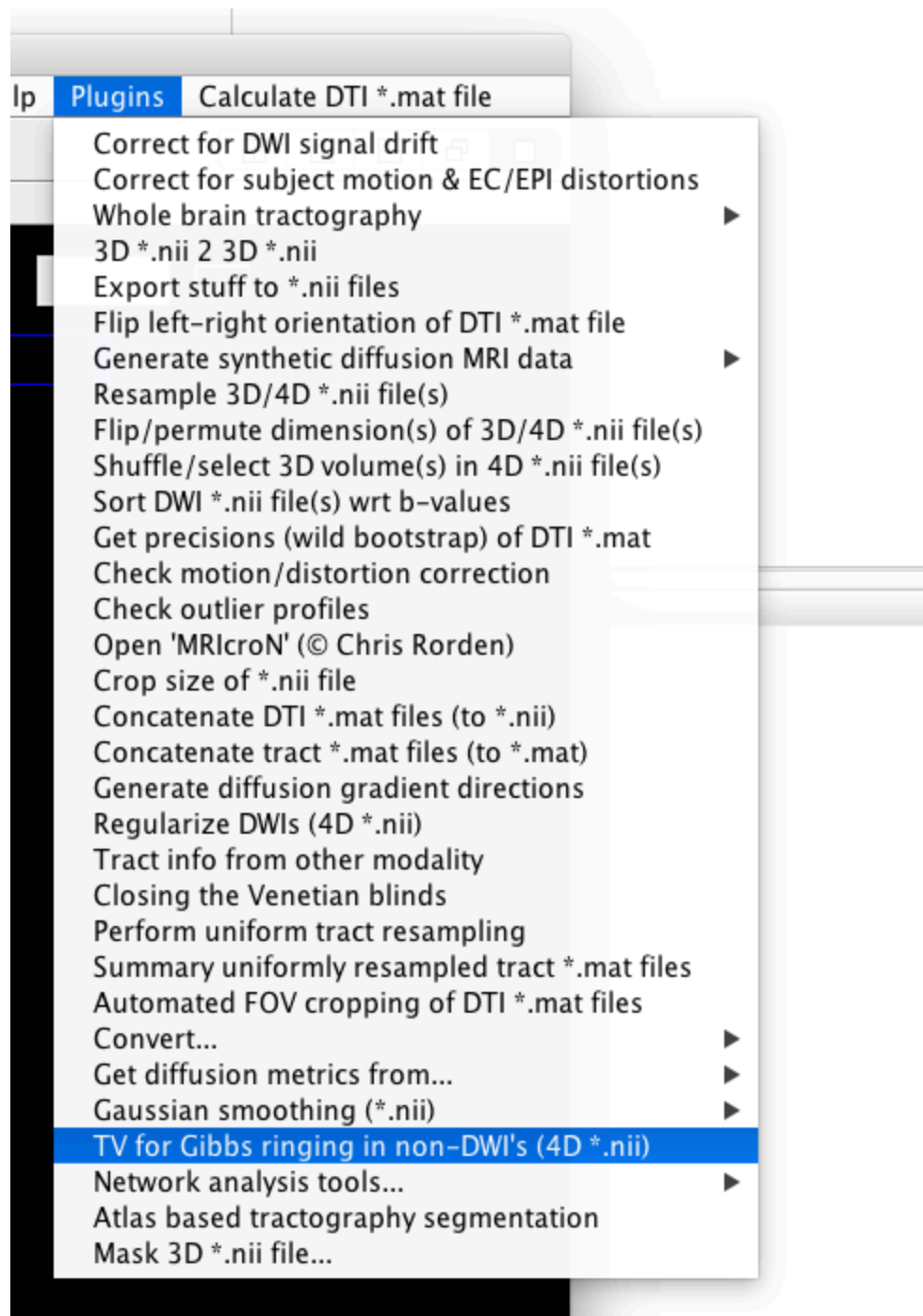


The output will be PS001_FP_sdc_sorted.nii and PS001_FP_sdc_sorted.txt.

Tips: Please delete PS001_FP_sdc.nii.

6) Correct Gibbs Ringing effect

This step will correct Gibbs Ringing effect in b0 images. The input is from the previous step, PS001_sdc_sorted.nii and PS001_sdc_sorted.txt. Select “Plugin” then “TV for Gibbs ringing in non-DWIs (4D *.nii)”:



Change the “Number of non-DWIs” to the number of b0 images in your data, in this case is “5”, you can check it in the text file, there are 5 rows which are zero at the top:

PS001_FP_sdc_sorted_GR_TV.txt						
1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
2	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
3	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
4	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
5	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
6	142.98832884	-187.81640655	-674.45950527	61.67461858	442.95419658	795.33698299
7	997.31506567	-96.25974027	-38.00909660	2.32272075	1.83429785	0.36214519
8	122.13406454	-104.53297541	646.48446815	22.36710739	-276.65887181	855.49876917
9	620.08700802	-690.70133739	682.09087442	192.33927308	-379.88304309	187.57366101
10	590.82418977	-151.43339732	-971.63350395	9.70342541	124.51907437	399.47233134
11	125.54504156	620.93628692	231.45634032	767.77598626	572.38278294	106.67891940
12	224.33687738	492.60226259	-673.33671743	270.41584954	-739.26140538	505.24722055
13	627.68505478	930.64864806	-262.06706500	344.96070071	-194.27924712	27.35414283
14	44.17295755	-4.79422683	-410.93045813	0.13008304	22.29977274	955.69693977
15	42.59721074	-349.53835522	-202.37017826	717.04848532	830.29073992	240.35428806
16	218.12792271	-820.13671923	97.81760797	770.90570281	-183.89166111	10.96636816
17	635.46829377	-577.67396159	769.99100226	131.28397167	-349.98107460	233.24772824

Other parameters could be the default settings:

Gibbs ringing removal with...

Number of non-DWIs
5

Lambda ([1 200])
100

Number of iterations ([1 200])
100

Step size ([0.001 - 0.1])
0.01

Imaging plane (coronal:1, sagittal:2, axial:3)
3

File name suffix (\".nii\" will be added after this suffix!)
_GR_TV

OK Cancel

Then click “OK” and select “PS001_FP_sdc_sorted.nii” and click “Open”, then click “Save” with the default naming.

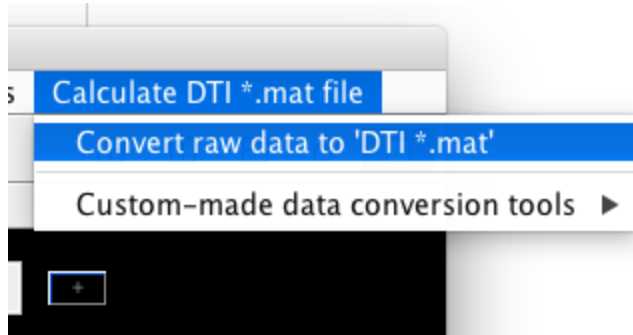
The output will be PS001_FP_sdc_sorted_GR_TV.nii, **again change the “PS001_FP_sdc_sorted.txt” filename to match for the NIFTI file as “PS001_FP_sdc_sorted_GR_TV.txt”.**

Tips: Please delete PS001_FP_sdc_sorted.nii.

7) Convert NIFTI/text file to a single mat-file

This step will convert the NIFTI and text file to a single mat-file in ExploreDTI format. The input is from the previous step, PS001_FP_sdc_sorted_GR_TV.nii and PS001_FP_sdc_sorted_GR_TV.txt.

Select “Calculate DTI *.mat file” then “Convert raw data to ‘DTI *.mat’” :



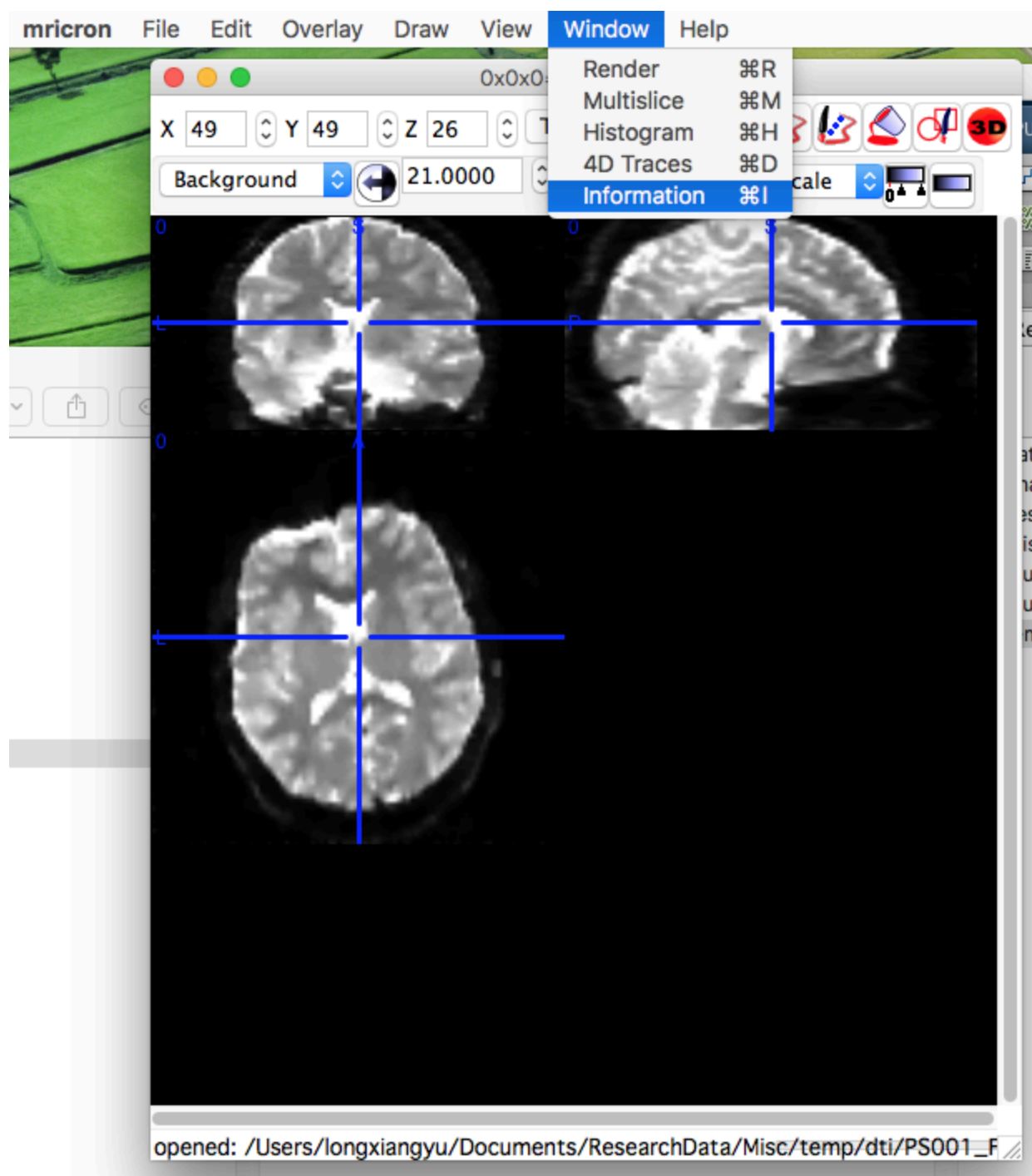
A window will pop up:

ExploreDTI's data converter to create DTI *.mat files

Format diffusion weighted data	4D Nifti (*.nii)
Permute spatial dimensions	AP RL IS
Flip spatial orientations	AP RL IS
Perform visual data check	No
Diffusion tensor estimation	Linear
Format diffusion information	Text file (*.txt)
Background masking approach	Automatic
Permute gradient components	x y z
Flip sign of gradient components	x y z
Data processing mode	Multiple data sets
b-value in units s/mm ²	1000
Voxel size [AP RL IS] (in mm)	2.21 2.21 2.2
Number of non-DW images	5
Number of DW images	30
Matrix size [AP RL IS]	96 96 50

Start converting to DTI *.mat file(s) ...

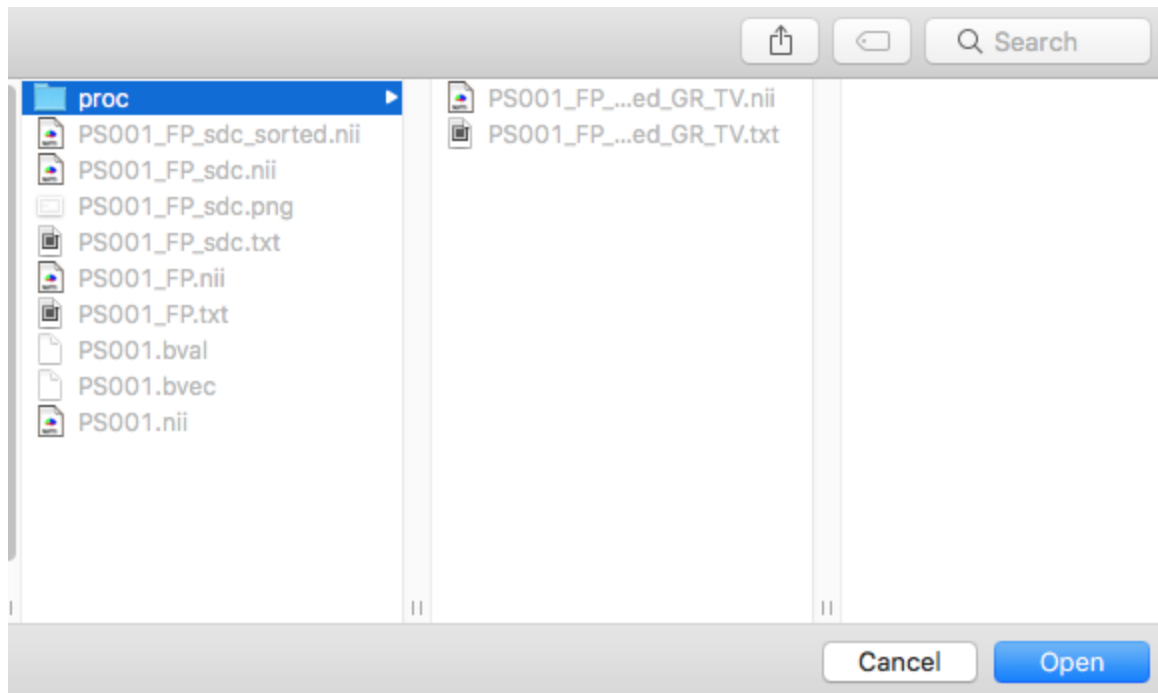
Then input the correct b-value (“NaN for multishell DWI data”), voxel size, number of non-DW images (i.e., number of b0 images, in this example there were 5 b0 volumes), number of DW images (i.e., the rest of the volumes, the total volume number of this example is 35, so the number of DW images is $35 - 5 = 30$) and Matrix in this window. The voxel size can be obtained from MRICron. Open the “PS001_FP_sdc_sorted_GR_TV.nii” image with MRICron, then click “Window” then “Information”:



A small window will pop up, the voxel size of x, y, z is in the “Spacing” column, which is “2.21 2.21 2.2”. The Matrix size is in the “Length” column, which is “96 X 96 X 50”:

Dimension	Length	Spacing
I Space	96	2.21
J Space	96	2.21
K Space	50	2.20

After typing the related parameters for the DTI data, click “Start converting to DTI *.mat file(s)”, then select the folder that contains the NIFTI image and text file, here you’d better create a new folder than only contains the latest processed image which is “XXX_FP_sdc_sorted_GR_TV.nii” and “XXX_FP_sdc_sorted_GR_TV.txt”, because this step will process all data in the folder:



Then click “Cancel” when asking to select the related text file, because ExploredTI will automatically detect the text file that shares the same prefix name with the NIFTI file, then click “Open” to select the output directory.

When finished, check the command window in MATLAB. In the command window, a recommended setting for “x y z” will be displayed like this:

```
Processing data: '/Users/longxiangyu/Documents/ResearchData/Misc/temp/dti/proc/PS001_FP_sdc_sorted_GR_TV.nii' (1 of 1)...
*****
Warning: there seems to be a coordinate systems mismatch!
Use the following conversion settings:
Permute gradient components: y x z
Flip sign of gradient components: -x y z
*****
Elapsed time is 10.754467 seconds.
File: '/Users/longxiangyu/Documents/ResearchData/Misc/temp/dti/proc/PS001_FP_sdc_sorted_GR_TV.nii' successfully processed!
```

Then delete the created mat file, run the converting again with the recommended setting of Permute gradient components and Flip sign of gradient components (Also if the “Warning” above didn’t show up, it means the current setting of those two options are the correct ones):



Generally, the combination of “x y z” should be the same for all data which use the same scanning parameters. Sometimes ExploreDTI might suggest different “x y z” for the data that was crooked or twisted, though still try to make this option the same with all data, then the processing procedure will not be different across the dataset.

The output will be a single mat file, for example, “PS001_FP_sdc_sorted_GR_TV.mat”. This time there will be no “Warning” in the command window of MATLAB.

7.5) Optional, Using pre-processed T1 image to correcting for EPI/susceptibility distortions during motion & EC/EPI distortions correction

This step is recommended if you had well preprocessed T1 images by ANTs.

The code for preprocessing T1 images by ANTs are below (need ANTs installation):

```
N4BiasFieldCorrection -d 3 -i T1.nii.gz -s 2 -c [50x50x50x50,0.0000000001] -o T1_new.nii.gz
```

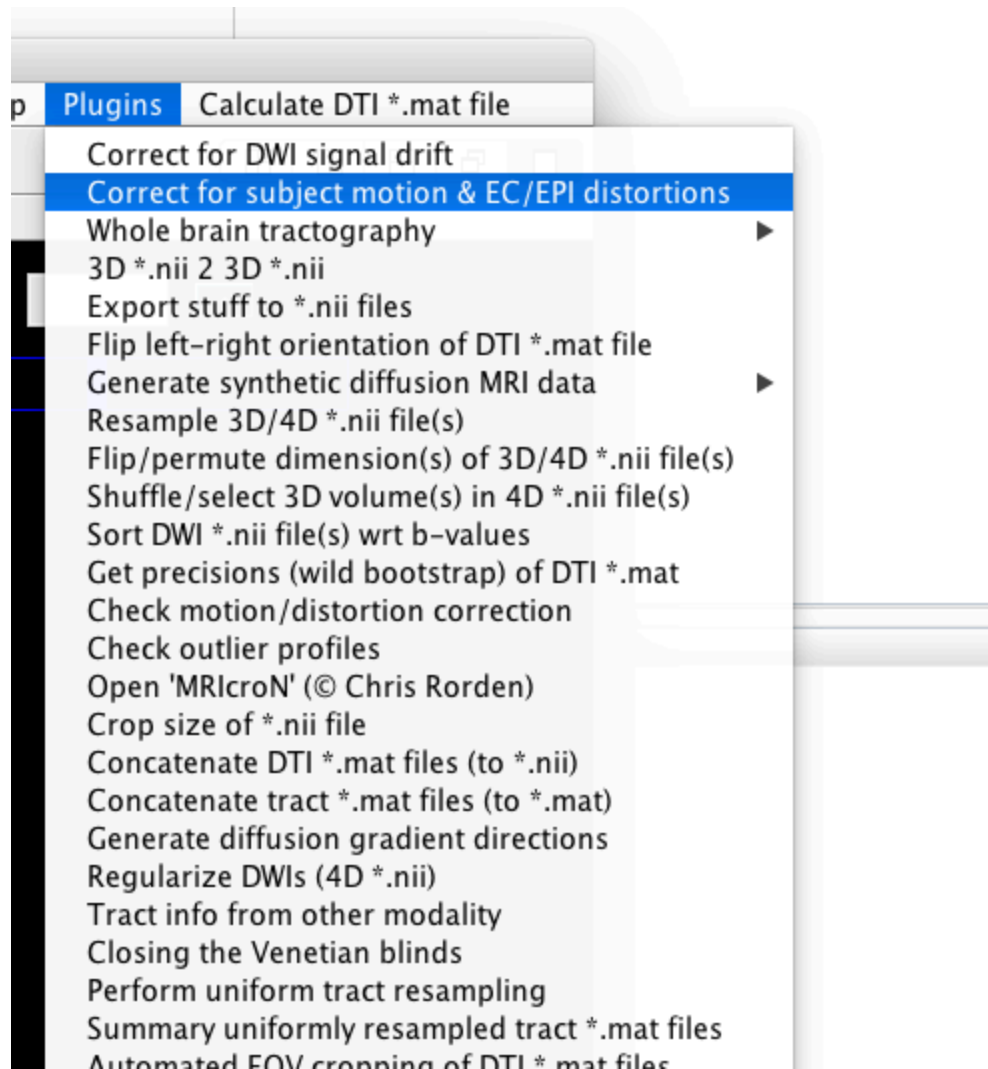
```
# Reslice to 1mm voxels (this one is in AFNI)
```

```
3dresample -dxyz 1 1 1 -prefix T1_1mm.nii -input T1_new.nii.gz
```

Then you can follow the instructions from “11. Correcting for EPI/susceptibility distortions” in the Explore DTI official tutorial which could be found in the Explore DTI installation folder. Then go to the 8th step below.

8) Correct for subject motion and eddy current distortions

This step will correct some nuisance signals from the DTI images. The input is from the previous step, PS001_FP_sdc_sorted_GR_TV.mat. Select “Plugins” then “Correct for subject motion & EC/EPI distortions”:



Then select the file PS001_FP_sdc_sorted_GR_TV.mat and click “Open” to select the output directory (this step will take around 5 minutes).

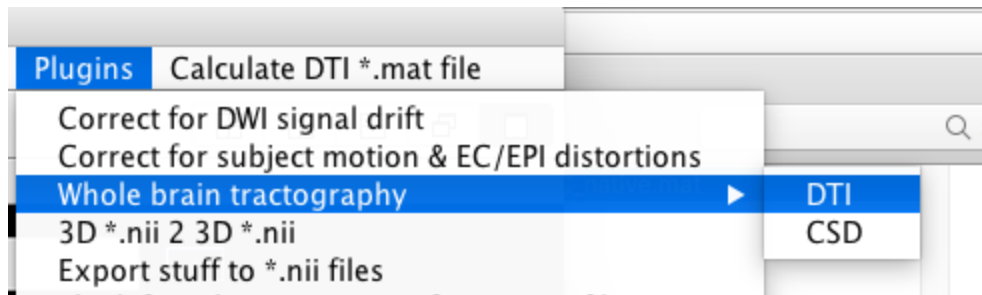
The output will be a single mat file, for example, PS001_FP_sdc_sorted_GR_TV_MD_C_native.mat (or PS001_FP_sorted_GR_TV_MD_C_native.mat if the signal drift correction was not performed). This file is final pre-processed DTI data and will be used for further fiber tractography and connectomic analysis. Load this mat-file in the ExploreDTI and please double check the orientation should be L/R flipped compared to the raw NIFTI images viewed in MRICron.

7. Create connectivity matrix with pre-processed DTI data

1) Run whole-brain fiber tractography

This step will create a file with whole-brain fiber tractography in ExploreDTI.

Select “Plugins” then “Whole brain tractography” then “DTI”:



A window will pop up:

Deterministic streamline tracto...

Seedpoint Resolution (in mm)
2 2 2

Seed FA threshold
0.2

FA tracking threshold range (limits: [0 1])
0.2 1

MD tracking threshold range (limits: [0 realmax])
0 Inf

Linear geometry tracking threshold range (limits: [0 1])
0 1

Planar geometry tracking threshold range (limits: [0 1])
0 1

Spherical Geometry tracking threshold range (limits: [0 1])
0 1

Fiber Length Range (in mm)
50 500

Angle threshold (in degree)
30

Step size
1

Interpolation method (NN: 0; Linear: 1; Cubic: 2)
1

Random perturbation of seed points? (No = 0; Yes = 1)
0

OK Cancel

“Seed FA threshold” and “FA tracking threshold range” determines how deep fibers could reach grey matter, so lower the threshold more fibers could touch the grey matter. This value could be a bit lower than the default settings (which is 0.2 in the screenshot). I recommend “0.15”. It is necessary for the connectomic analysis, because all grey matter regions should connect to each other physiologically.

Then press “OK” and select the pre-processed file

“PS001_FP_sdc_sorted_GR_TV_MD_C_native.mat” then click “Open” to select the output directory. The output file is “PS001_FP_sdc_sorted_GR_TV_MD_C_native_Tracts_DTI.mat”.

2) Create individual space atlas (optional)

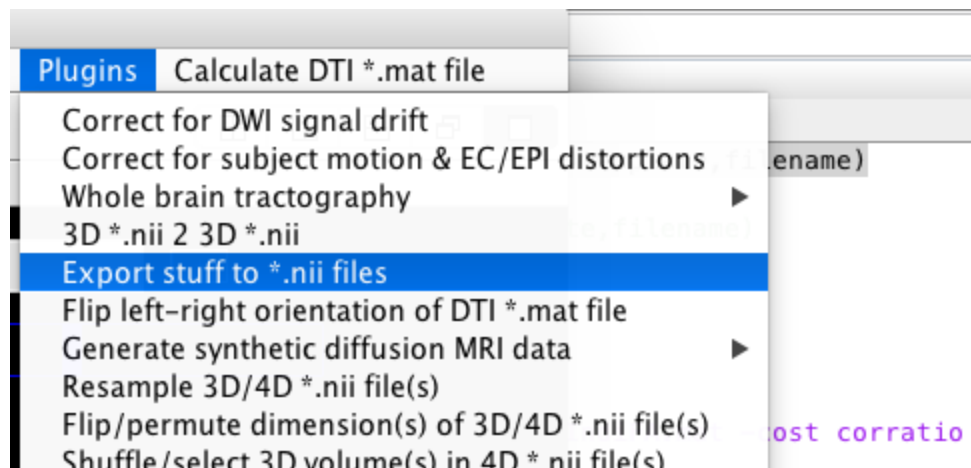
This step will create an individual space AAL atlas, because the connectomic analysis is done in the individual space of the DTI images. This step could be also done by ExploreDTI (please check the details in ExploreDTI's manual), though sometimes the registration of the AAL atlas in ExploreDTI looks bad.

There are two ways to create an individual space AAL atlas instead of using ExploreDTI:

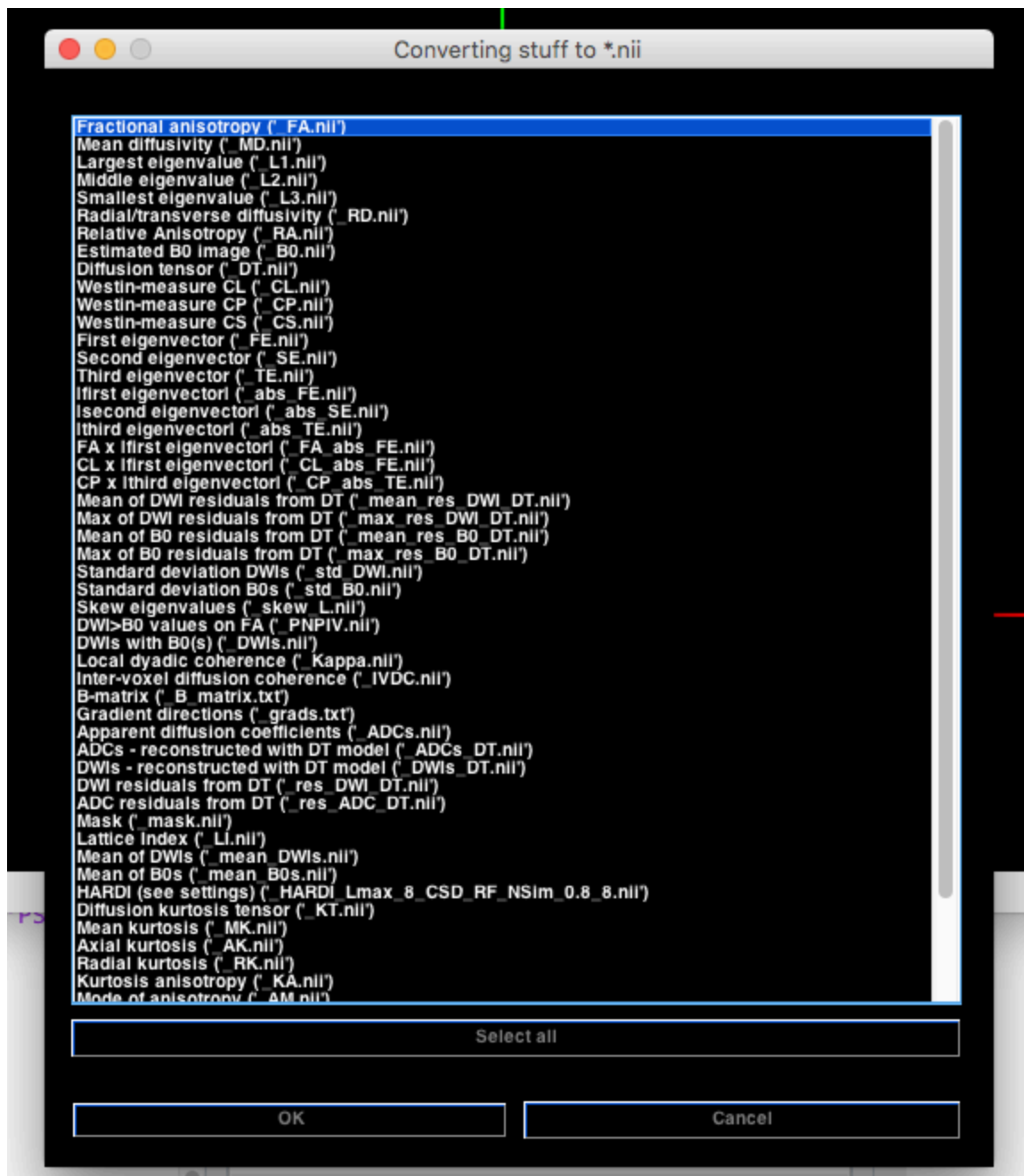
The first way is simply using FSL with the “-interp” option “nearestneighbour”:

```
flirt -ref individual_FA_map.nii -in FMRIB58_FA_1mm.nii.gz -out MNI_FA2_individual_FA -omat MNI_FA2_individual_FA.mat -cost corratio -dof 12 -interp trilinear
```

The individual FA or B0s image can be extracted from the pre-processed ExploreDTI mat file. Open ExploreDTI in MATLAB and select “Plugins” then “Export stuff to *.nii files”:



A window will pop up:



Select “Fractional anisotropy” then click “OK” and select the pre-processed mat file, for example, “PS001_FP_sdc_sorted_GR_TV_MD_C_native.mat”, then click “Open” to select the output directory. A FA image “PS001_FP_sdc_sorted_GR_TV_MD_C_native_FA.nii” will be created.

The FMRIB58_FA_1mm.nii.gz is a simple MNI space standard FA image, which can be obtained from the FSL installation folder under /data/standard/ directory. Then register the atlas as a whole to the individual DTI space as FA map as below:


```
flirt -ref individual_FA_map.nii -in MNI_FA_AAL_template.nii -out individual_AAL -applyxfm -init MNI_FA2_individual_FA.mat -dof 12 -interp nearestneighbour
```

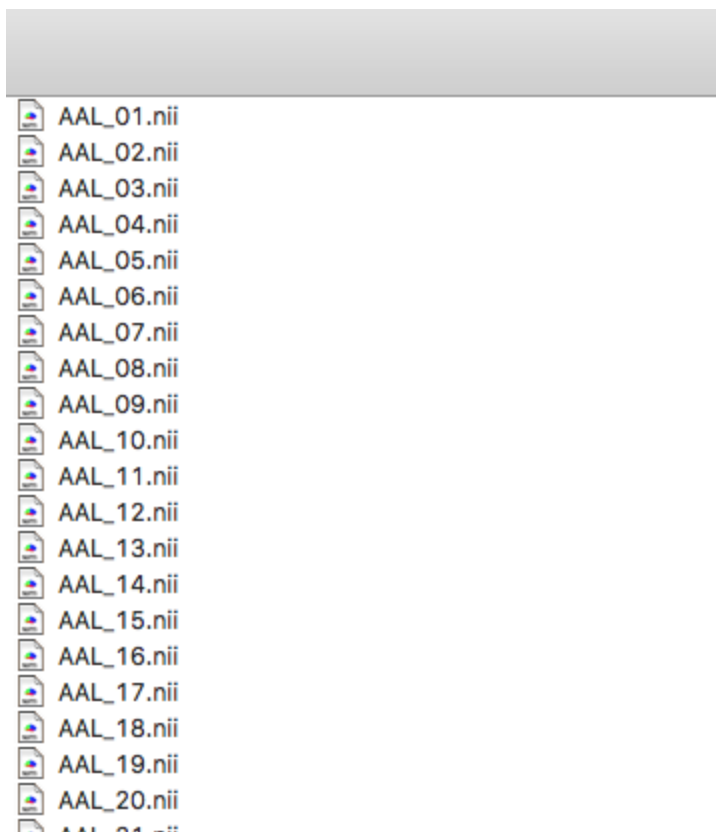
The MNI_FA_AAL_template can be found on my Github page. So please check which template (ExploredTI, my MATLAB code, the FSL version) looks better, and make sure to use the same way to create the template for all data for your study.

Also if an AAL atlas doesn't have a FA map such as the infant AAL atlas. This registration could be used on infant T1 and the mean B0s image from the DTI data, which means the mean B0 image will be the intermediate image instead of the FA image.

Tips: If you would like to use another atlas for this analysis, make sure the values in the atlas are a series of numbers, just like 1 to 90 manners in the AAL atlas. For instance, if there are 8 regions in the atlas, make sure the values/indices in the Nifti file is from 1 to 8, not some random numbers such as 99, 200, 150 in the atlas, because the ExploredTI might output the wrong connectivity matrix if the values/indices in the atlas was not started from 1.

The second way: Several files and codes are required in this step:

a) A folder that contains the separated AAL regions:



- b) A FA image in a standard MNI space that is provided by FSL: FMRIB58_FA_1mm.nii.gz. You can find it at “/your_fsl_directory/data/standard/”
- c) The MATLAB code to create individual space AAL atlas: gbip_create_native_aal90.m. This code will launch some FSL and AFNI commands.
- d) The individual FA image. It can be extracted as mentioned above.

After all required data are set, now run “gbip_create_native_aal90” in MATLAB in a new script (p.s., you can also use FSL instead of my script, the instruction was right after this one), something look like this below:

```
aalcheck = gbip_create_native_aal90(individualfolder,...
aalfolder,individualFA,FSLFAtemplate,filename);
```

Here, “individualfolder” is a folder that will be created to store the individual AAL files;

“aalfolder” is the folder contains separated AAL files (please use the files from the lab server: “/catherine_team/XYLong/BrainTemplates/FA_AAL/”);

“individualFA” is the individual FA image;

“FSLFAtemplate” is “FMRIB58_FA_1mm.nii.gz”;

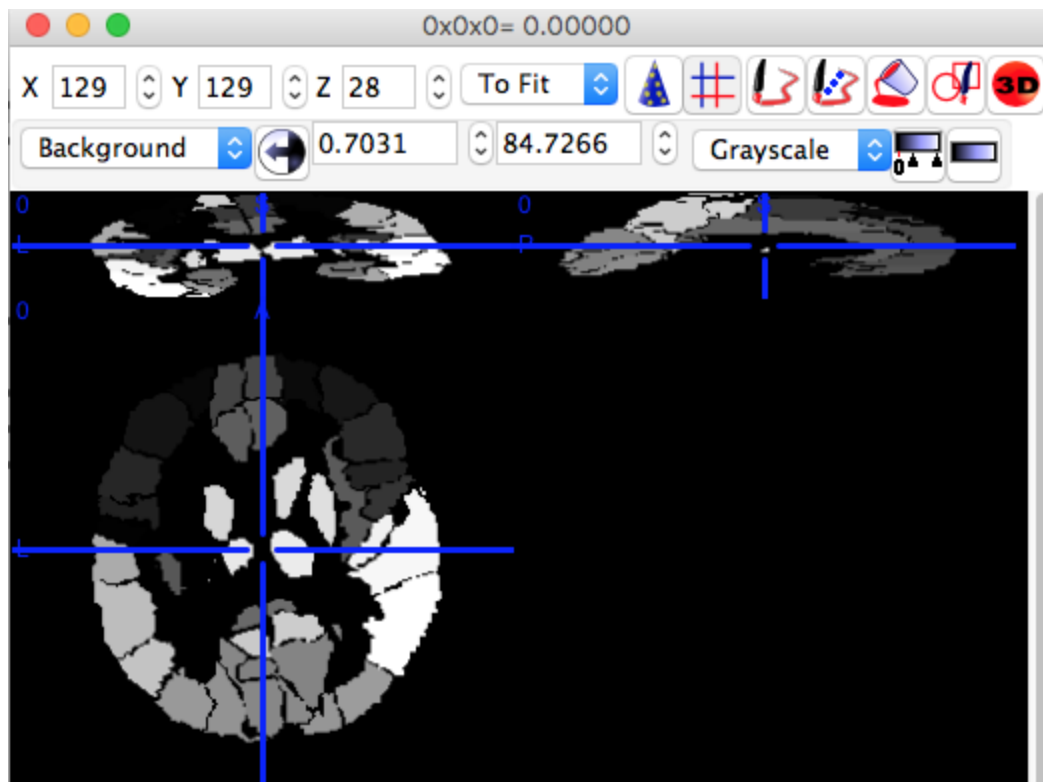
“filename” is the file name of the individual space AAL atlas;

“aalcheck” is a binary value output from the program “gbip_create_native_aal90”. It should be “1”, which means there’s no overlap between AAL regions.

Here is an example (please run it in a separate script):

```
clear;
aalfolder = '/your_data_directory/FA_AAL/';
FSLFAtemplate = '/your_data_directory/FMRIB58_FA_1mm.nii.gz';
datadir = '/your_data_directory/dti/';
individualfolder = [datadir,'aals/PS001'];
individualFA = [datadir,'PS001_FP_sdc_sorted_GR_TV_MD_C_native_FA.nii'];
filename = [datadir,'aals/PS001_FP_sdc_sorted_GR_TV_MD_C_native_labels.nii'];
aalcheck = gbip_create_native_aal90(individualfolder,...
aalfolder,individualFA,FSLFAtemplate,filename);
```

The file PS001_FP_sdc_sorted_GR_TV_MD_C_native_labels.nii will be created. It’s an individual space AAL atlas including 90 AAL regions without cerebellum. It looks like this:



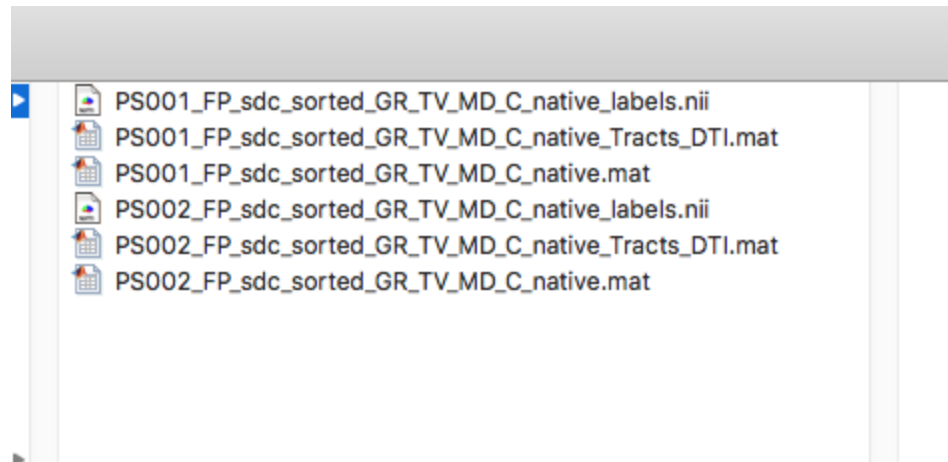
Tips: If the output matrix had lots of isolated regions, this could be due to the fibers not reaching the ROIs since the ROIs are gray matter based and only include a little part of white-matter/gray-matter boundary. The ROIs could be "dilated" a bit to let them touch the white matter with FSL, the command looks like this:

```
fslmaths aal_80_aa.nii -kernel sphere 3 -dilD aal_80_aaa
```

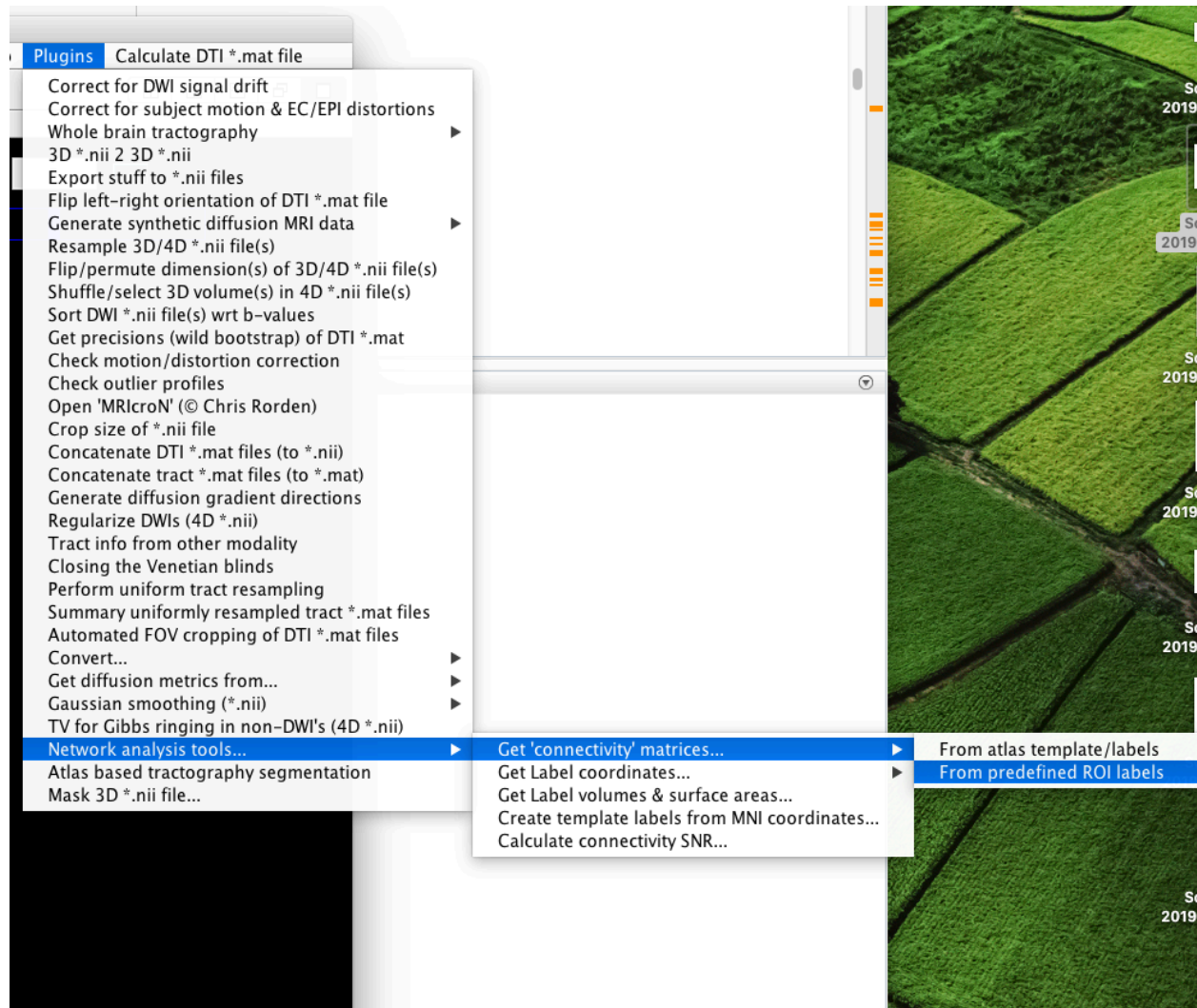
"3" here means 3 voxels around the ROI. This command should run for each ROI separately then combine them back to a single template image. This procedure would let fibers reach the white matter boundary of the grey matter ROIs. Also that's why I set the FA threshold kind of low (0.15 or 0.1) when running the tractography, to let the fibers reach out further to the brain surface.

3) Calculate the connectivity matrix

Before doing anything, move the tractography file, individual atlas and DTI mat file to a new folder that only contains these three files. The folder can store multiple participants/dataset, as long as each participant/dataset has their own prefix name, something like this:

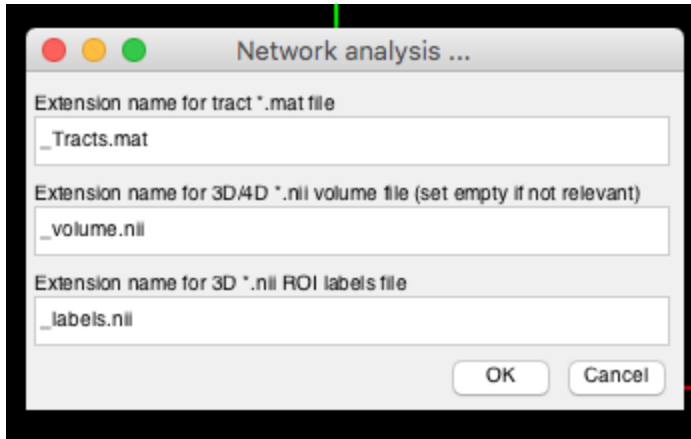


Now open ExploredTI in MATLAB and click “Plugins” then “Network analysis tools...” then “Get ‘connectivity’ metrics...” then “From predefined ROI labels”.

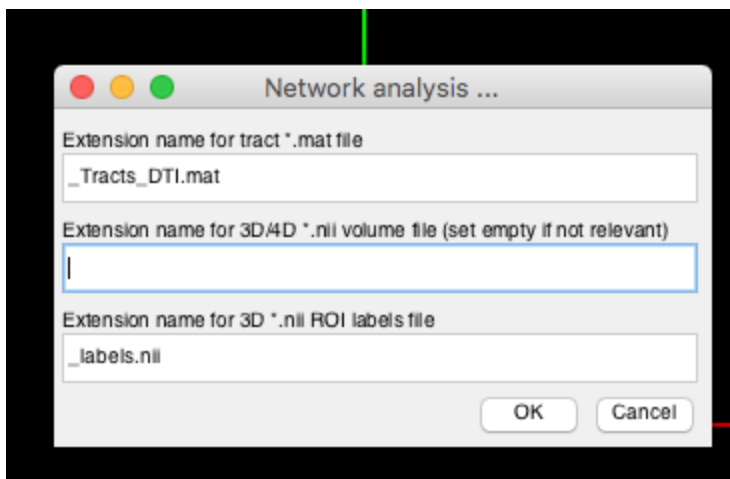


The option “From atlas template/labels” will create the individual space AAL atlas by ExploreDTI, however sometimes the atlas is distorted, therefore I recommend to create the individual AAL atlas separately as introduced above.

Then select the directory that contains the pre-processed DTI data
 “PS001_FP_sdc_sorted_GR_TV_MD_C_native.mat” and whole-brain tractography data
 “PS001_FP_sdc_sorted_GR_TV_MD_C_native_Tracts_DTI.mat”. After that, a window will pop up:

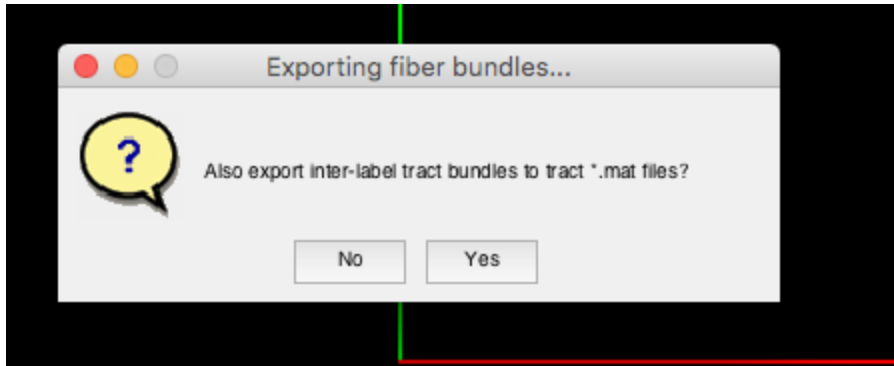


The first blank is the suffix name of the track file, in this example, is “_Tracts_DTI.mat”. The second blank can be empty. The last blank is the suffix name of the individual space AAL atlas, which is “_labels.nii”, so this window should be like this:

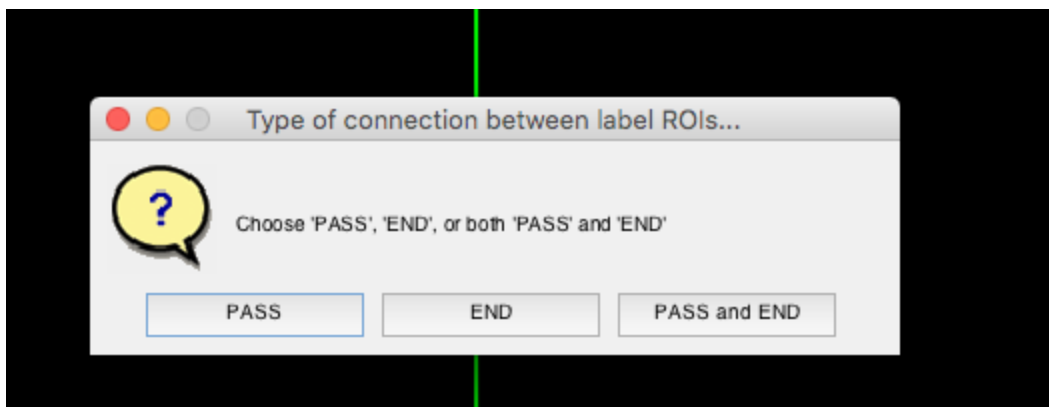


Then select the AAL index file which can be found at “/your_ExploreDTI_install_folder/AAL_Template/AAL_Label_Names.txt”, click “Open”.

Then click “No” in “Also export inter-label bundles” window

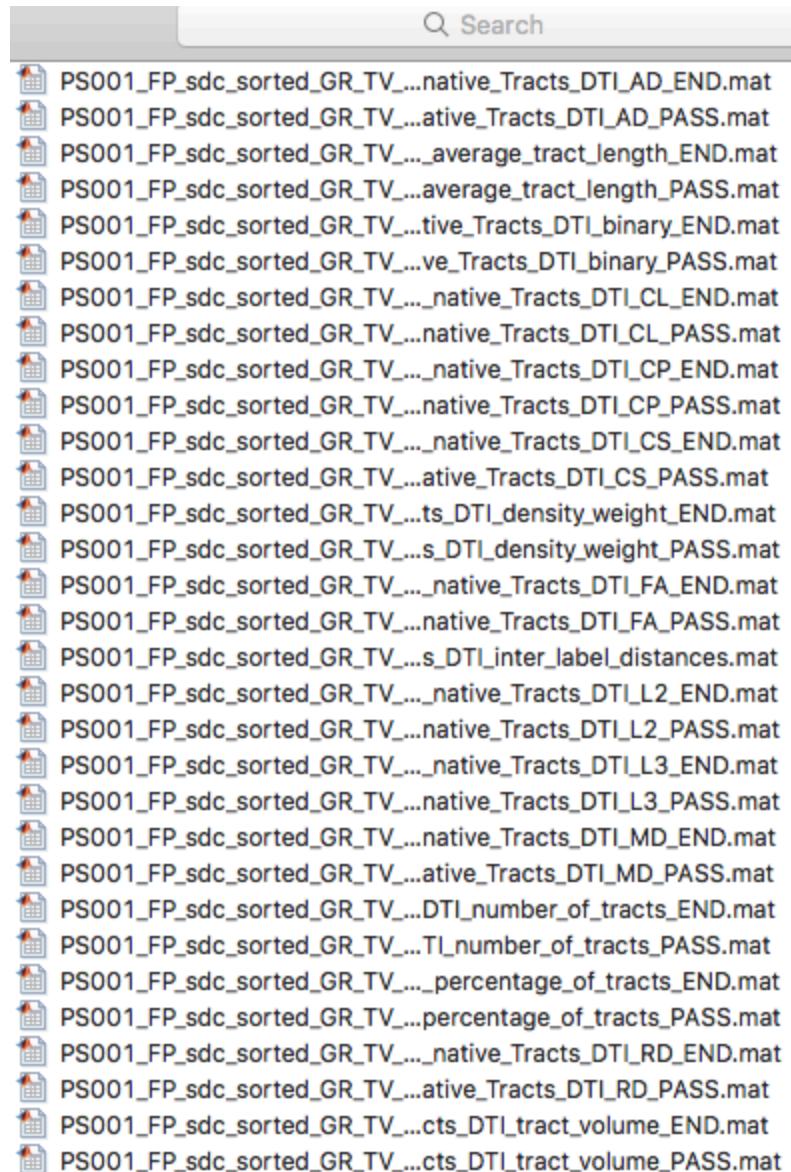


Click "PASS and END" in the next window:



"PASS" means if two regions could be connected via other regions then these two regions were considered connected. "END" means only if two regions were directly connected without passing through other regions. So "PASS" option will give more connections in the connectivity matrix.

Then select the output directory. The output includes both "PASS" and "END" results:



Generally the file “PS001_FP_sdc_sorted_GR_TV_MD_C_native_Tracts_DTI_FA_PASS.mat” that contains a 90 x 90 connectivity matrix with FA values will be used for the further connectomic analysis. You can also use “number of tracts” mat-file since some researchers used this one to represent structural connectivity strength as well.

The last thing is that there are NAN values within those connectivity matrices, please make sure to remove them before doing any follow-up analysis. You can load them into MATLAB and run a code like:

```
CM(isnan(CM)) = 0;
```


CM is the variable in Workspace of MATLAB after you load any of those connectivity matrices into MATLAB.

8. Calculate graph-theory based metrics

This step is fit for both fMRI and DTI based connectivity matrices. In this tutorial. The binary matrix that only contains 0 and 1 is used for the analysis. To binarized connectivity matrix, use the following code in MATLAB (DTI mat file as an example):

```
load('PS001_FP_sdc_sorted_GR_TV_MD_C_native_Tracts_DTI_FA_PASS.mat');
CM(isnan(CM)) = 0; % remove the NAN value in the matrix
CM(logical(eye(size(CM)))) = 0; % This step will set the diagonal element to 0.
CM(CM>0) = 1; % You could skip this step to create the weighted matrix as well
save('PS001_bin.mat','CM');
```

This code is only for one participant/dataset. You can use the “for” loop in MATLAB for multiple participants as well. PS001_bin.mat contains the binarized matrix.

You can also check if the connectivity matrix was a fully connected matrix which means any pair of AAL regions has a pathway to connect with each other. This can be done by gbip_graph_fullyconnected.m:

```
[a b] = gbip_graph_fullyconnected(CM);
```

The output “a” should be “1” and “b” should be empty if the connectivity matrix is a fully connected matrix. If not, “a” will be “0” and “b” contains the AAL regions that are isolated in the matrix. Generally some graphic measures can only be measured from the well-connected matrix.

For the MAC computer users, if GRETNA didn’t work properly, please use Brain Connectivity Toolbox: <https://sites.google.com/site/bctnet> instead.

Before opening the GRETNA toolbox in MATLAB

Important: If the GRETNA toolbox didn’t run on your macOS, Please delete the matlab_bgl folder in the GRETNA toolbox folder and download/unzipped with the latest MATLAB bgl version as below:

<https://www.mathworks.com/matlabcentral/fileexchange/10922-matlabbbgl>

Or you could re-compile the C files using “mex” in MATLAB, the C files are under: `./GRETNA_2.0.0_release_20171121/matlab_bgl/private/`, some compiler such as Xcode is needed (size is around 10Gb).

(Update June, 2022: tested on macOS Catalina with MATLAB R2020b, only betweenness centrality didn't work, you could use Brain Connectivity Toolbox to get the betweenness centrality: <https://sites.google.com/site/bctnet/>. Betweenness centrality is not crucial for graph theory analysis though, you can also just skip this measure.)

Another solution: change the `gretna_distance_weight.m` file as follows:

```
W = abs(W);
W = W - diag(diag(W));
W(logical(W)) = 1./W(logical(W));

% [D] = all_shortest_paths(sparse(W),struct('alname','auto'));

N = length(W);
D = zeros(N); D(~eye(N)) = inf;    %distance matrix

% Dijkstra's algorithm
for i = 1:N
    S = true(1,N);                %distance permanence (true is temporary)
    G1 = W;
    V = i;
    while 1
        S(V) = 0;                %distance u->V is now permanent
        G1(:,V) = 0;             %no in-edges as already shortest
        for v = V
            Nei = find(G1(v,:));  %neighbours of shortest nodes
            for w = Nei
                D(i,w) = min(D(i,w),D(i,v)+G1(v,w));
                %the smallest of old (if exist) and current path lengths
            end
        end

        minD = min(D(i,S));
        if isempty(minD)||isinf(minD), break, end;
        %isempty: all nodes reached; isinf: some nodes cannot be reached
        V = find(D(i,:)==minD);
    end
end
return
```

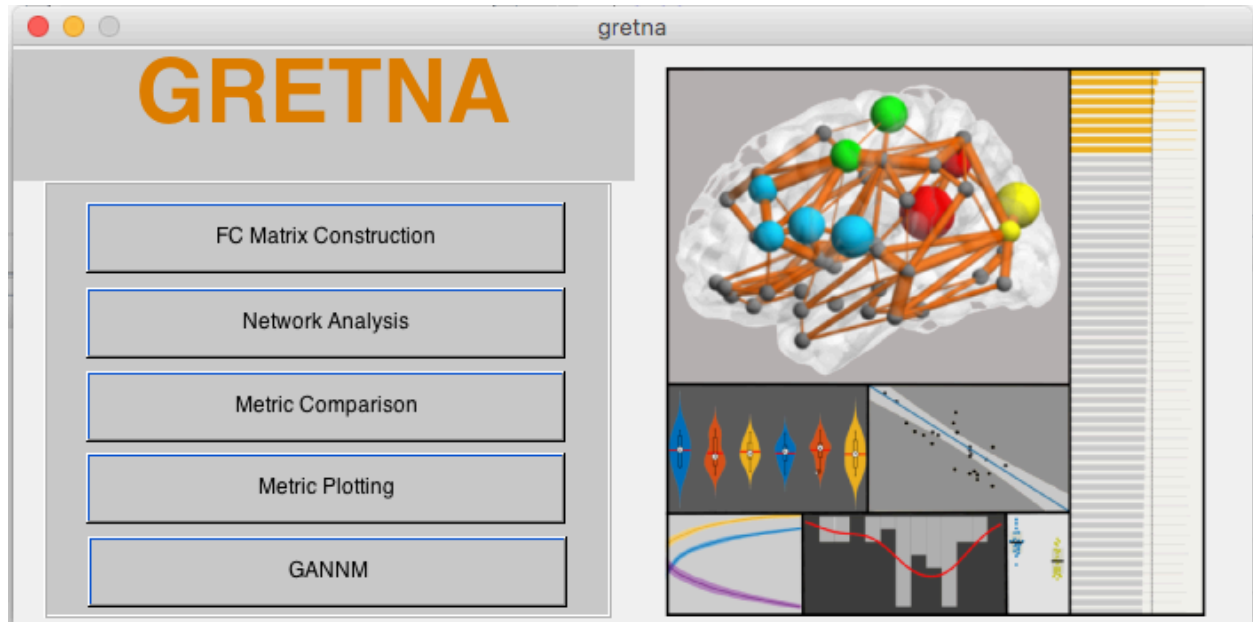
This change will let GRETNATM not use `matlab_bgl` files.

Tips: Here are some MATLAB codes to calculate the graphic measures in scripts rather than GUI using weighted matrices, if the GUI didn't work properly:

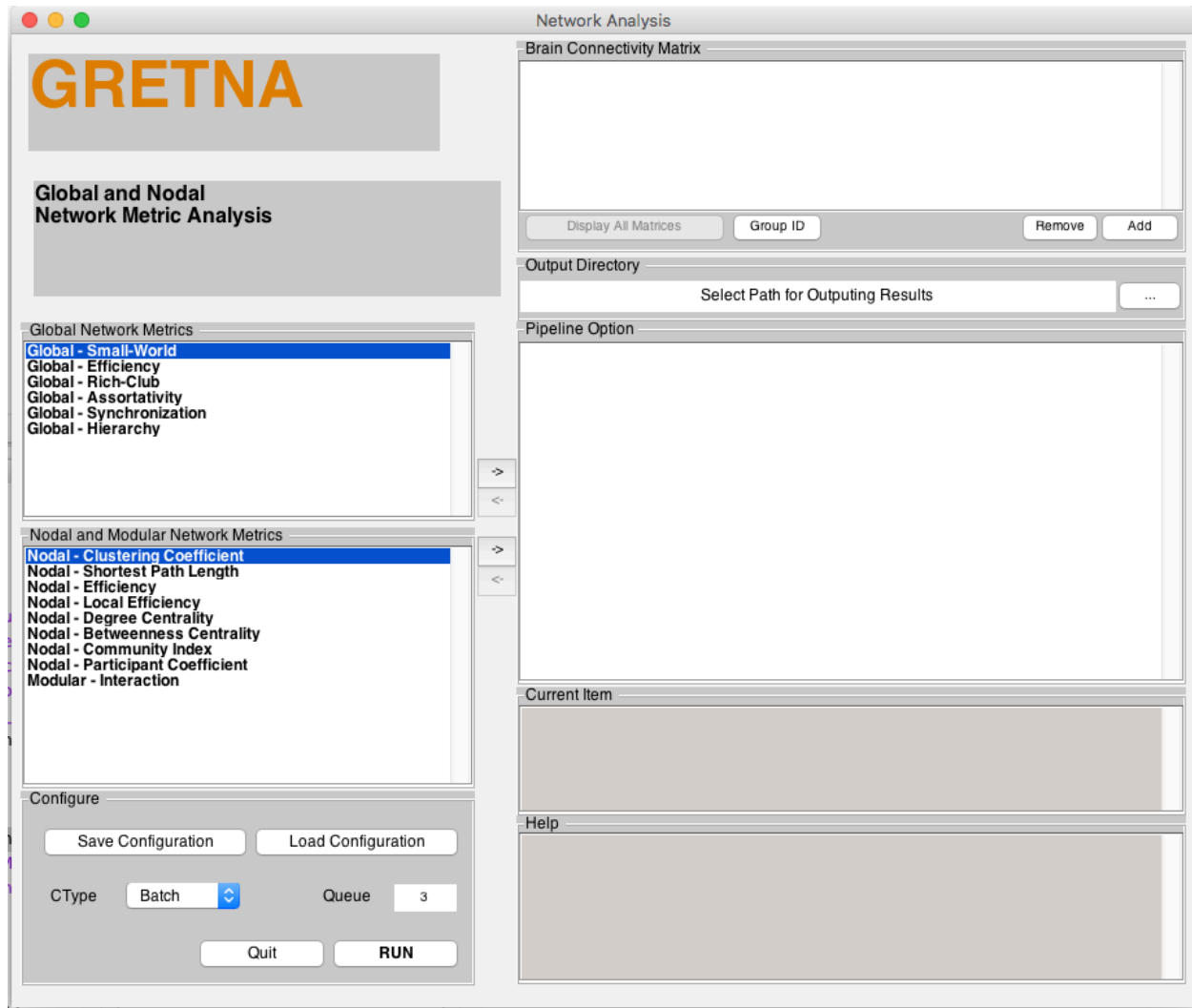
```
load('Subject_CM.mat');% change the mat file name accordingly
[avergEloc nodal_Eloc] = gretna_node_local_efficiency_weight(CM);
[avergEg nodal_Eg] = gretna_node_global_efficiency_weight(CM);
[averccl nodal_cc] = gretna_node_clustcoeff_weight(CM,1);
[avercd nodal_dc] = gretna_node_degree_weight(CM);
[averglp, nodal_lp] = gretna_node_shortestpathlength_weight(CM);
save('Subject_graph','avergEloc','nodal_Eloc',...
    'avergEg','nodal_Eg','averccl','nodal_cc','averglp','nodal_lp','avercd','nodal_dc');% change the
output file name accordingly
```

1) Start the GREYNA toolbox in MATLAB

To start GREYNA, simply type “gretna” in a MATLAB command window, a GUI will pop up:

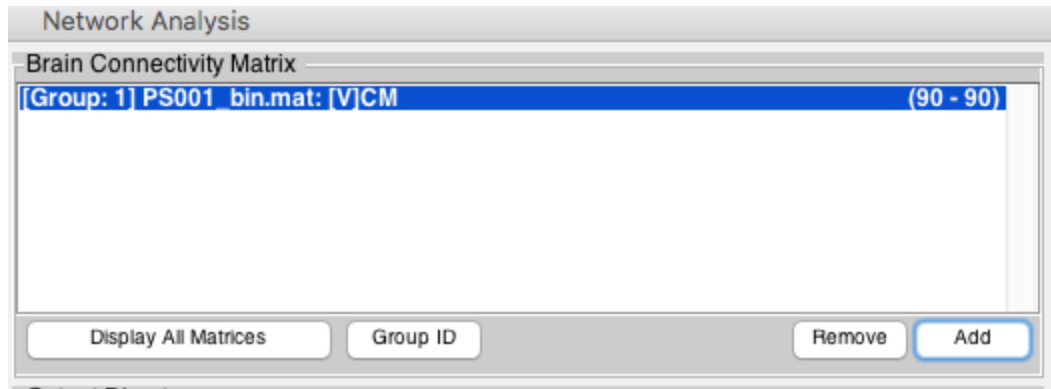


Then select “Network Analysis”, then another window will pop up:



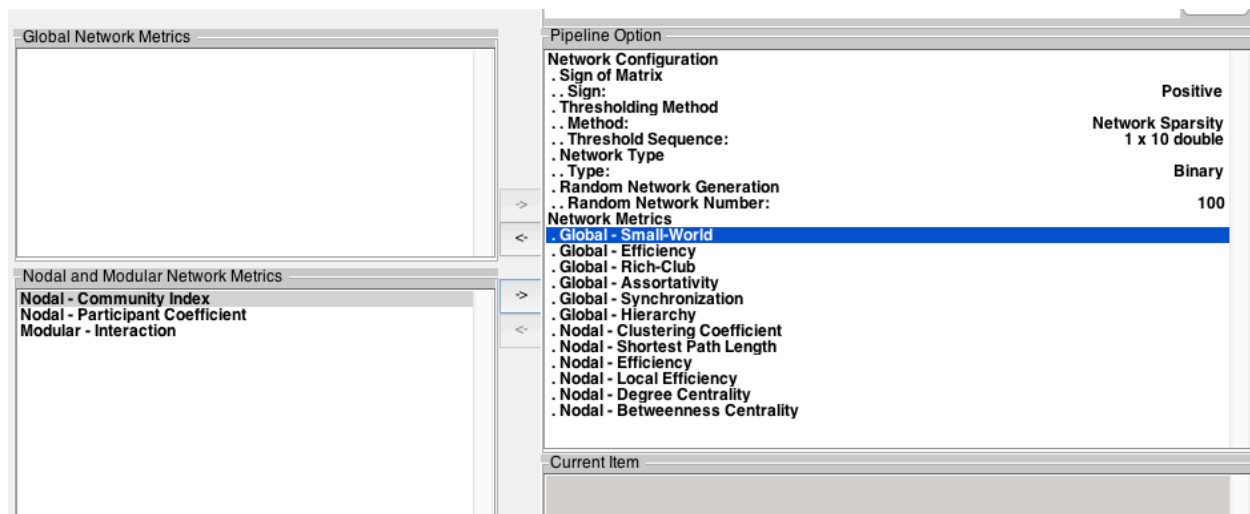
2) Input the mat files that contains connectivity matrix

Simply click “Add” at the upper right under “Brain Connectivity Matrix” and select the mat file that contains a 90 x 90 connectivity matrix. You can select multiple mat files as inputs. For each matrix, there is a “(90 - 90)” in the GUI:



3) Select graph-theory based metrics

This step will choose which metrics will be calculated. Generally we choose as much as possible. Click “->” at the left side of the GUI for both global and nodal metrics:



For the community analysis, there are two ways to get the community index for the 90 AAL regions. The community index assigns the 90 AAL regions into several separated “communities” or “sub-networks”. One way is simply input a pre-defined community index. Here we use the pre-defined community index from Yeo et al., 2011 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174820/>), which contains 7 sub-networks. This community index (a 1 X 90 array) is below:

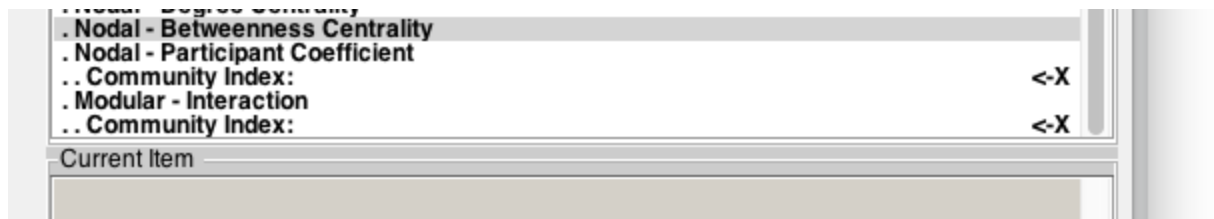
2	2	3	3	5	5	6	6	6	6	6	6	6
6	7	7	2	2	2	2	5	5	7	7	7	7
5	5	4	4	7	7	4	4	7	7	7	7	5
5	5	5	1	1	1	1	1	1	1	1	1	1
1	1	1	1	2	2	3	3	6	6	4	4	7

7	7	7	2	2	8	8	8	8	8	8	8	8
2	2	2	2	5	5	7	7	5	5	3	3	

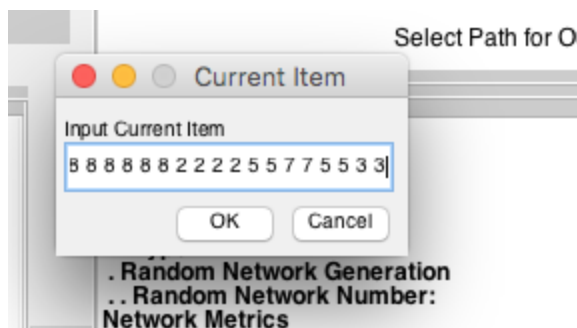
It's a 90 numbers array and each number represents one AAL region. Each number represents one network as below:

- 1: Visual network
- 2: Somatomotor network
- 3: Dorsal Attention network
- 4: Ventral Attention network
- 5: Limbic network
- 6: Fronto Parietal network
- 7: Default Mode network
- 8: Deep gray matter network

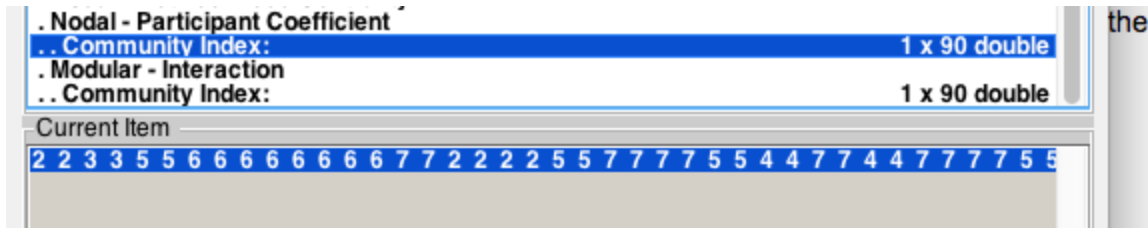
So, in this case, **please only** click “->” on “Nodal-Participant Coefficient” and “Modular-Interaction” at the nodal metrics window:



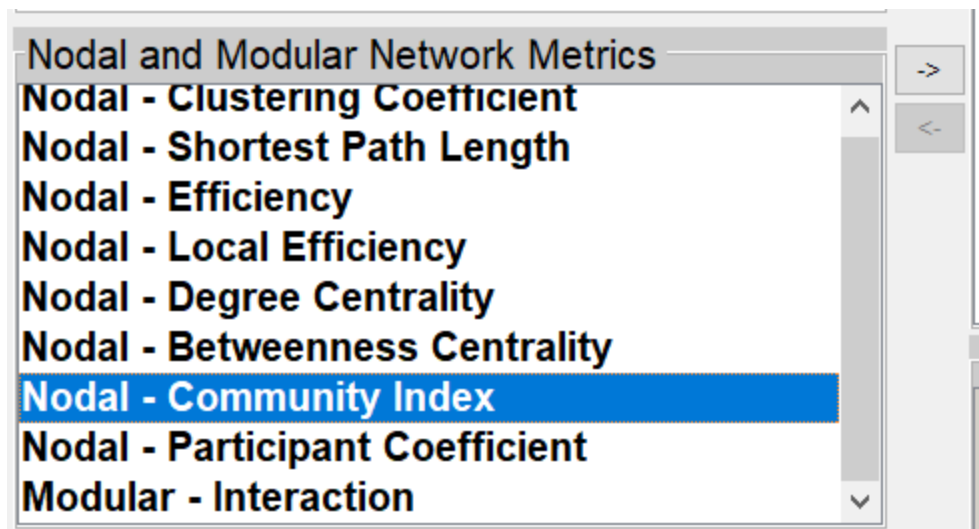
Then click “<-X” at the first “Community index” on the right window below the “Nodal - Participant Coefficient” option to input the community index. Copy the 90 numbers community index from above and paste them in the “Input Current Item”:



After press “OK”, the GUI should look like this:

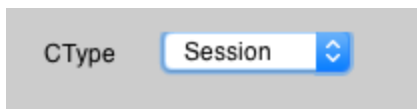


If you don't want to pre-defined the community index, you could also use "Nodal - Community index" under the "Nodal and Modular Network Metrics" panel to estimate the number of communities, but sometimes the output only includes 2 or 3 communities. **The "Nodal-Community Index" option can not be estimated at the same time with the predefined index (the Yeo version mentioned above).**



4) Configure other parameters:

In the Network Analysis GUI, change "CType" from "Batch" to "Session" because "Batch" option might have some bugs in Mac OS:



Choose an output directory to store outputs:



Configure the calculation settings:

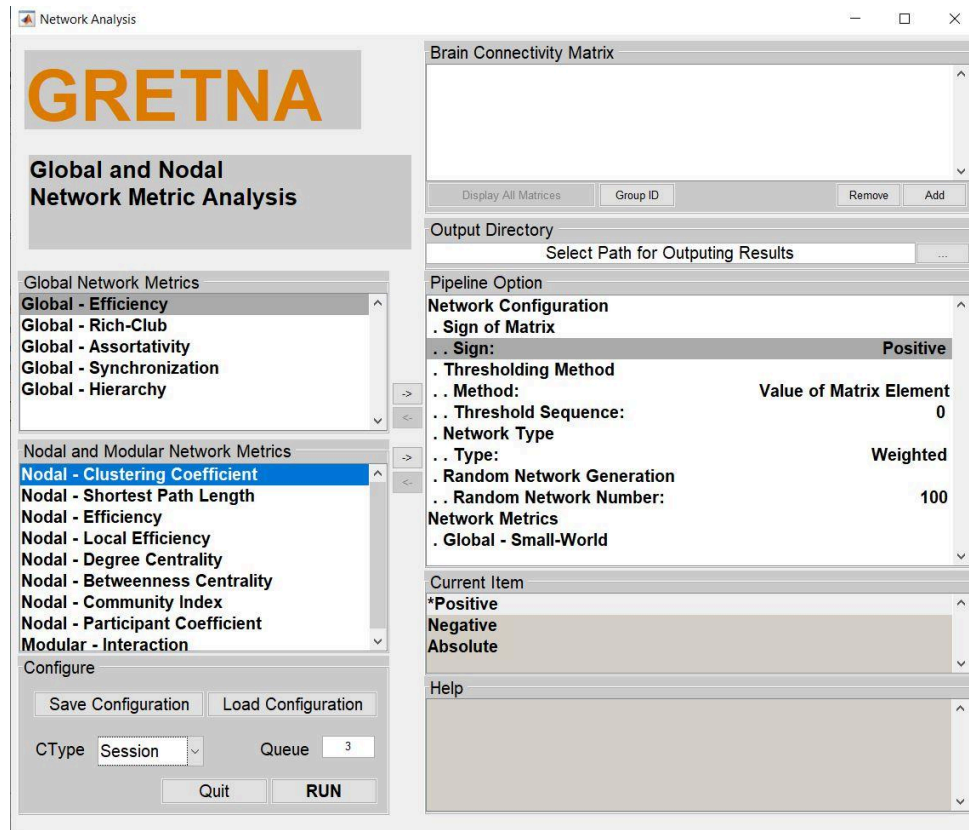
Pipeline Option	
Network Configuration	
. Sign of Matrix	
.. Sign:	Absolute
. Thresholding Method	
.. Method:	Value of Matrix Element
.. Threshold Sequence:	0
. Network Type	
.. Type:	Binary
. Random Network Generation	
.. Random Network Number:	100
Network Metrics	

The “Sign of Matrix” means which value should be included. Generally we used absolute or positive value for the connectivity matrix.

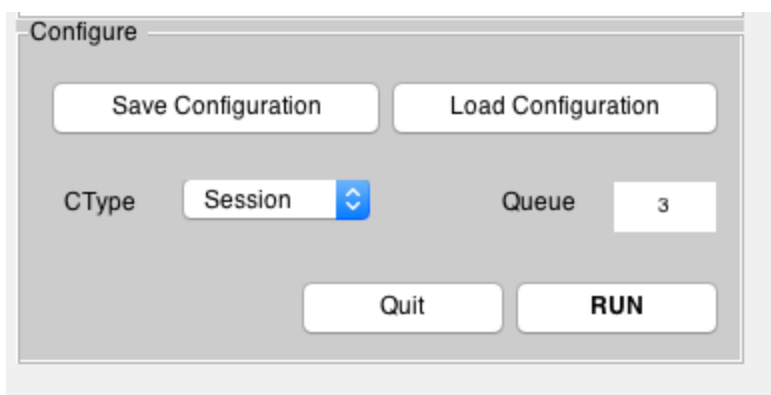
The “Thresholding Method” is not available for the binarized matrix, so we choose “Value of Matrix Element” under the “Method” option and set the “Threshold Sequence” to “0”. You can also set certain thresholds for the weighted matrix (which means each element in the connectivity matrix is a quantitative value).

Choose “Network Type” to “Binary” since the current input connectivity matrix is a 0 & 1 matrix.

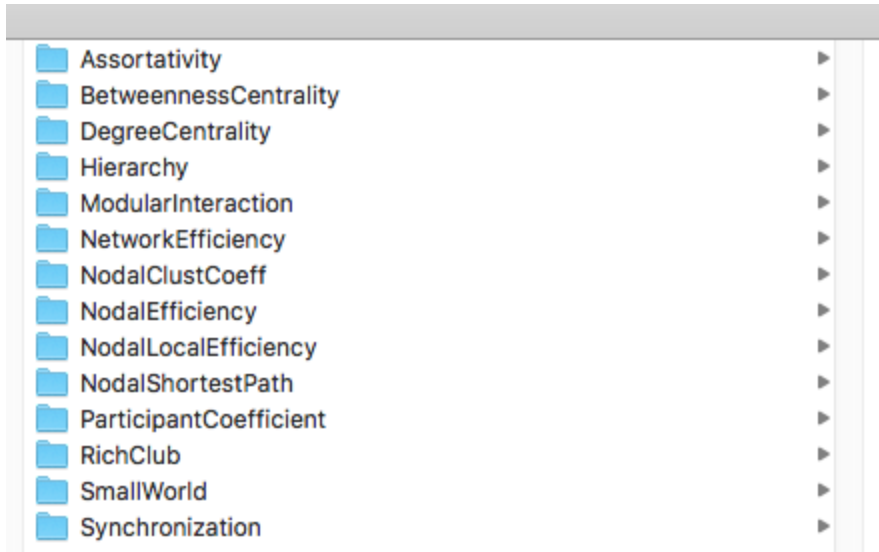
Here are the settings for the weighted matrix if needed:



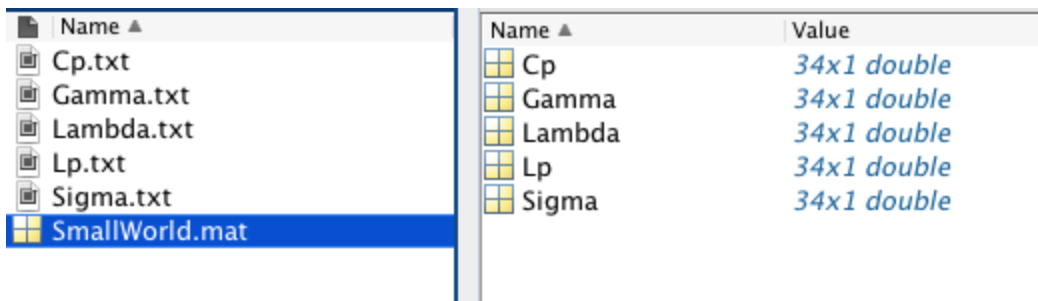
Then click “Run” in the GUI:



The output includes several folders:



Each folder stores the specific metric for all participants/datasets. You can load them in MATLAB, such as typing “load(‘SmallWorld.mat’)” or double clicking “SmallWorld.mat” in the “SmallWorld” folder, and run second-level analysis, something like this:



Here the “Cp” contains the clustering coefficient of all participants/dataset, in this case, 34 participants. The order of the value is the same as when the matrices were input in the GRETNA GUI.

The GRETNA toolbox provides a way to do permutation tests for the group-comparison analysis. It looks like this:

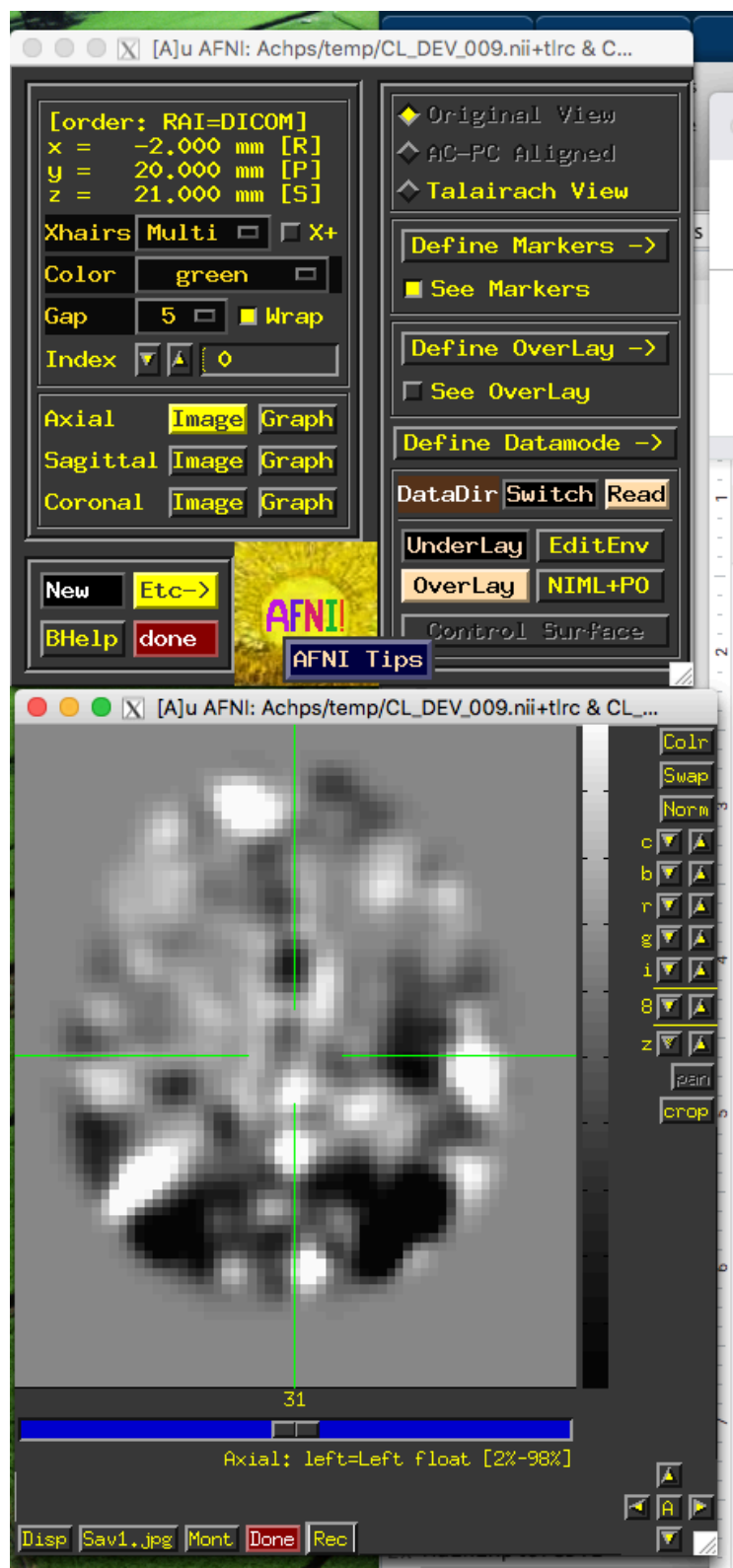
```
[perm, P] = gretna_permutation_test(alldata, group1id, group2id, 10000, allcovariates);
```

Here “alldata” includes the data from both group 1 and group 2; “group1id” and “group2id” indicates the index of each group in the “alldata”; “10000” means the permutation will run 10000 times; “allcovariates” includes all covariates such as age and sex for both groups, this option can be empty. The output “perm” includes the difference of mean value from the raw data and the permutation data; “P” indicates the p-value based on permutation analysis.

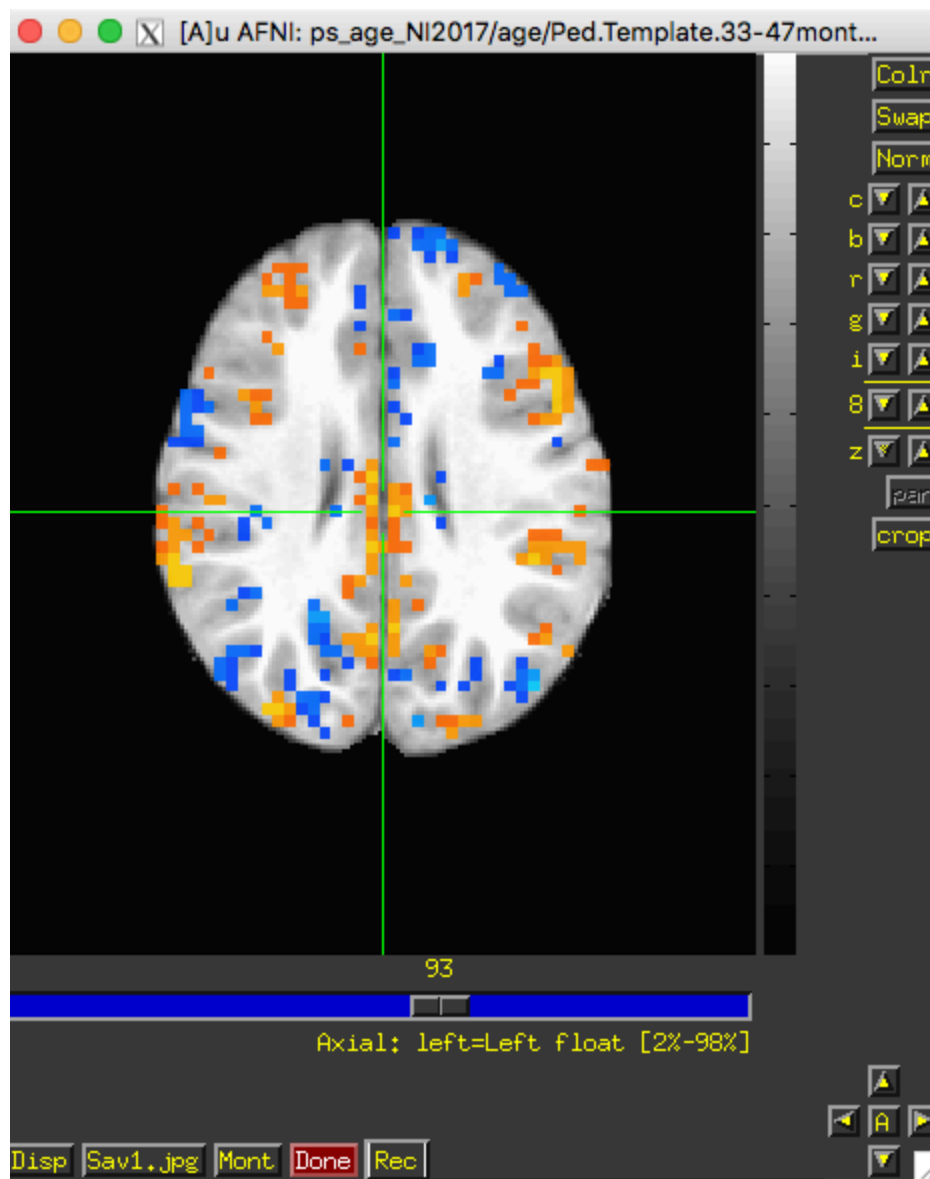
9. Display the results

9.1 Display the results in volumetric space

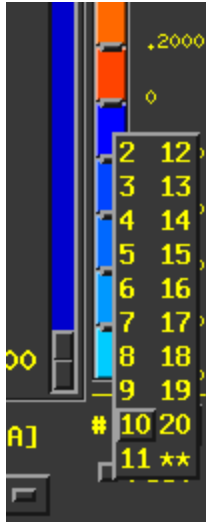
Generally I use AFNI to display results in volumetric space. Type afni in the Terminal then a GUI will pop up:



Click "Underlay" to select a background image, for instance, a T1 image, then click "Overlay" to choose the image you'd like to merge on the background image:



You can change the colour bar by clicking “***” and change to “10”, so the warm colour will be values above zero and the cold colour will be values below zero:



You can also move the bar at the left side to change the thresholds of Overlay image.

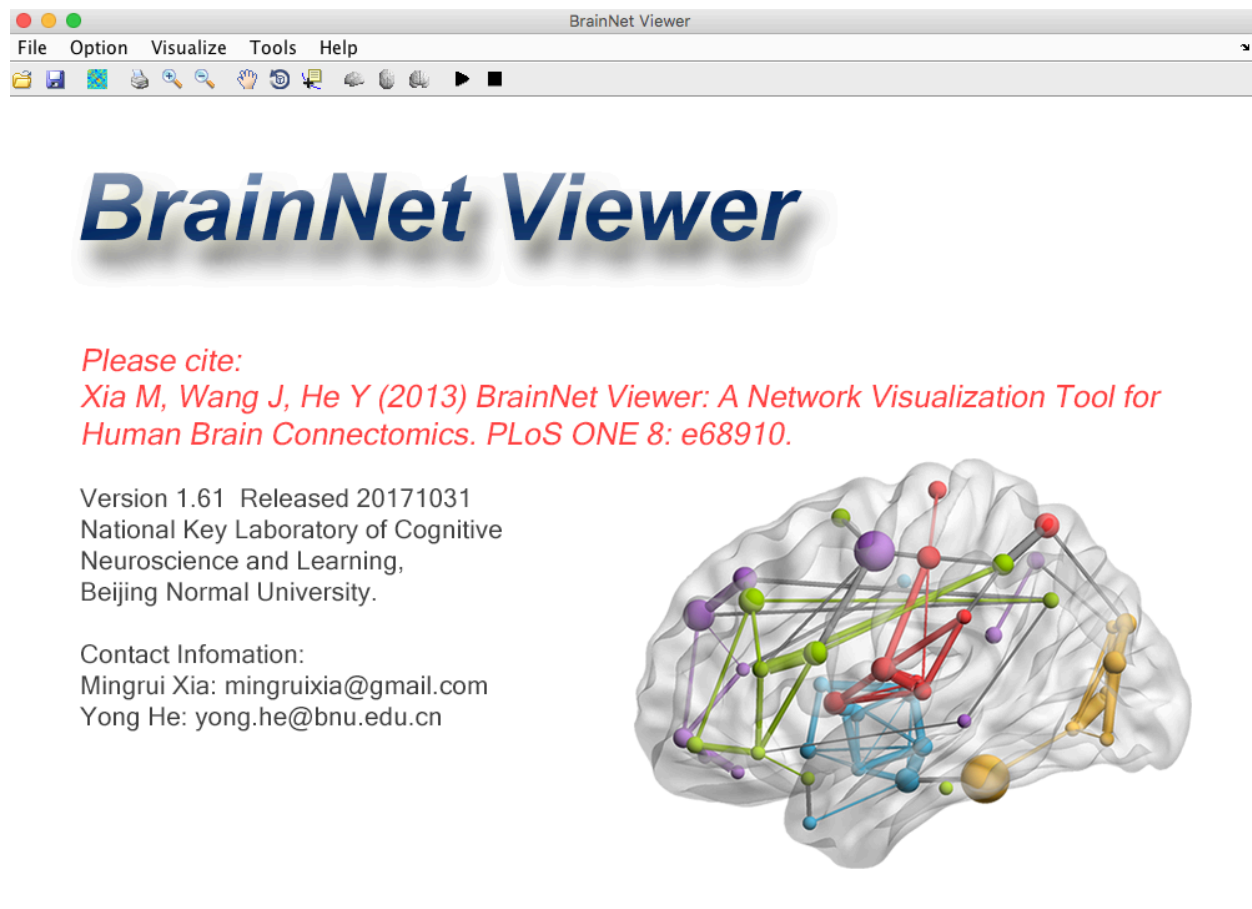
AFNI sometimes incorrectly display MNI images in Talairach space. To fix this, run the following code in Terminal on the Underlay image, for example, oldT1.nii:

```
nifti_tool -mod_hdr -prefix newT1.nii -infile oldT1.nii -mod_field sform_code 2
```

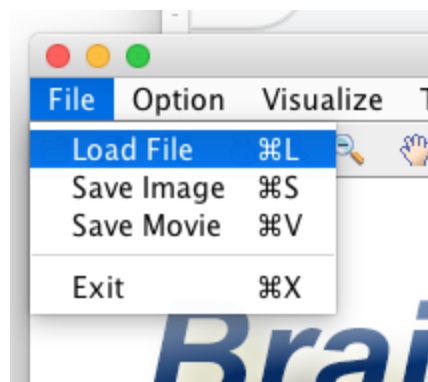
Then the output newT1.nii will display correctly in AFNI.

9.2 Display the results on brain surface

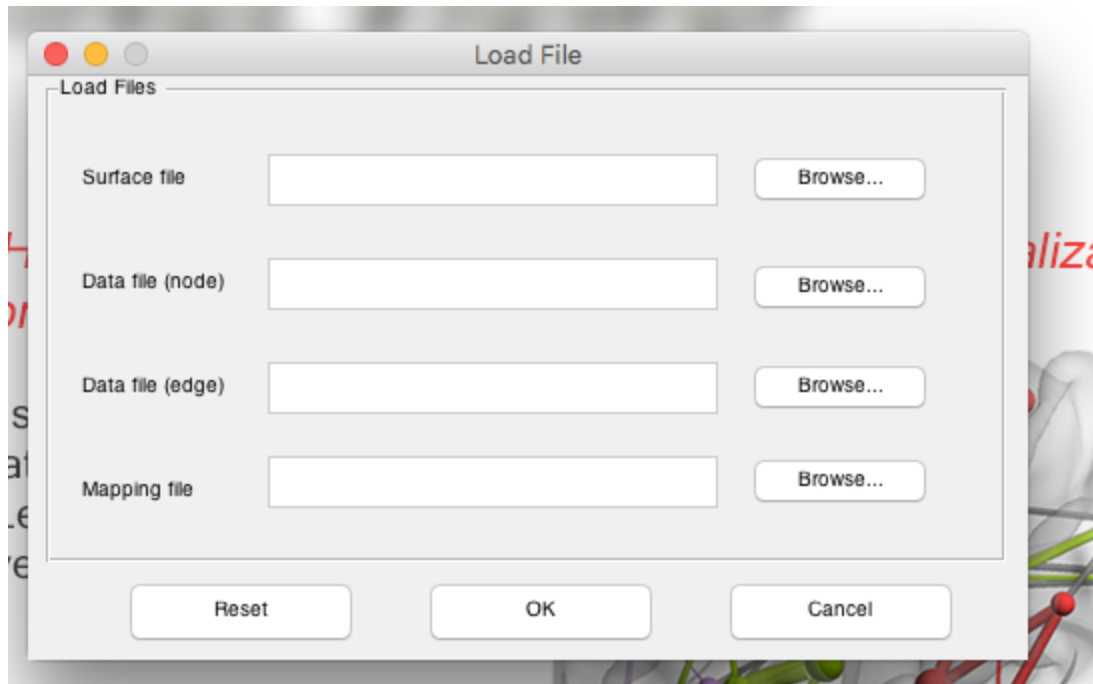
The volumetric images in standard MNI space can be also displayed on a surface map by a MATLAB toolbox named BrainNet Viewer. Type “BrainNet” in MATLAB command window, a GUI will pop up:



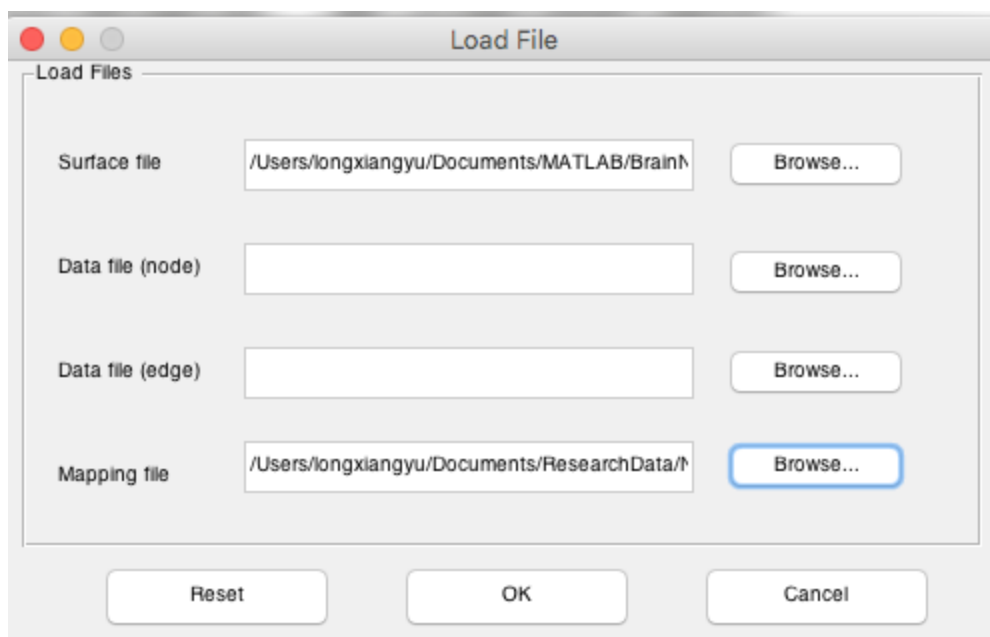
Select “File” then “Load File”:



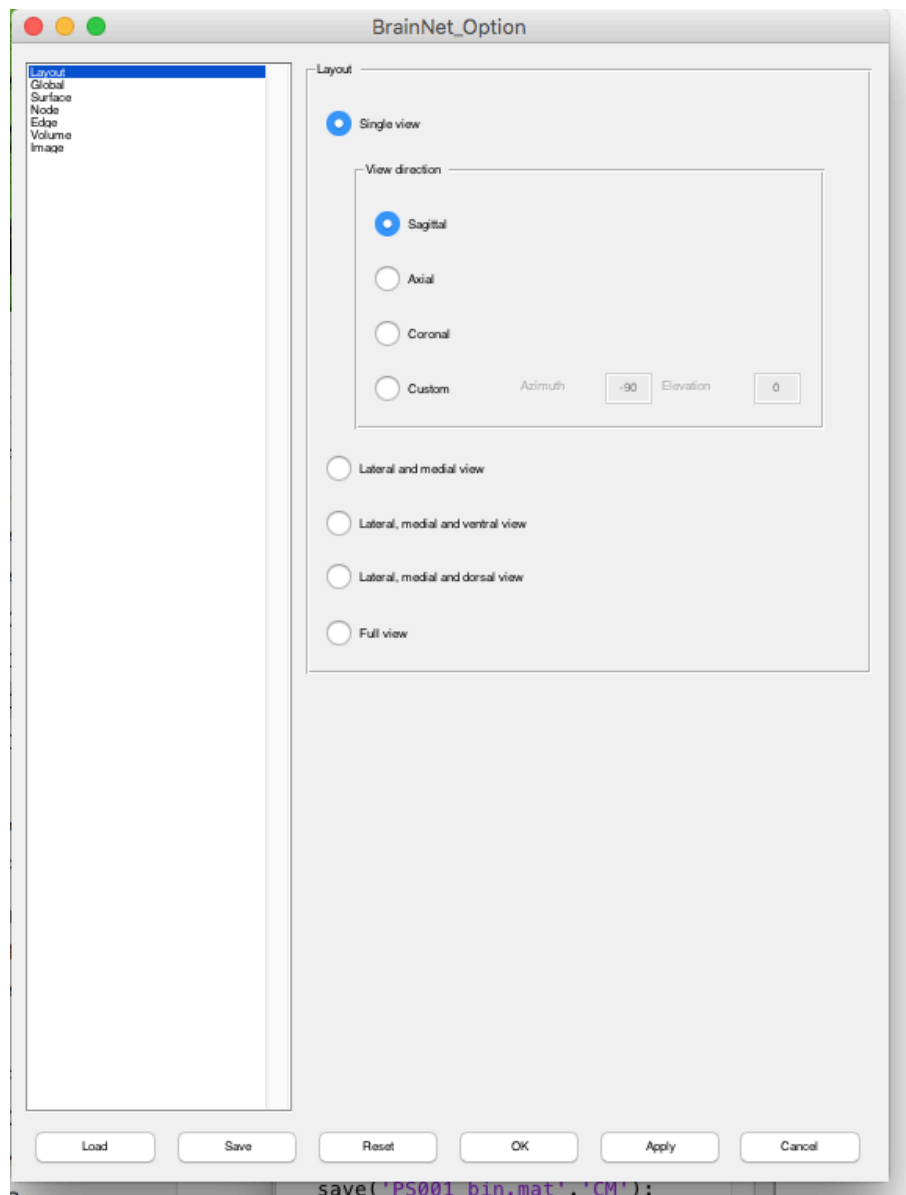
A window will pop up:



Click “Browse...” at the “Surface file” row, and choose “BrainMesh_ICBM152_smoothed.nv”. This template is in the standard adult MNI space, so make sure your NIFTI image is also registered to the adult MNI space. The toolbox will automatically select this file when clicking the “Browse...” button, if not, this nv file is located at /BrainnetViewer/Data/SurfTemplate/. Then click “Browse...” at the “Mapping file” row to select a volumetric image which is in standard MNI space:

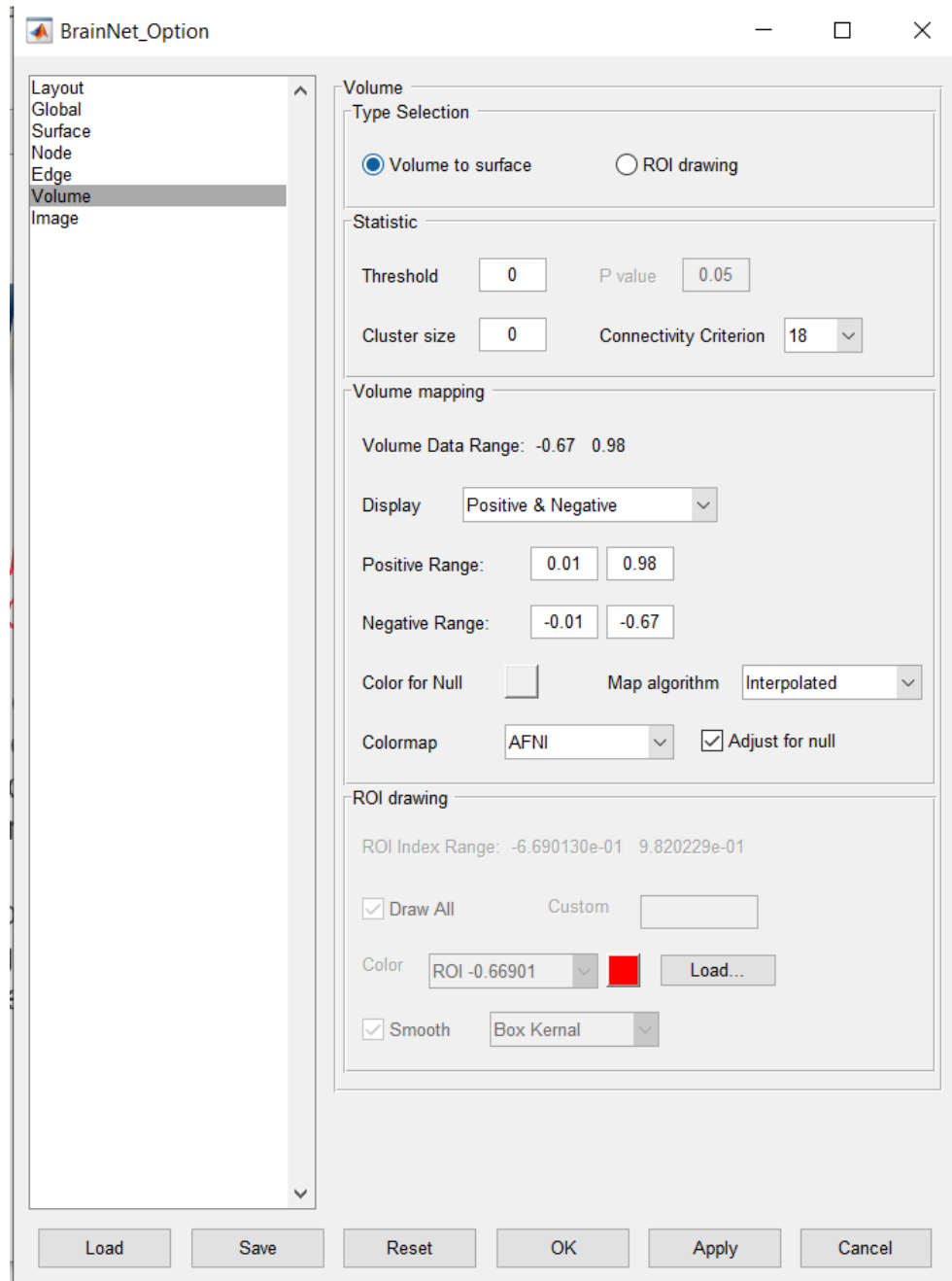


Then click “OK”, a window will pop up:

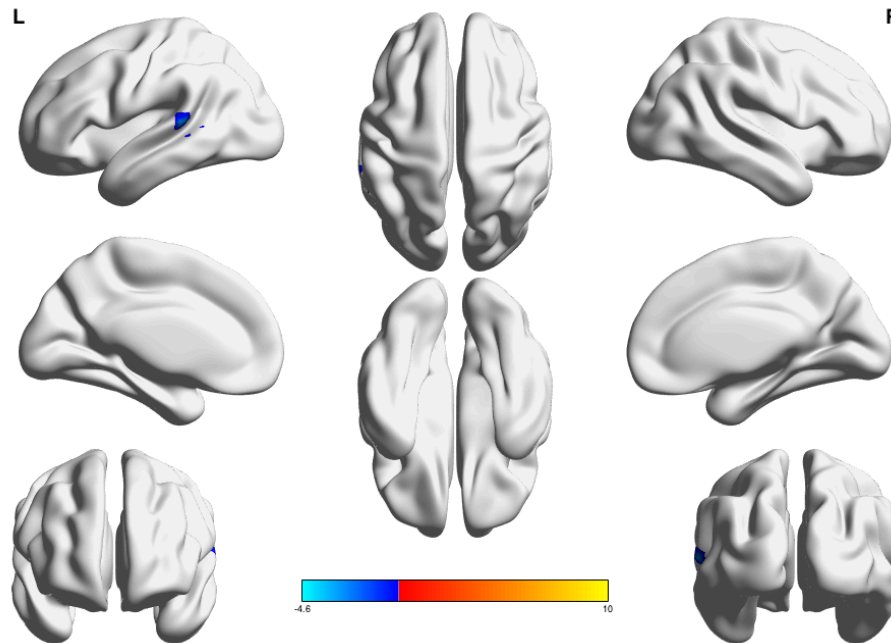


The first option is to select the view. You can try “Full view” which displays multiple views of the surface or “Lateral and medial view” to just display two views.

Then select “Volume” to define colors and thresholds. You can choose “Positive & Negative” in the “Display” option, and change the “Colormap” to “AFNI”. Also check the “Positive Range” and “Negative Range” to make sure they fall into the correct ranges of your data. Then click “OK”:



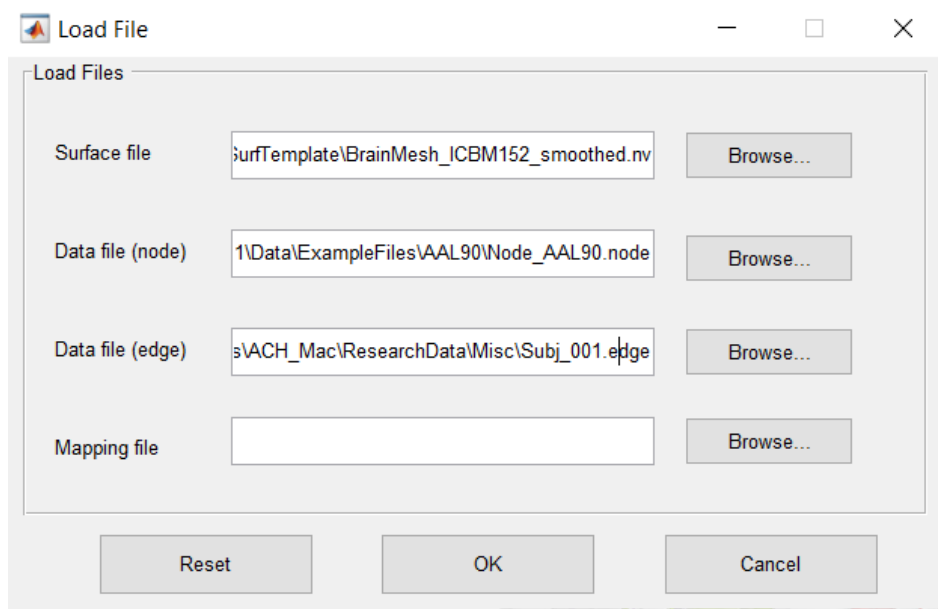
GUI will display the results like this (with “Full view”):



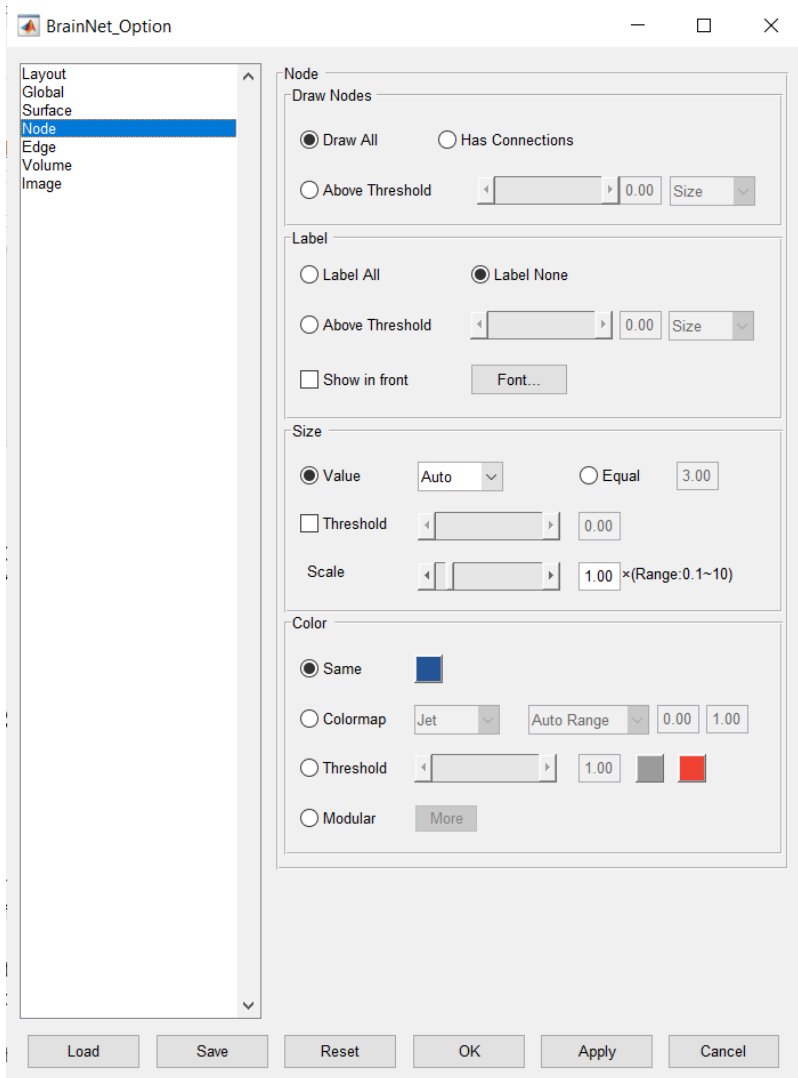
9.3 Display the connectivity matrix on brain surface

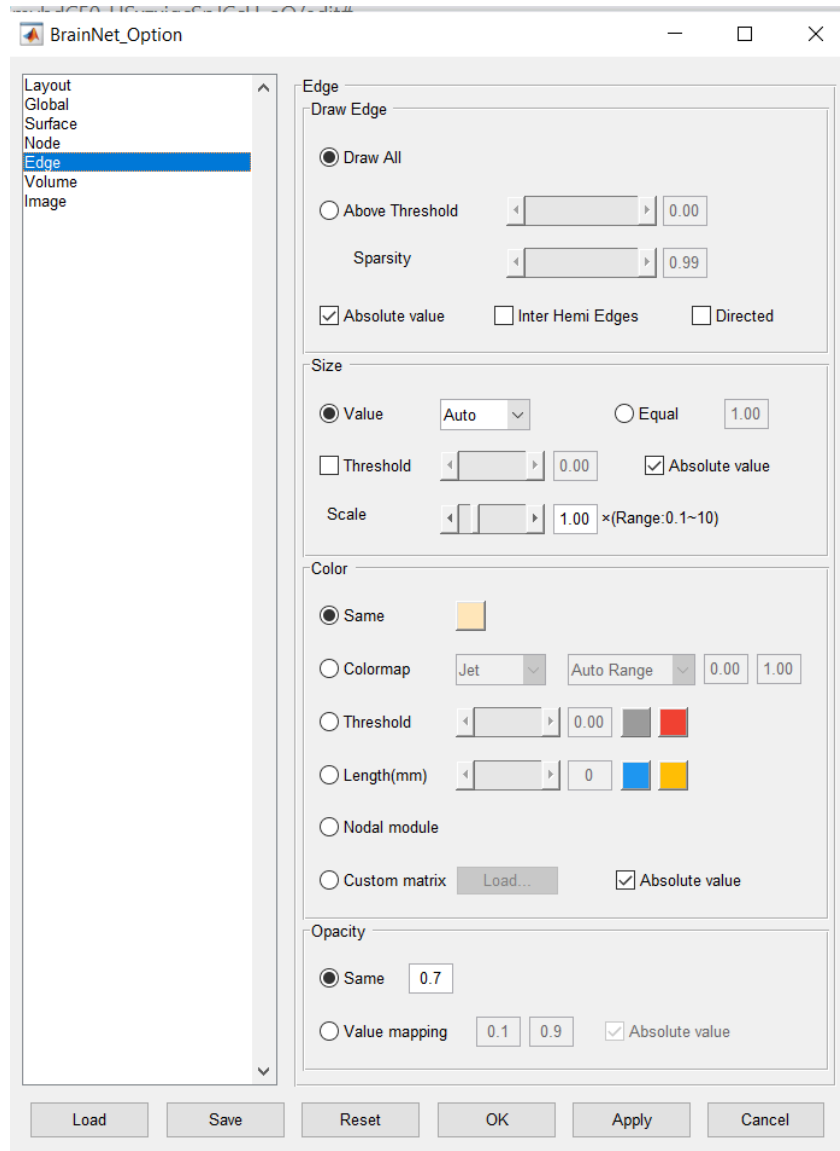
The first step would be creating an “edge” file for the toolbox BrainNet Viewer. It’s basically a text file with “edge” as the suffix, so you could first create a text file such as “subj_001.txt”, then, load the mat-file which only includes one connectivity matrix (generally 90 X 90 from the AAL template) in MATLAB, and copy the contents of the matrix from Workspace in MATLAB to the text file “subj_001.txt”, then change the suffix to “subj_001.edge”.

Then just like the procedure in 9.2, after selecting the surface file, then click “Data file (node)” and choose “Node_AAL90.node” in the directory of “/BrainNetViewer_20171031/Data/ExampleFiles/AAL90/”, then click “Data file (edge)” and choose the edge file you just created (e.g., “subj_001.edge”):

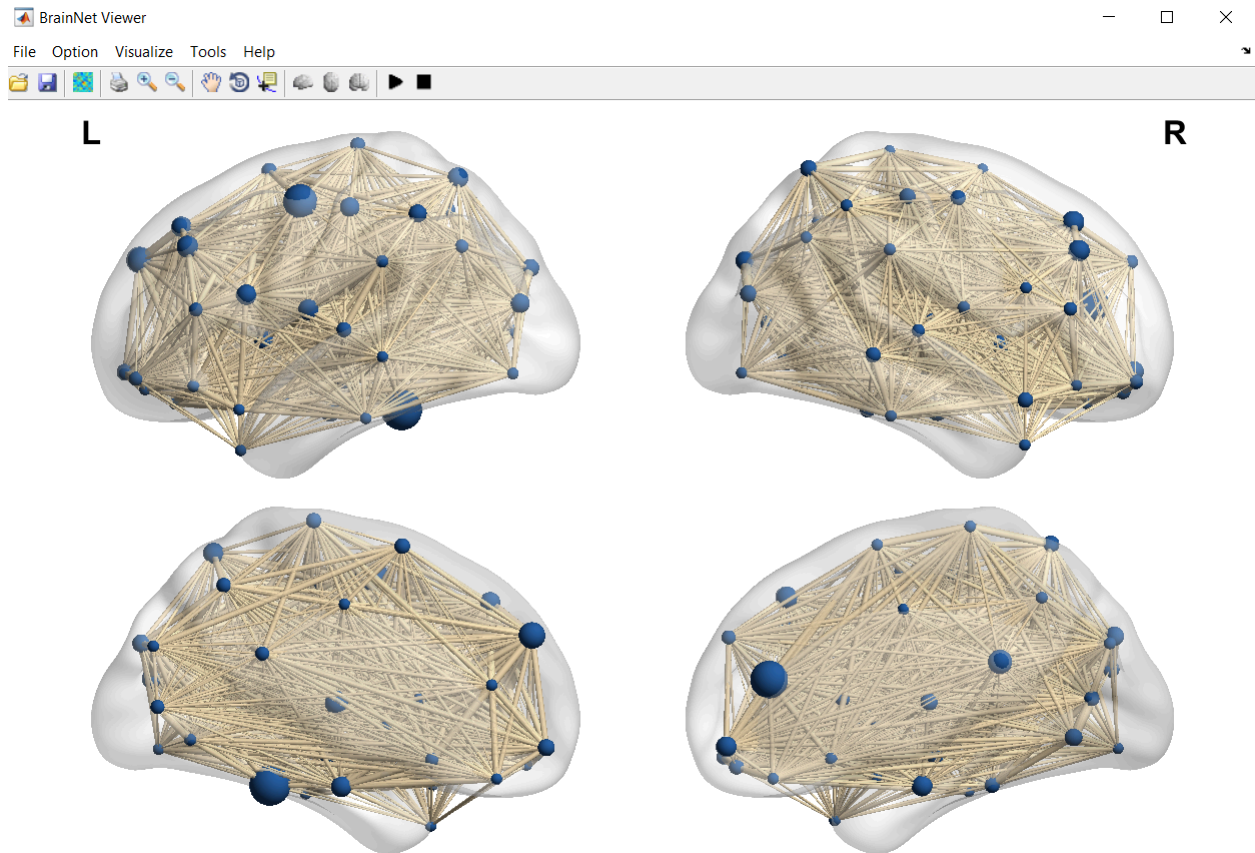


Then click “OK” and an Option menu will pop up. You could play around the settings in the “Node” and “Edge” tabs to change the colors or sizes of nodes and edges, such as thresholds of the matrix or width of the edges:





After finishing setting the properties of nodes and edges, then click “OK” and the connectivity matrix will be displayed like this (it might take a few minutes), you can always go back to the Option in the top menu to adjust nodes and edges.



10. Possible problems you might encounter

1) “File not found” problem:

This type of problem generally happens if a program/code/script can not locate the input of the data. The programs can not “automatically” find the location of the input if the input only contains a “filename”. Therefore, either run the program at the directory of the input data:

```
cd /your_data_directory/
3dcalc -a input.nii -expr 'a*1' -prefix output.nii
```

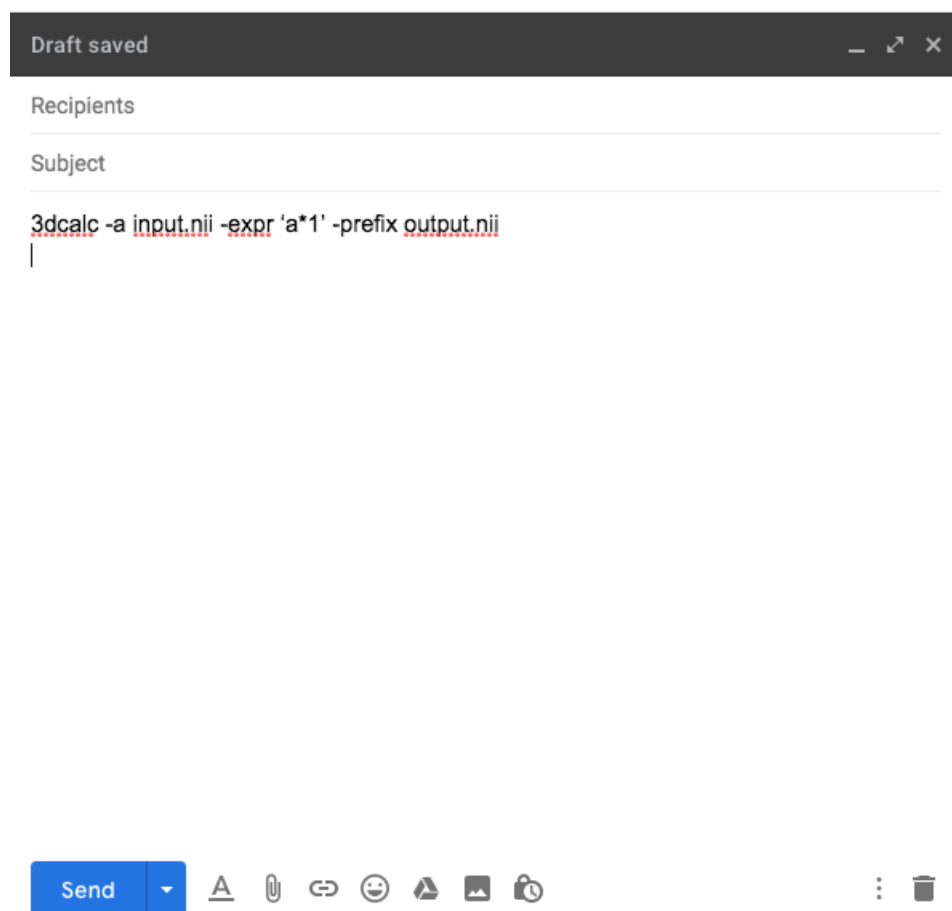
or put the whole directory with the data in the input:

```
3dcalc -a /your_data_directory/input.nii -expr 'a*1' -prefix /your_output_directory/output.nii
```

You can always use the “pwd” command in Terminal or MATLAB to see where you are running the command at the moment. Also always check if the data was “corrupted”, such as the data can not be opened by other software or has little storage space.

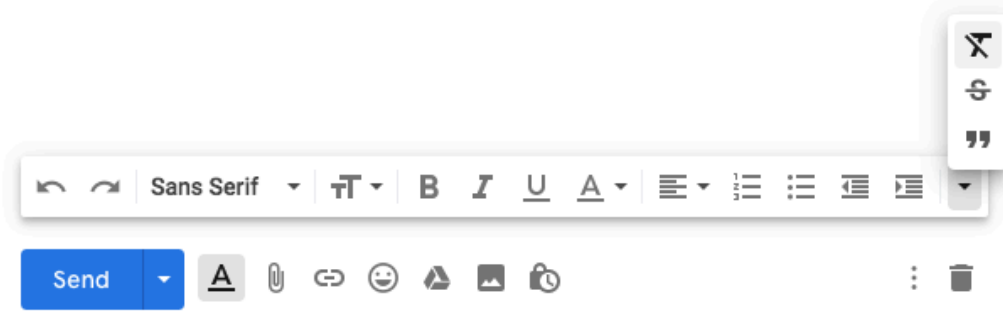
2) Formatting problem:

This type of problem happens when the command line or a script includes some letters or symbols with formats that can not be identified in the Terminal. For example, symbol “ ’ ” and symbol “ ‘ ”, only symbol “ ' ” can be identified as a letter in a command. To avoid the potential of adding formats, either use a simple text editor, or copy the command in a Gmail draft:



Then select the command line, and click “A” with an underline and select “T” with a slash at the bottom menu to remove the format of the command:


```
3dcalc -a input.nii -expr 'a*1' -prefix output.nii
```



This procedure will remove all formats in the command and you can copy the command and paste in Terminal to run it.

3) Use Google to search the exact error message

Sometimes there might be bugs in the software and there might already be some discussions and problem solving in the forum of the software. You can just Google the error message you saw in Terminal or MATLAB and find solutions from the internet.

4) The final straw: my (personal) email address: longspace82@gmail.com